APPENDIX 6. WIRELINE LOGS (IMAGE)

Weatherford

MLL- SLL - DLL - SONIC
DENSITY NEUTRON
1:100

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA
LOCATION	PL198

FIELD PRINT

Latitude	27°23'18.4689" S	Other Services
Longitude	151°6'16.9779" E	BOREHOLE NAVIGATION
UTM Easting	312 589.65	
UTM Northing	6 969 111.4	

Permanent Datum M.S.L., Elevation 0.00 metres
Log Measured From GL, 345.10 metres above Permanent Datum
Drilling Measured From KB

Elevations:	metres
KB	349.55
DF	349.55
GL	345.10

Date	18-OCT-2019		
Run Number	1		
Service Order	90518		
Depth Driller	565.55	metres	
Depth Logger	560.75	metres	
First Reading	560.23	metres	
Last Reading	0.00	metres	
Casing Driller	164.55	metres	
Casing Logger	164.40	metres	
Bit Size	215.900	inches	
Hole Fluid Type	KCL		
Density / Viscosity	1.01 g/c3	26.00 sec/qt	
PH / Fluid Loss	8.50	0.00 ml/30Min	
Sample Source	MUD TANK		
Rm @ Measured Temp	0.302 @ 25.0	ohm-m	
Rmf @ Measured Temp	0.275 @ 25.0	ohm-m	
Rmc @ Measured Temp	0.453 @ 25.0	ohm-m	
Source Rmf / Rmc	CAL	CAL	
Rm @ BHT	0.218 @ 43.0	ohm-m	
Time Since Circulation	8 HRS 37 MIN		
Max Recorded Temp	43.00	deg C	
Equipment / Base	13354	ROMA	
Recorded By	K. SHIBESHI		
Witnessed By	M. GALBRAITH		
Stop Circulation	06:00 / 18 OCT 2019		

BOREHOLE RECORD

Last Edited: 18-OCT-2019 11:45

Bit Size millimetres	Depth From metres	Depth To metres
311.150	0.00	164.95
215.900	164.95	565.55

CASING RECORD

Type	Size millimetres	Depth From metres	Shoe Depth metres	Weight pounds/ft
SURFACE	244.475	0.00	164.55	36.00

REMARKS

RUN NUMBER 1 IS THE PRIMARY DEPTH REFERENCE LOG. ALL OTHER RUNS ARE CORRELATED BACK TO THIS LOG.

SOFTWARE ISSUE: VERSION 19.01.8879 RELEASED DATE APRIL 1, 2019.

CUSTOMER SCALES AND INTERVALS LOGGED.

RUN 1: MMR, MLE,MUG, MPD, MDN, MBN, MCG, MBE, MBE MTA, CBH TOOLS RAN IN COMBINATION.

HARDWARE
RUN 1:
- MBE: 1 X 0.5" STANDOFF ON EACH.
- MUG: 1 X 0.5" STANDOFF.
- MMR: 2 X 0.5" STANDOFF.
- MDN: 1 X DUAL ECCENTRALIZER.

TIME ON BOTTOM: 14:35 / 18 OCT 2019.

ALL DEPTH REFERENCE TO GROUND LEVEL.RT TO GL IS 4.45 M.

LOGGER TOTAL DEPTH: 560.75 mGL.

MMR CALIPER CLOSED 5 METRES BELOW CASING SHOE TO AVOID DAMAGE PAD.

MPD AND MDN CORRECTED FOR DENSITY CALIPER AND MUD DENSITY.

TOTAL HOLE VOLUME (HVOL) FROM TD M TO 9.625" SURFACE CASING SHOE = 14.2 CUBIC METRES.

TOTAL ANNULAR VOLUME (AVOL) FROM TD M TO CASING SHOE WITH 7" CASING = 4.66 CUBIC METRES.

MAXIMUM TEMPERATURE RECORDED 1.59 DEG AT 167.1 METRES.

REPEAT SECTION RECORDED FROM 545 M TO 385 M.

CONVEYANCE TYPE: WIRELINE.

BOREHOLE STATUS: OPEN HOLE.

RIG: SLR 167.

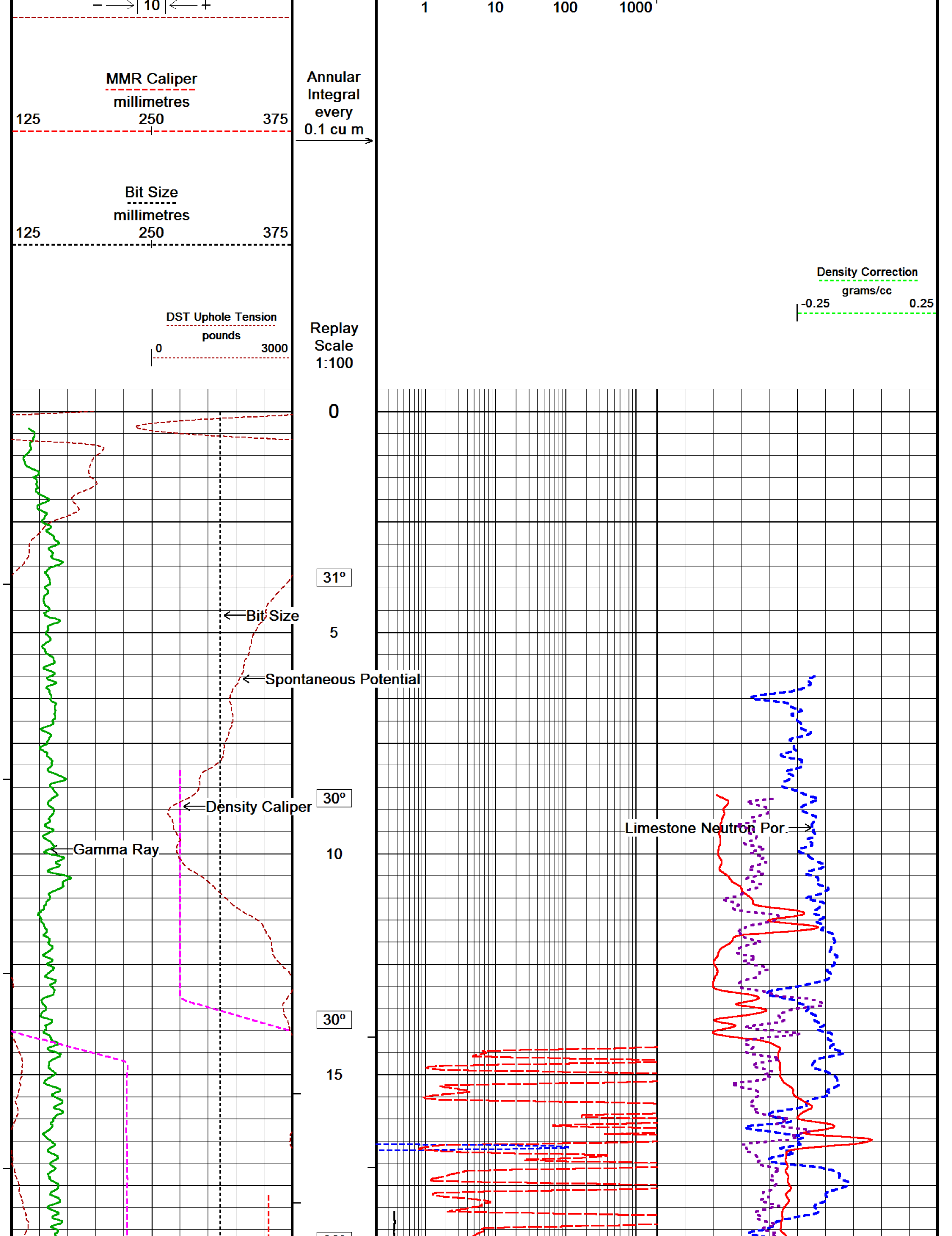
SERVICE REPORT NUMBER: 90518.

LOGGING CREW: ENGINEERS - K. SHIBESHI; OPERATOR: R. FRAMPTON, E. PAVIA

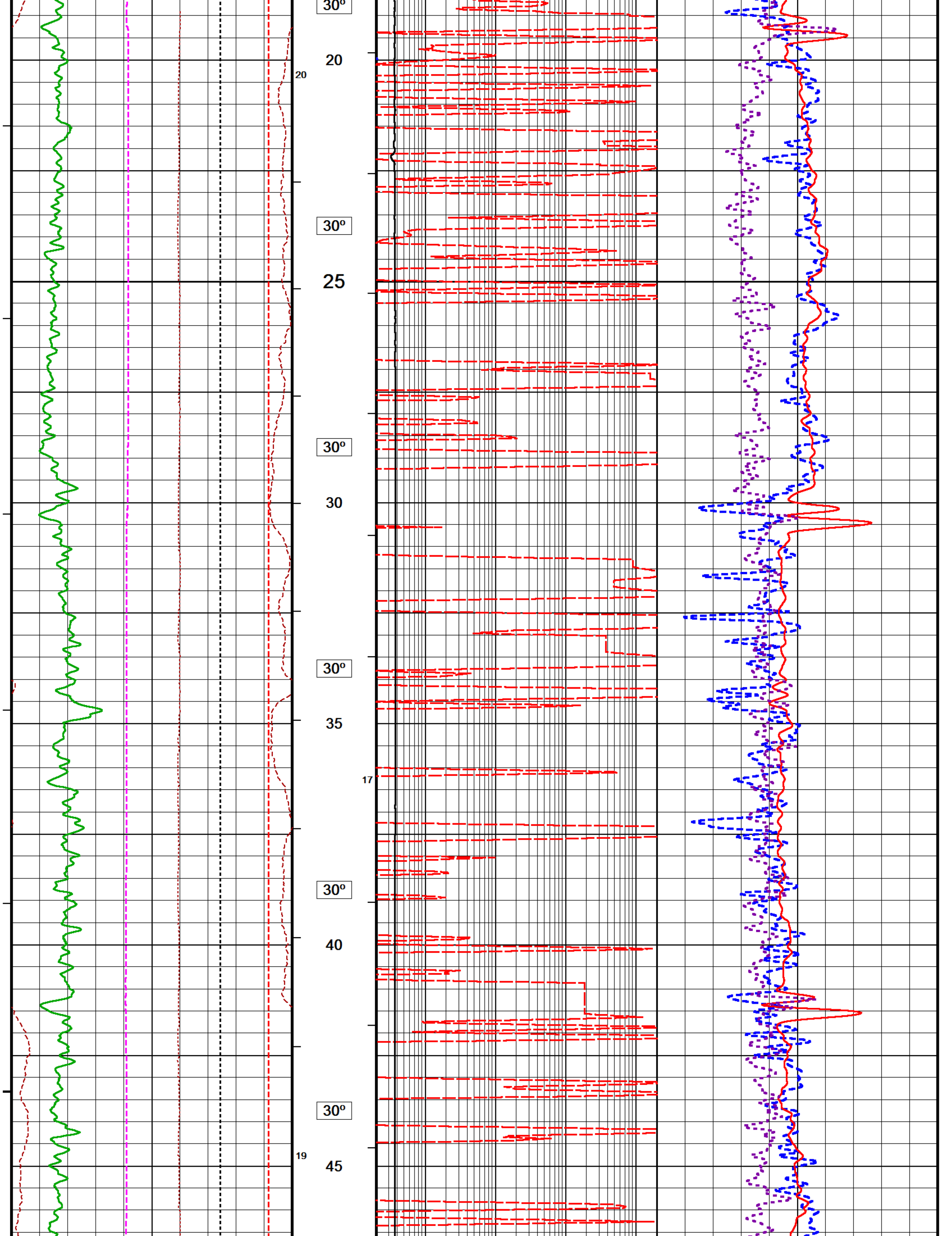
HIGH RESOLUTION 1:100

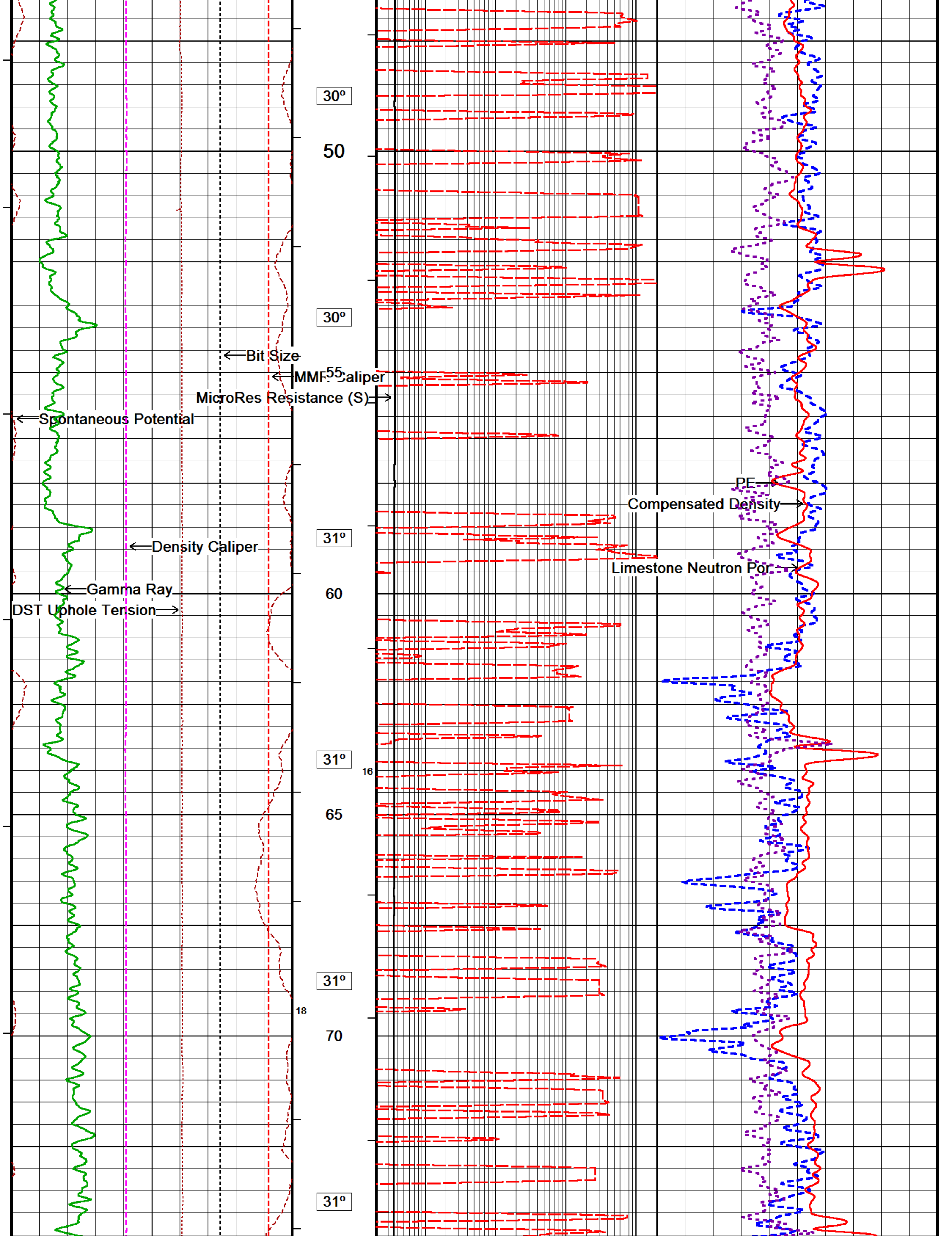
Depth Based Data - Maximum Sampling Increment 2.5cm — Plotted on 18-OCT-2019 17:46
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_2.5cm_CMP_OH_20191018.dta — Recorded on 18-OCT-2019 14:38
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

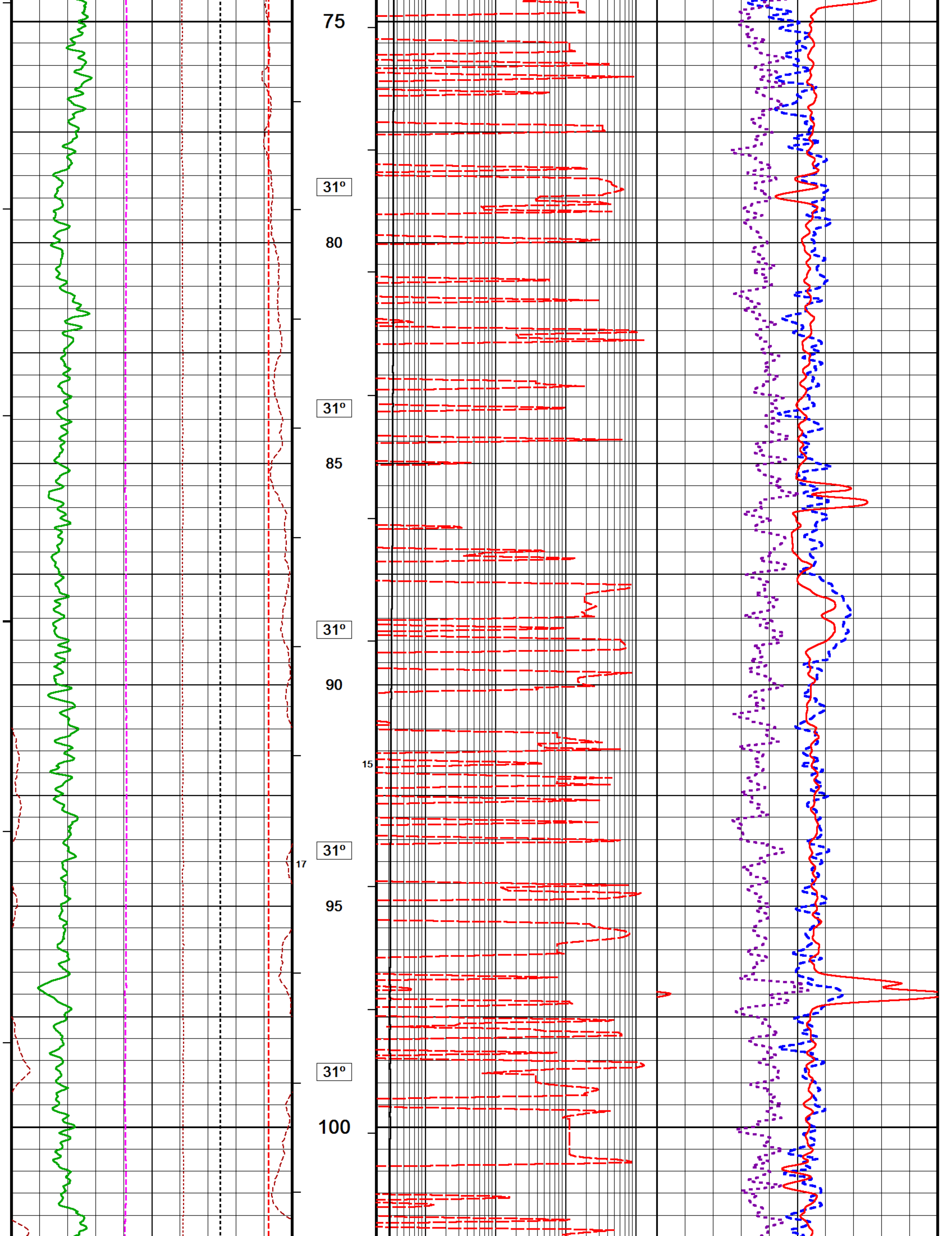
Track 1	Depth	Track 2/3	
Timing Marks every 60.0 sec	Depth in Metres	**Shallow Laterolog** ohm metres 0.20 – 2000 (1, 10, 100, 1000)	**Limestone Neutron Por.** v/v 1 – 0.40 – -0.20
Gamma Ray API 0 – 100 – 200; 200 – 300 – 400		**Deep Laterolog** ohm metres 0.20 – 2000 (1, 10, 100, 1000)	**Compensated Density** grams/cc 1 – 2 – 3
Density Caliper millimetres 125 – 250 – 375	Borehole Temp in deg C		
Spontaneous Potential millivolts	HVI every 0.1 cu m	**MicroRes Resistance (S)** ohm metres 0.20 – 2000	**PE** barns/electron 0 – 5 – 10

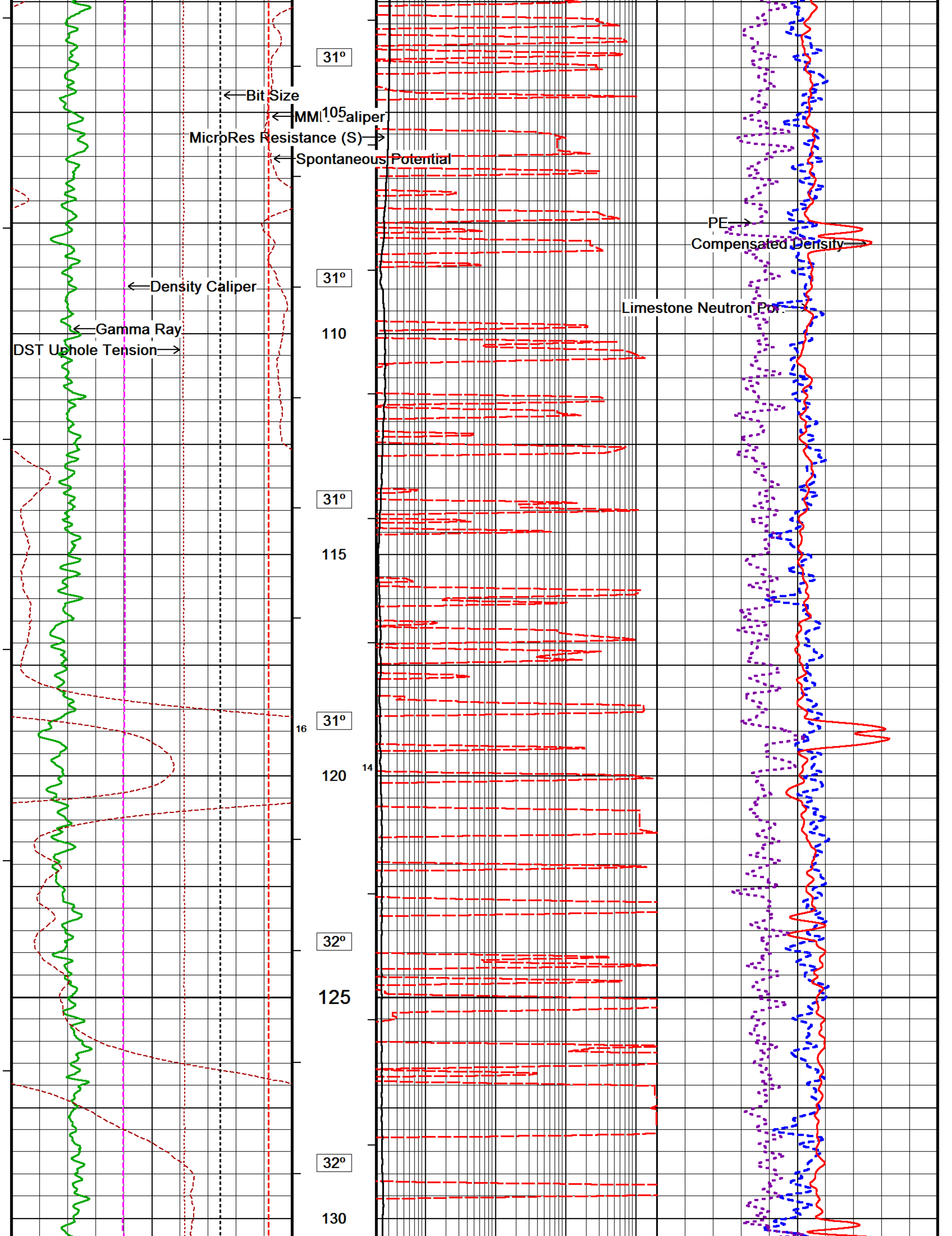


10
1
10
100
1000
MMR Caliper
millimetres
125
250
375
Annular
Integral
every
0.1 cu m
Bit Size
millimetres
125
250
375
Density Correction
grams/cc
-0.25
0.25
DST Uphole Tension
pounds
0
3000
Replay
Scale
1:100
0
31º
Bit Size
5
Spontaneous Potential
30º
Density Caliper
Limestone Neutron Por.
Gamma Ray
10
30º
15

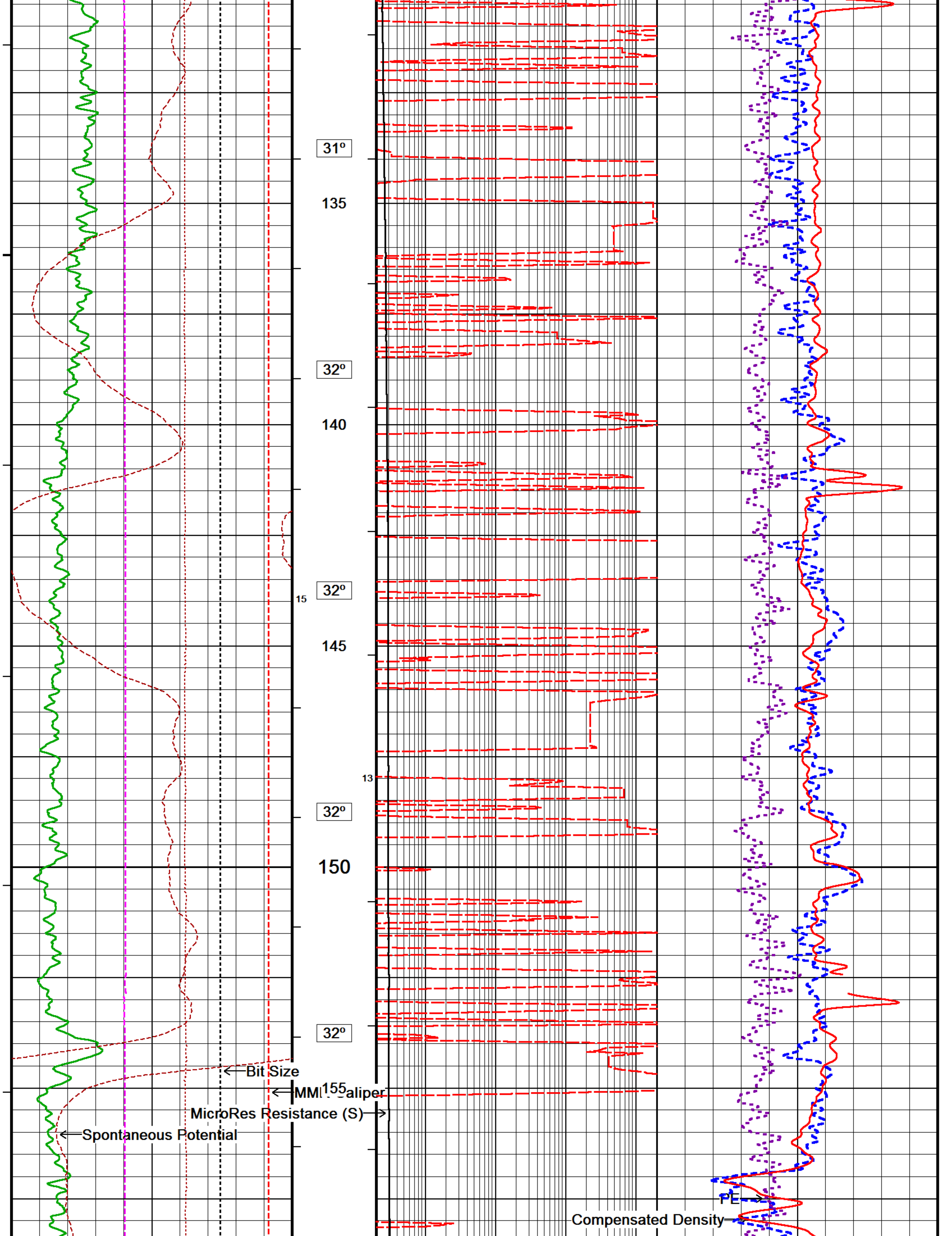

30°
20
20
30°
25
30°
30
30°
35
17
30°
40
30°
19
45


30º
50
30º
Bit Size
55
MMR Caliper
MicroRes Resistance (S)
Spontaneous Potential
PE
Compensated Density
31º
Density Caliper
Limestone Neutron Por
60
Gamma Ray
DST Uphole Tension
31º
16
65
31º
18
70
31º

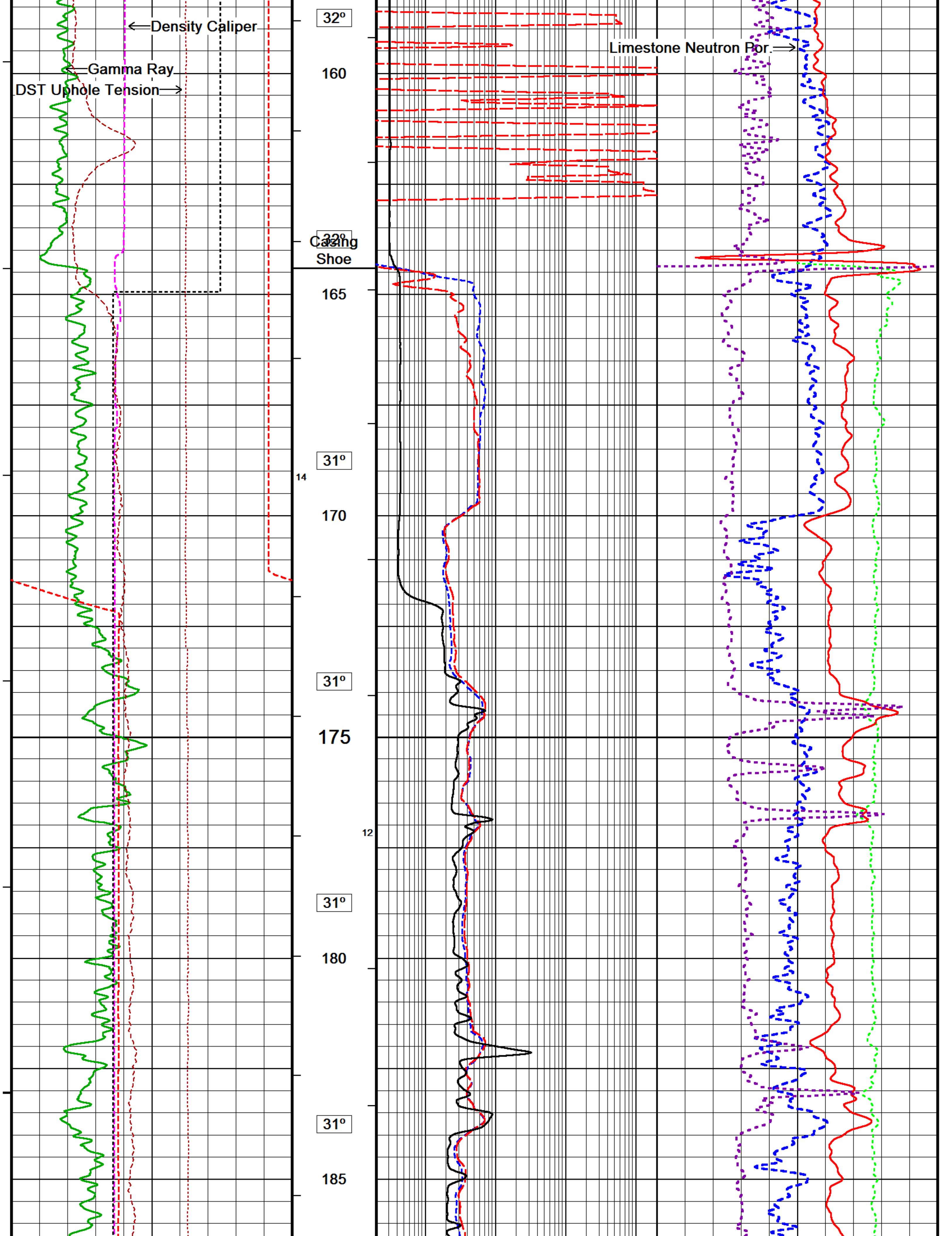

75
31°
80
31°
85
31°
90
15
31°
17
95
31°
100

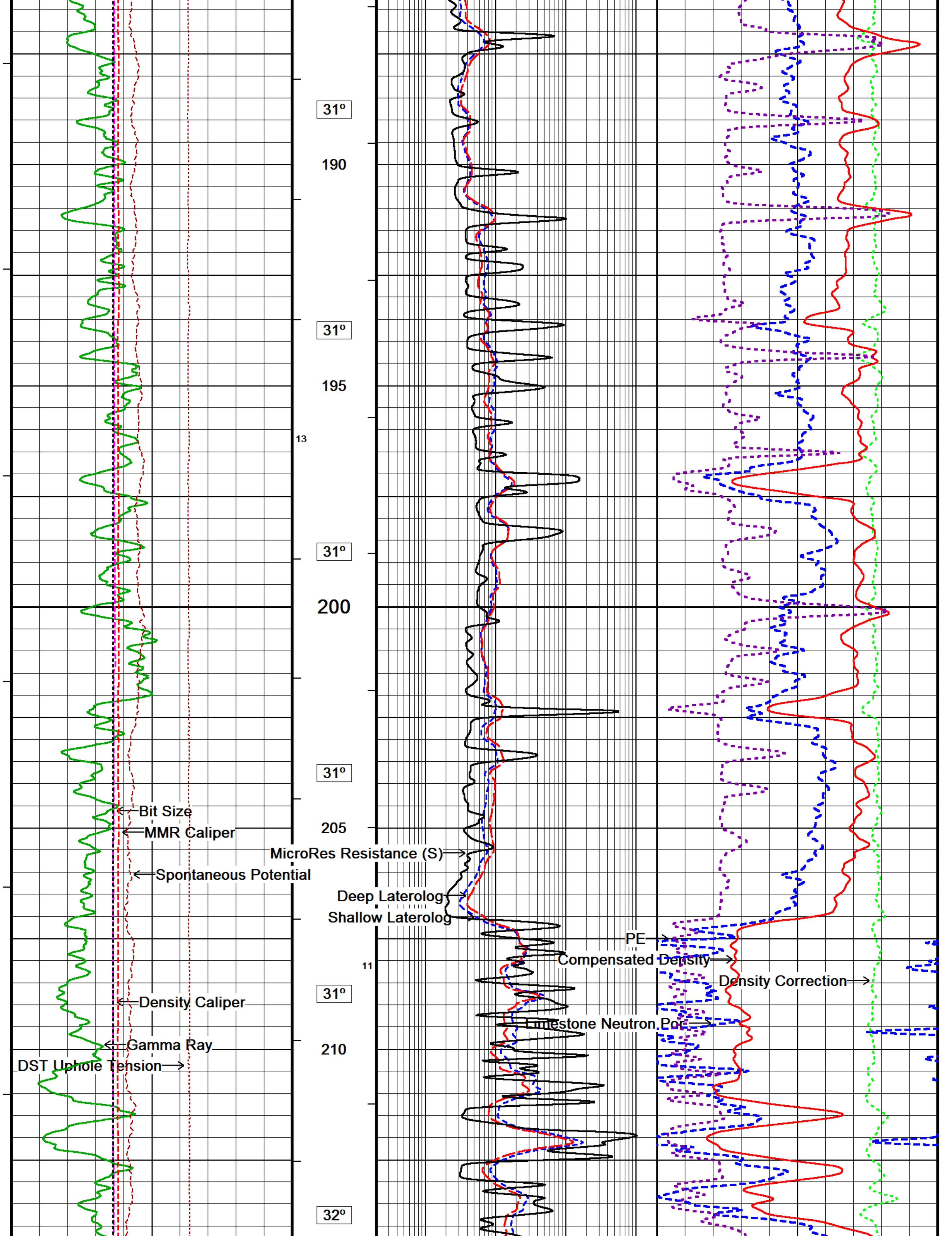


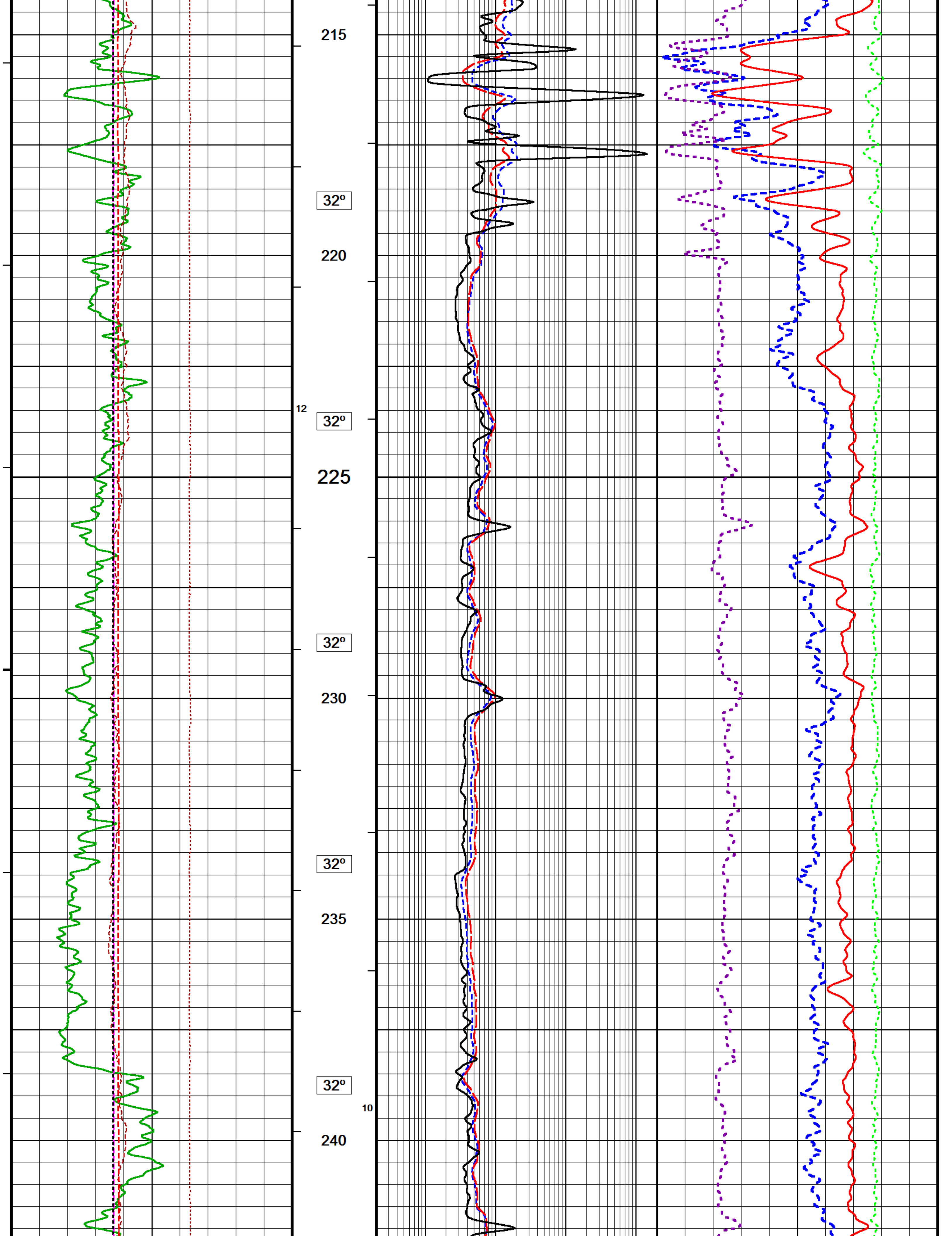
31º
Bit Size
105
MMI Caliper
MicroRes Resistance (S)
Spontaneous Potential
PE
Compensated Density
31º
Density Caliper
Limestone Neutron Por.
Gamma Ray
110
DST Uphole Tension
31º
115
16
31º
14
120
32º
125
32º
130

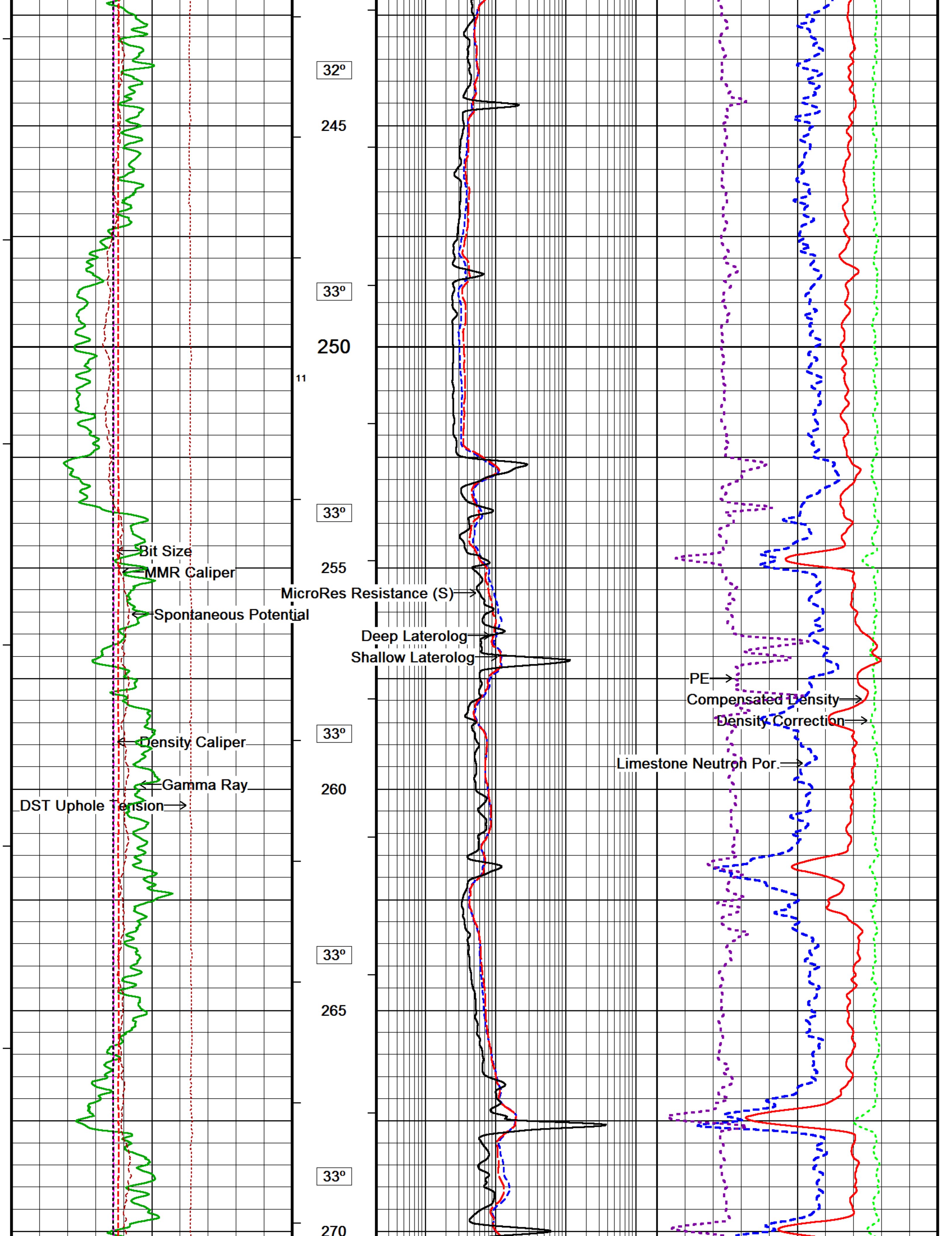


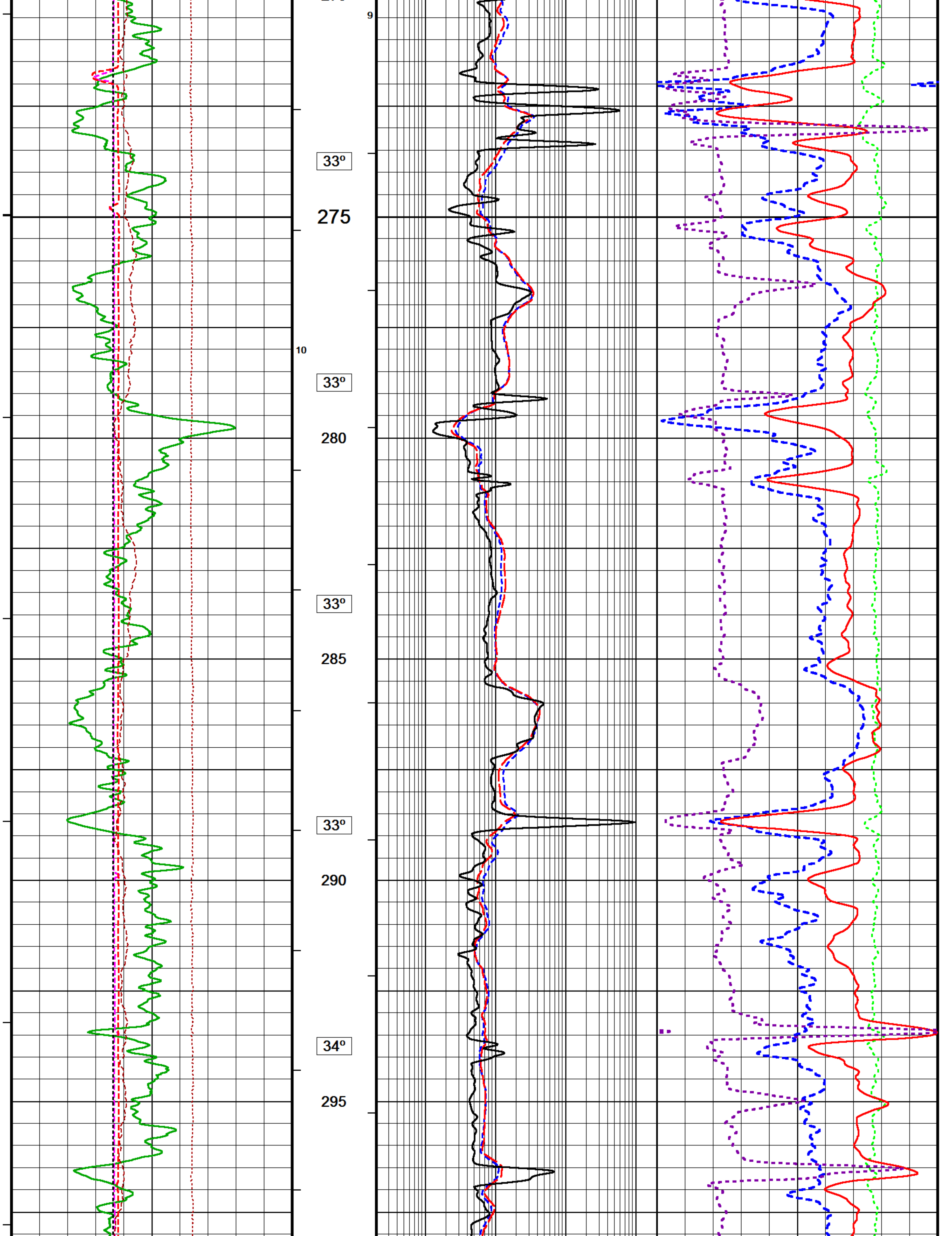
31º
135
32º
140
15
32º
145
13
32º
150
32º
Bit Size
155
MicroRes Resistance (S)
Spontaneous Potential
PE
Compensated Density

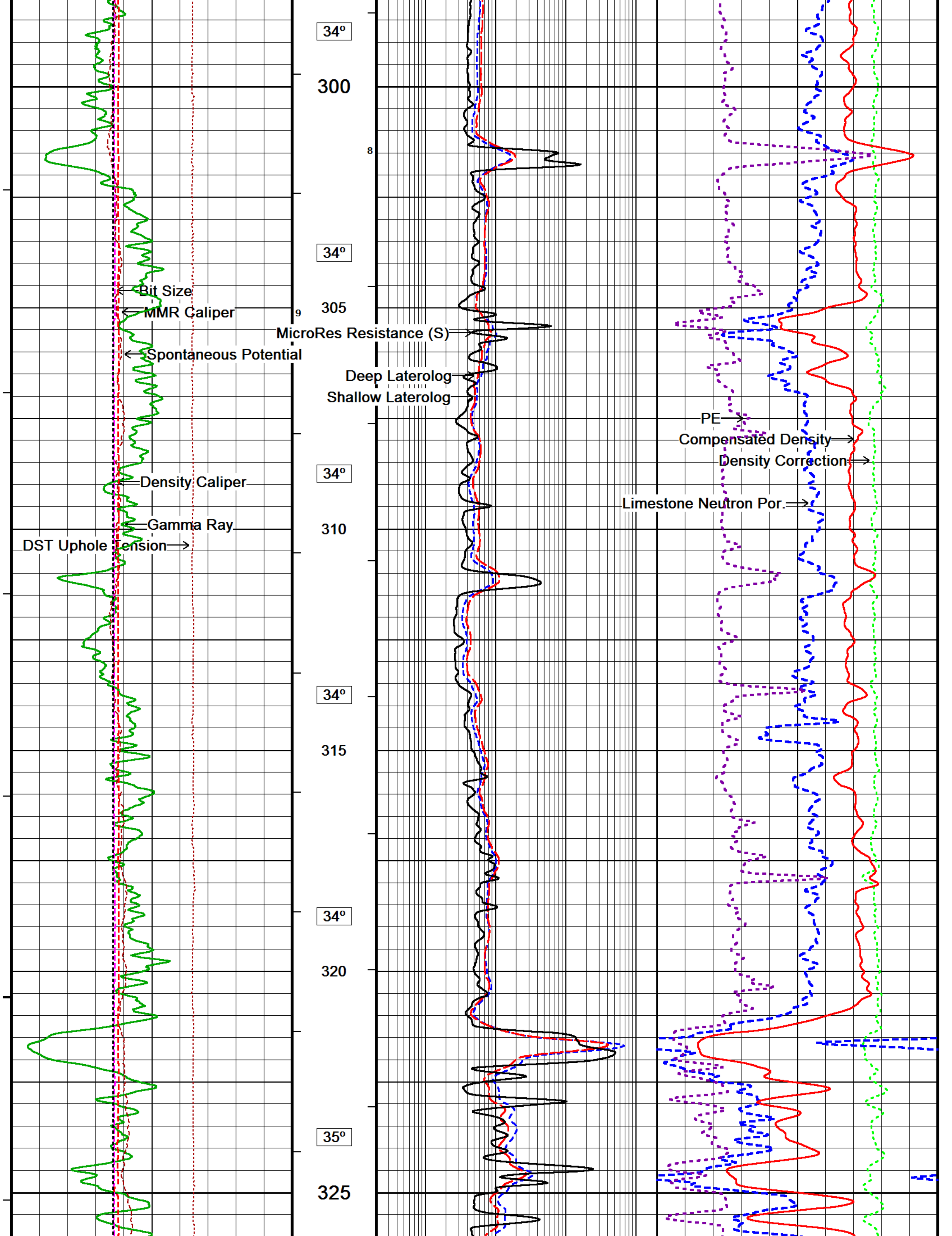

Density Caliper
Gamma Ray
DST Uphole Tension
Limestone Neutron Por.
32°
160
32°
Casing
Shoe
165
31°
14
170
31°
175
12
31°
180
31°
185

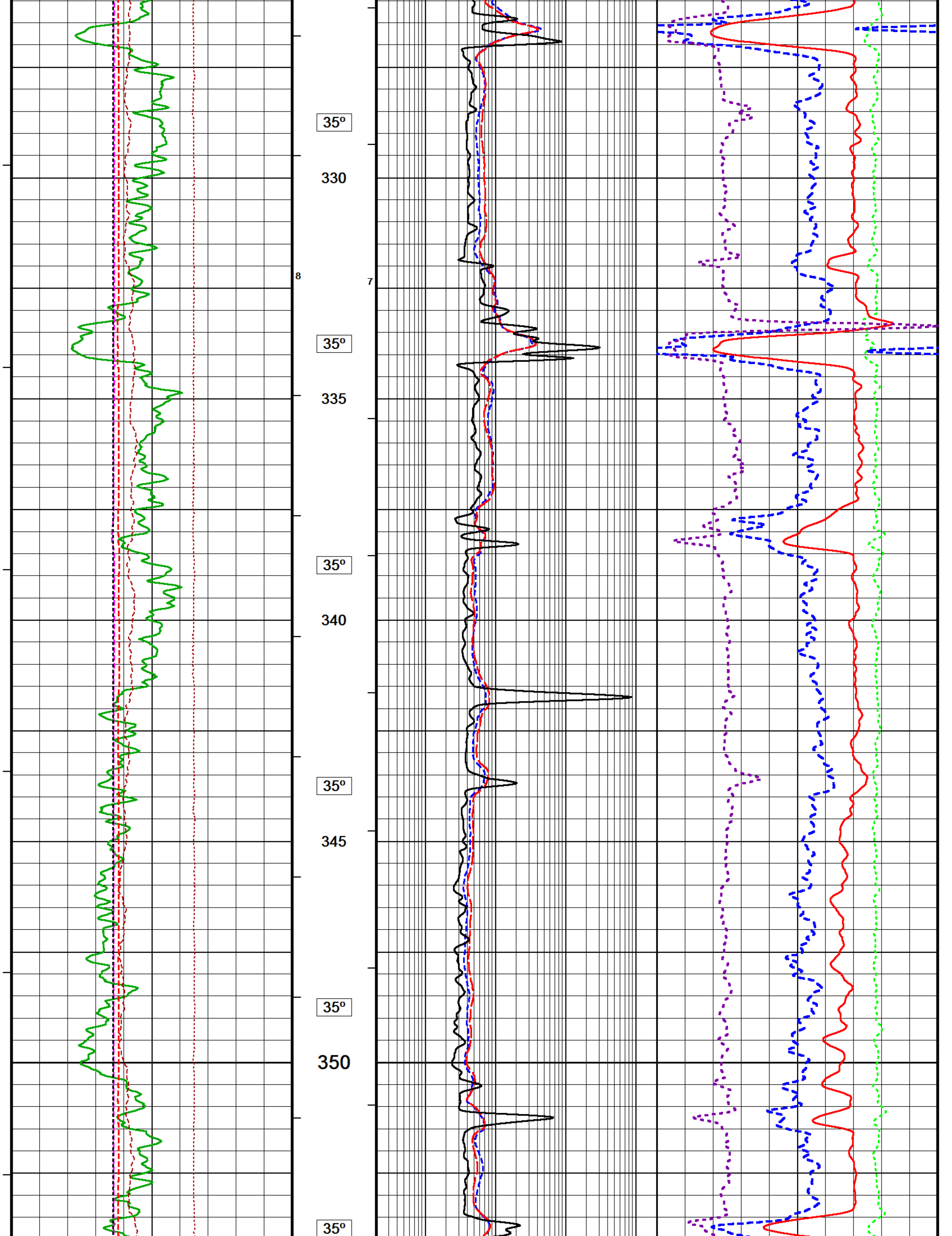

31°
190
31°
195
13
31°
200
31°
205
Bit Size
MMR Caliper
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
11
Density Correction
31°
Density Caliper
Limestone Neutron Por
Gamma Ray
210
DST Uphole Tension
32°

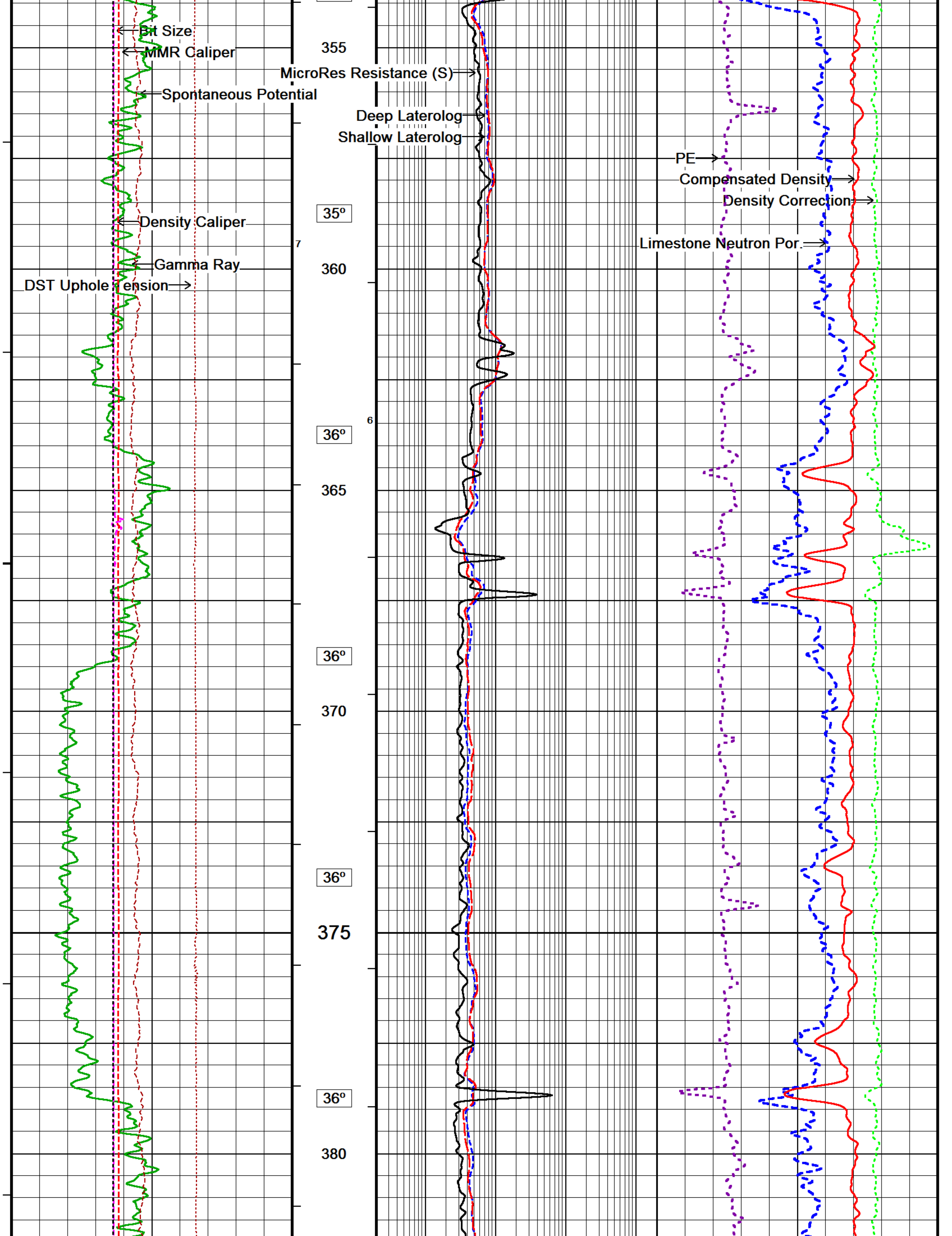

215
32º
220
12
32º
225
32º
230
32º
235
32º
240
10


32º
245
33º
250
11
33º
Bit Size
MMR Caliper
255
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
33º
Density Caliper
Limestone Neutron Por.
Gamma Ray
260
DST Uphole Tension
33º
265
33º
270

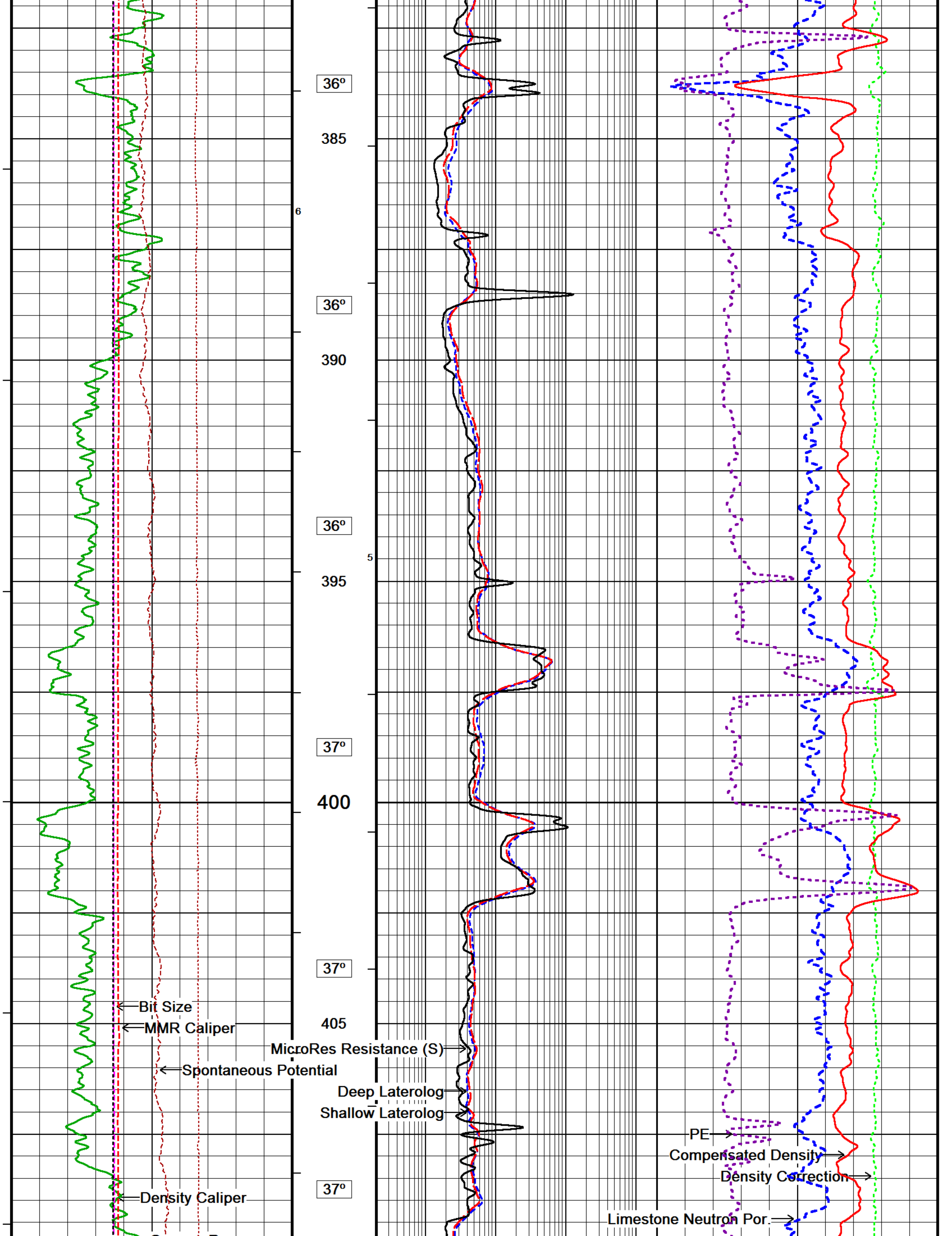

9
33º
275
10
33º
280
33º
285
33º
290
34º
295

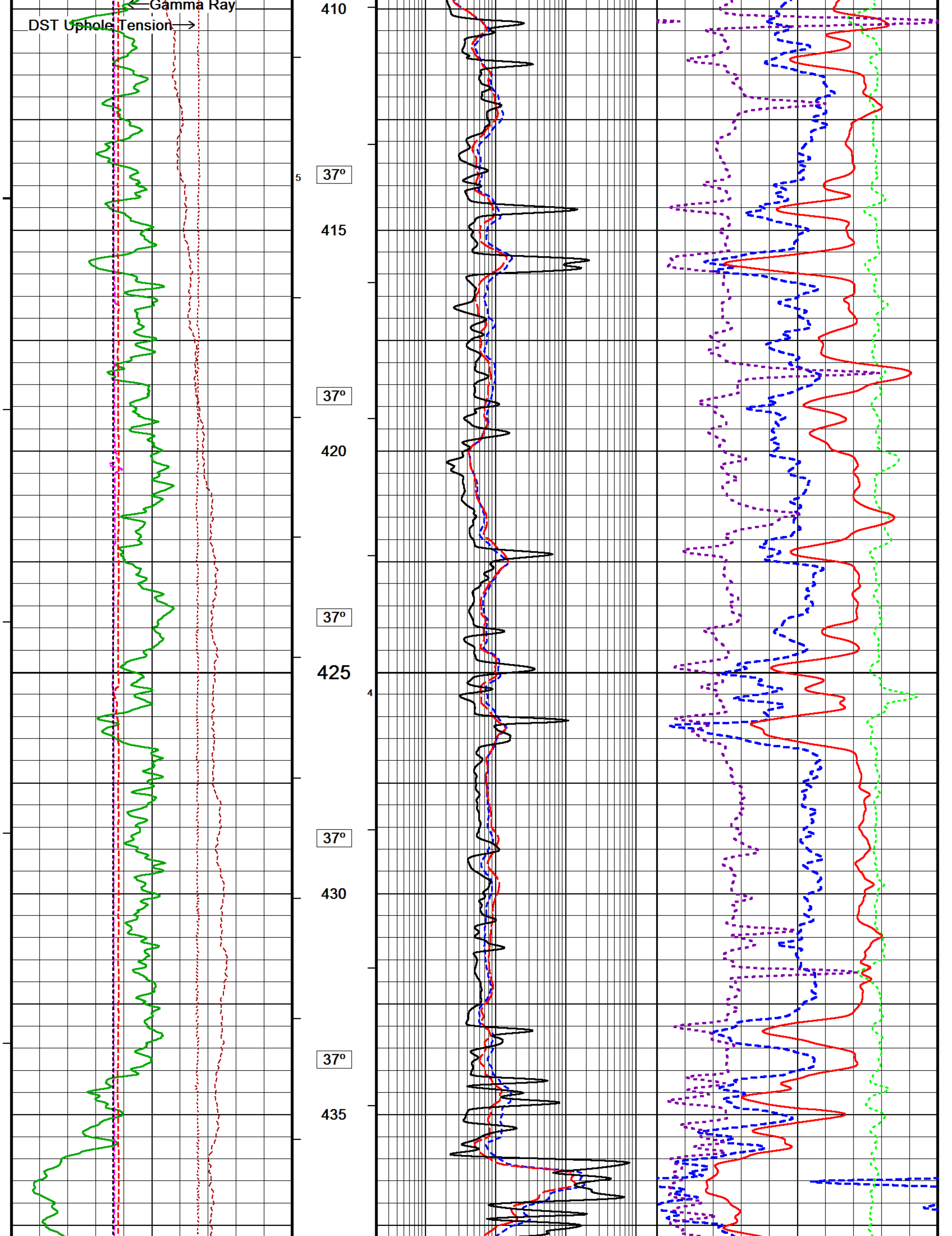

34°
300
8
34°
Bit Size
MMR Caliper
9
305
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
34°
Density Caliper
Limestone Neutron Por.
Gamma Ray
310
DST Uphole Tension
34°
315
34°
320
35°
325


35º
330
8
7
35º
335
35º
340
35º
345
35º
350
35º

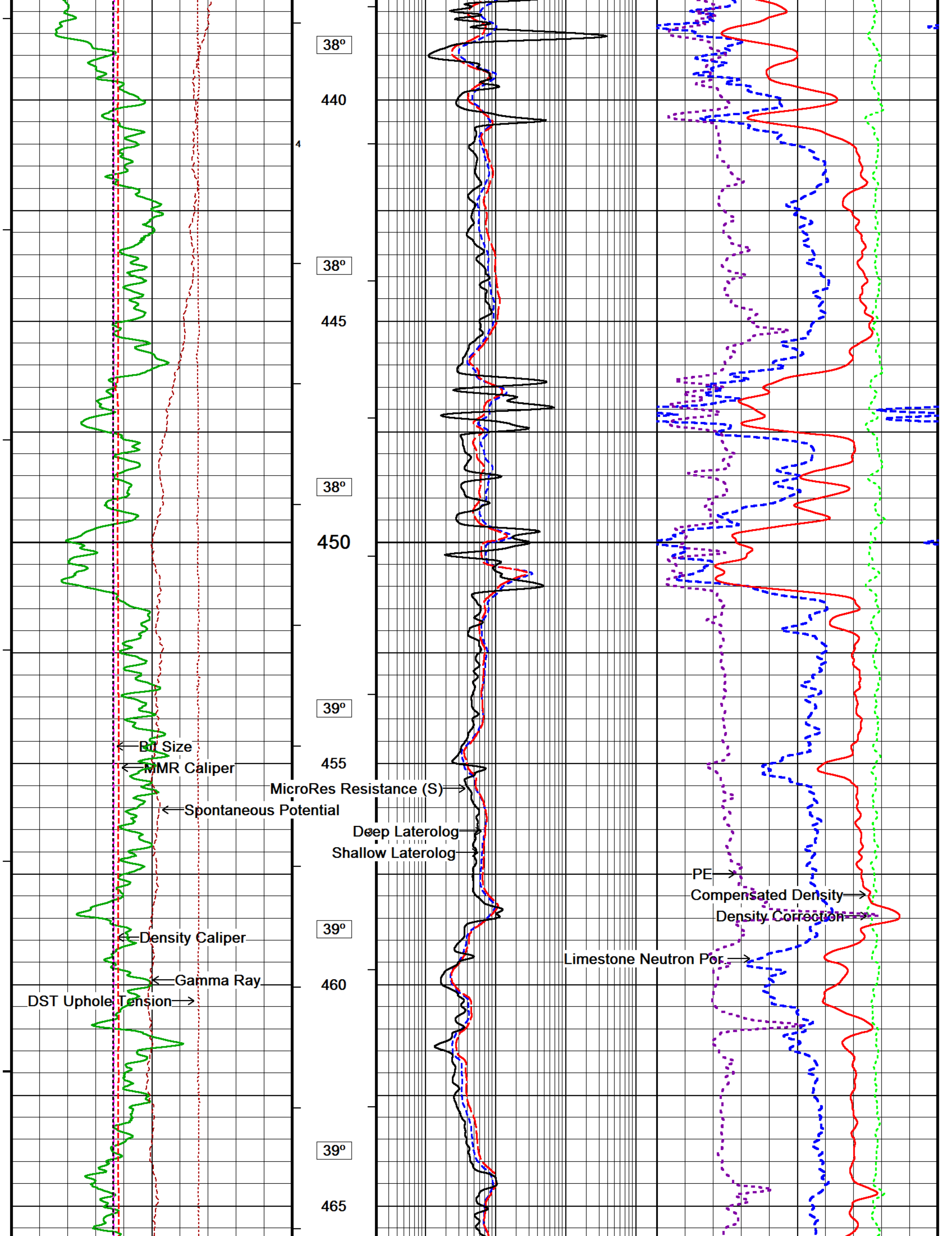


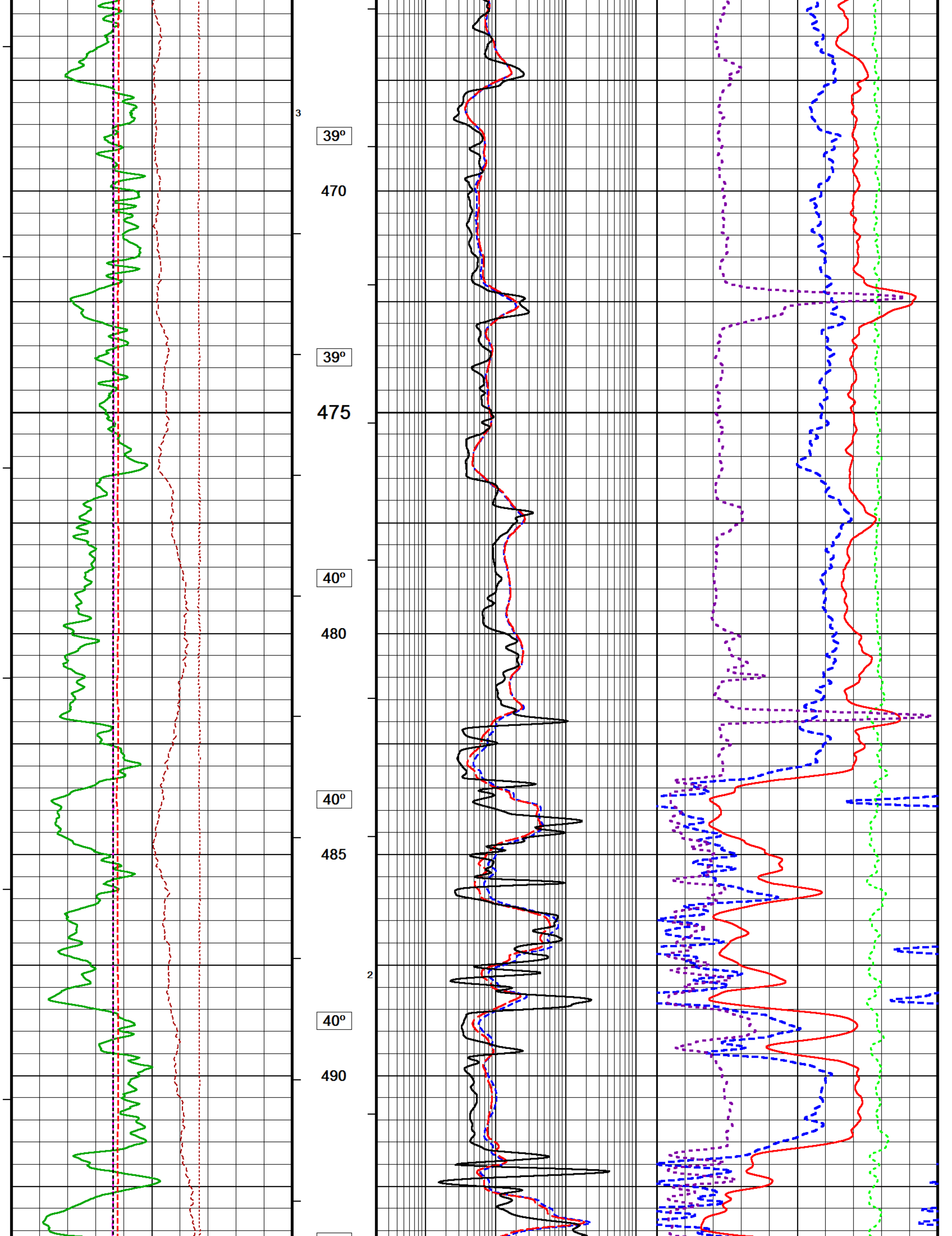
Bit Size
MMR Caliper
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
Density Caliper
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
355
35º
7
360
6
36º
365
36º
370
36º
375
36º
380

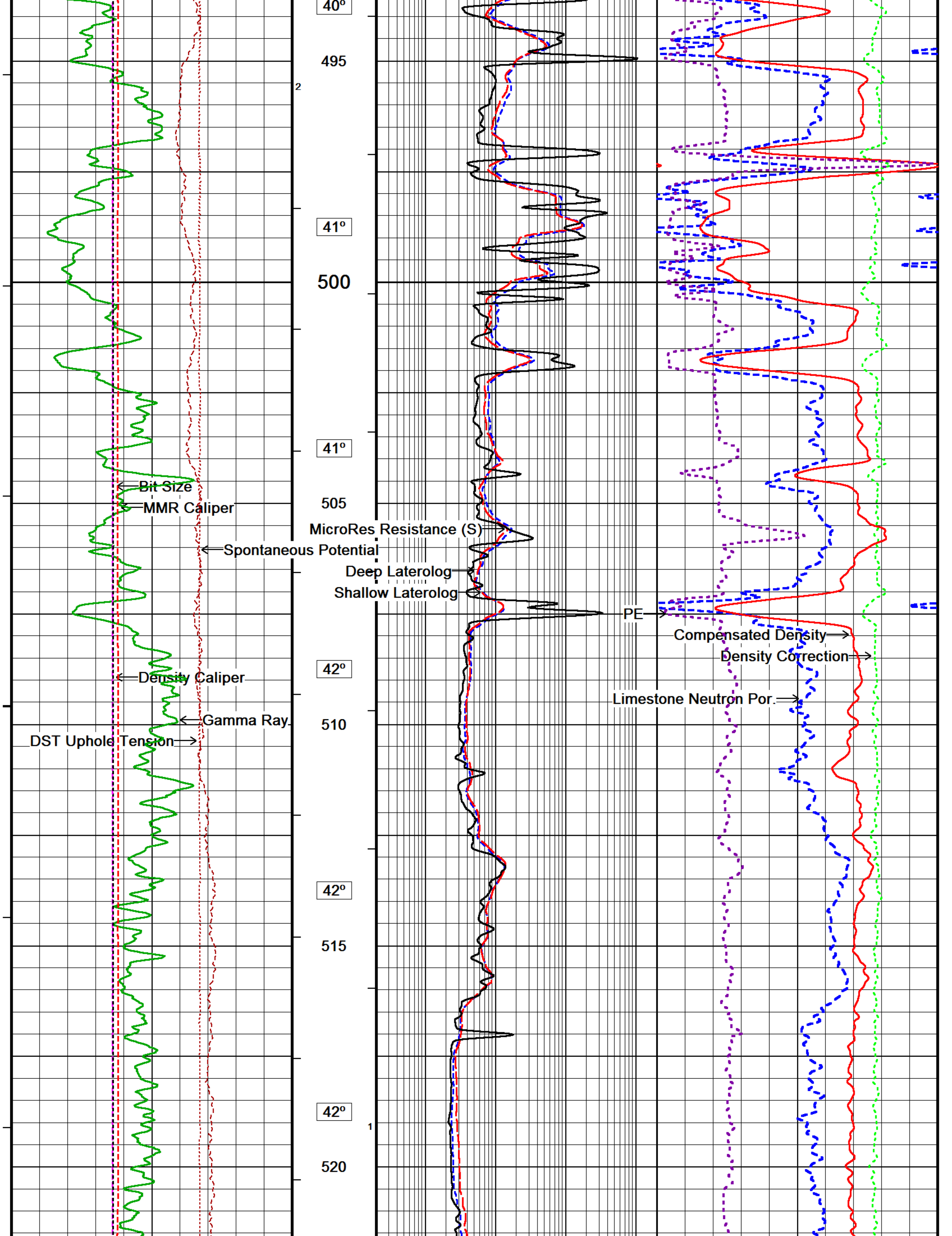

36º
385
6
36º
390
36º
5
395
37º
400
37º
Bit Size
405
MMR Caliper
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
37º
Density Caliper
Limestone Neutron Por.



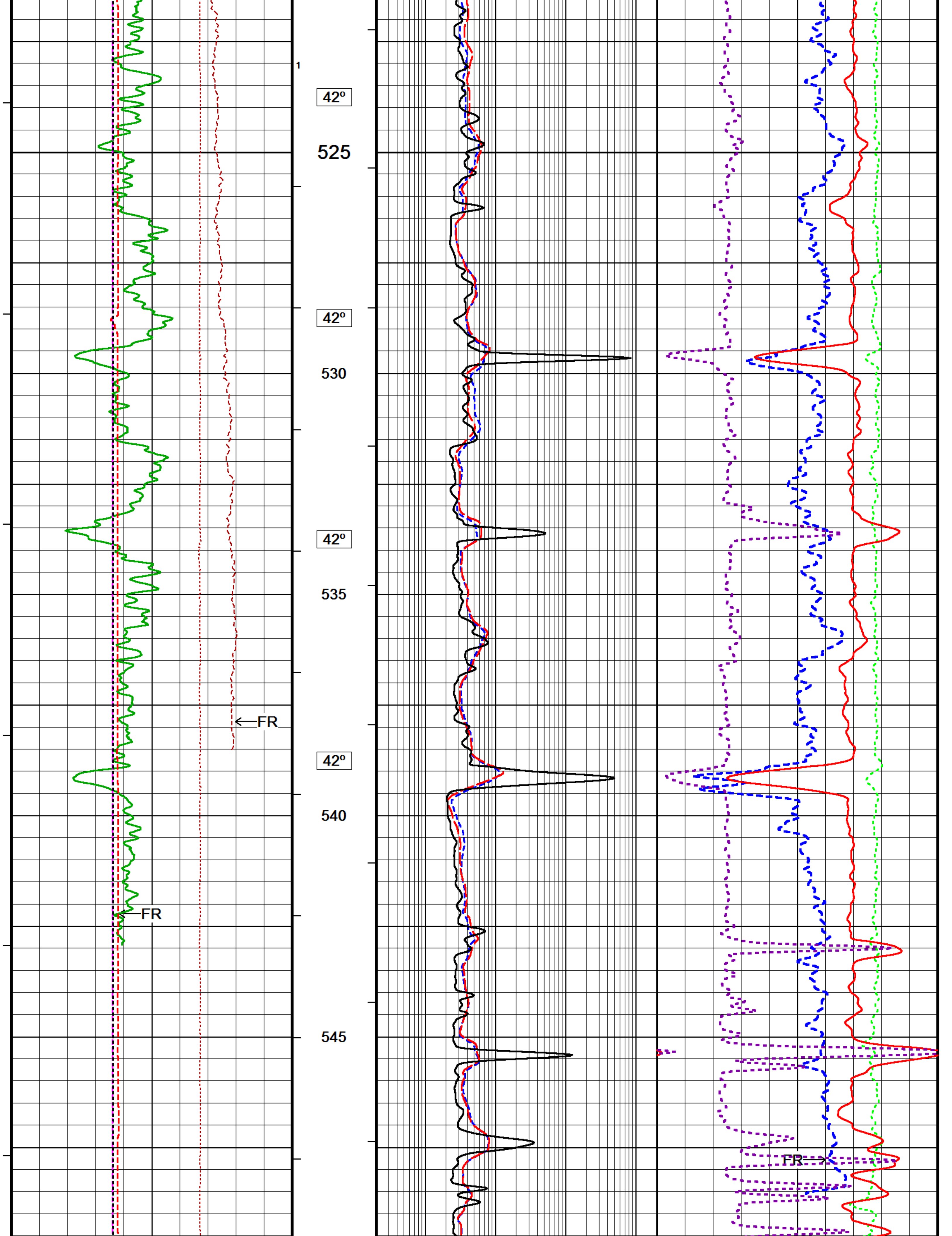
Gamma Ray
DST Uphole Tension
410
37º
415
37º
420
37º
425
37º
430
37º
435

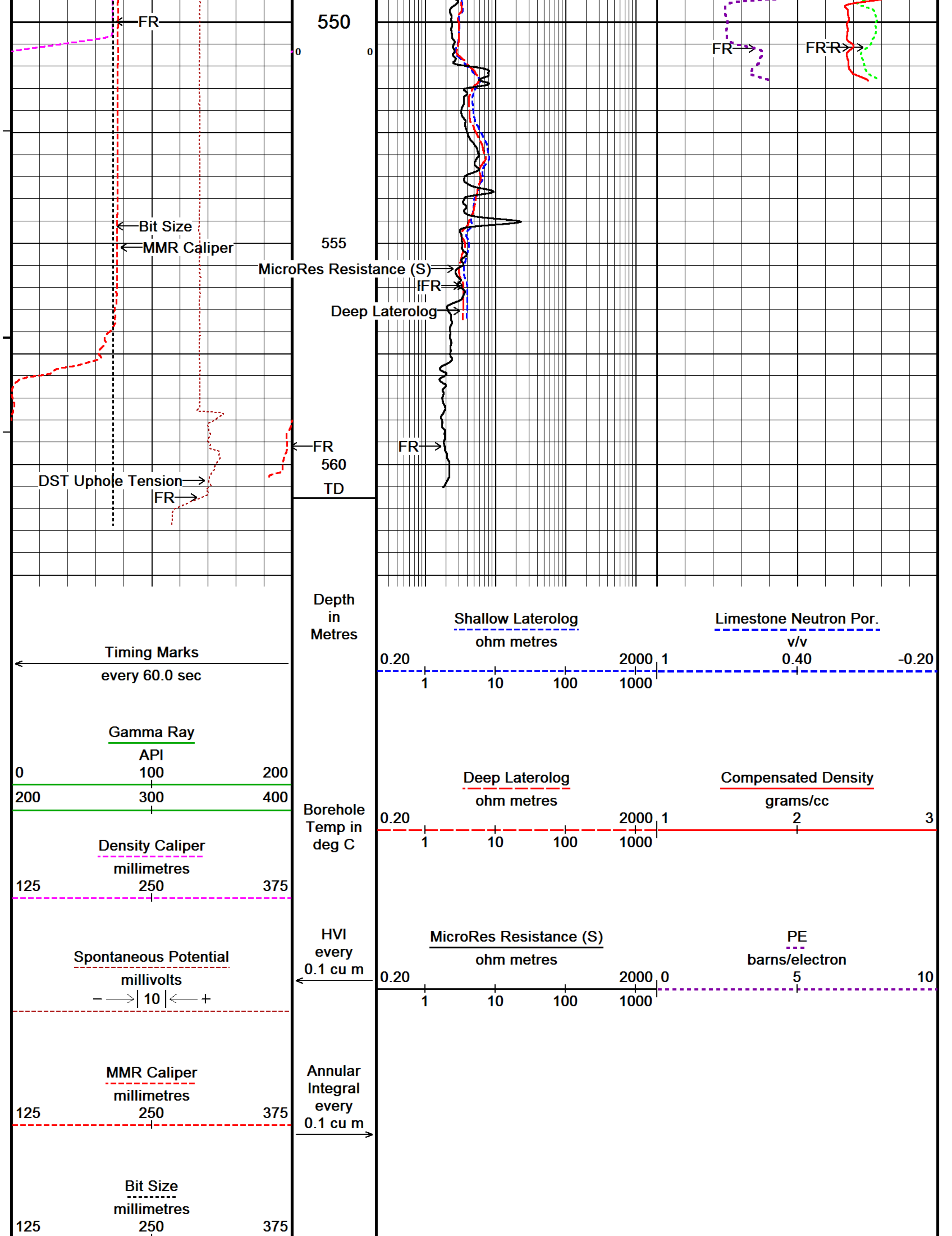

38º
440
4
38º
445
38º
450
39º
Bit Size
455
MMR Caliper
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
39º
Density Caliper
Limestone Neutron Por
Gamma Ray
460
DST Uphole Tension
39º
465


3
39°
470
39°
475
40°
480
40°
485
2
40°
490



40º
495
2
41º
500
41º
Bit Size
MMR Caliper
505
MicroRes Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
42º
Density Caliper
Limestone Neutron Por.
Gamma Ray
510
DST Uphole Tension
42º
515
42º
1
520


1
42º
525
42º
530
42º
535
←FR
42º
540
←FR
545
FR→



550
FR
Bit Size
MMR Caliper
555
MicroRes Resistance (S)
IFR
Deep Laterolog
FR
560
TD
DST Uphole Tension
FR
FR
FR
FR
Depth in Metres
Timing Marks
every 60.0 sec
Gamma Ray
API
0 100 200
200 300 400
Density Caliper
millimetres
125 250 375
Spontaneous Potential
millivolts
10
MMR Caliper
millimetres
125 250 375
Bit Size
millimetres
125 250 375
Borehole Temp in deg C
HVI every 0.1 cu m
Annular Integral every 0.1 cu m
Shallow Laterolog
ohm metres
0.20 2000
1 10 100 1000
Limestone Neutron Por.
v/v
1 0.40 -0.20
Deep Laterolog
ohm metres
0.20 2000
1 10 100 1000
Compensated Density
grams/cc
1 2 3
MicroRes Resistance (S)
ohm metres
0.20 2000
1 10 100 1000
PE
barns/electron
0 5 10

DST Uphole Tension pounds 0 3000	Replay Scale 1:100	Density Correction grams/cc -0.25 0.25

Depth Based Data - Maximum Sampling Increment 2.5cm — Plotted on 18-OCT-2019 17:46
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_2.5cm_CMP_OH_20191018.dta — Recorded on 18-OCT-2019 14:38
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

HIGH RESOLUTION 1:100

BEFORE SURVEY CALIBRATION

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018a.dta

General Constants All 000

Last Edited on 18-OCT-2019,12:12

General Parameters		
Mud Resistivity	0.302	ohm-metres
Mud Resistivity Temperature	25.000	degrees C
Water Level	0.000	metres
Borehole Fluid Processing	Water Level Switch	
Hole/Annular Volume and Differential Caliper Parameters		
HVOL Method	Single Caliper	
HVOL Caliper 1	Density Caliper	
HVOL Caliper 2	N/A	
Annular Volume Diameter	76.200	mm
Caliper for Differential Caliper	Density Caliper	
Rwa Parameters		
Porosity used	Base Density Porosity	
Resistivity used	Deep Laterolog	
RWA Constant A	0.620	
RWA Constant M	2.150	
SW/APOR Tool Source	0.000	

Gamma Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:35

	Measured	Calibrated (API)
Background	65	43
Calibrator (Gross)	849	565
Calibrator (Net)	784	522

Gamma Calibration Tolerances MCG-E.A 580

1.40 1.475 1.55

Ratio 1.502 Counts/API

Gamma Constants MCG-E.A 580

Last Edited on 18-OCT-2019,12:13

Gamma Calibrator Number	120	
GRC-M Calibrator Jig in Use?	NO	
Inactive Background Jig in Use?	NO	
Mud Density	1.01	gm/cc
Caliper Source for Processing	Density Caliper	
Tool Position	Eccentred	
Potassium Equivalence	Chloride	
K Mud Concentration	0.00	%

High Resolution Temperature Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:36

	Measured	Calibrated(Deg C)
Lower	0.00	0.00
Upper	100.00	100.00

High Resolution Temperature Constants MCG-E.A 580

Last Edited on 12-OCT-2019,13:36

Pre-filter Length	11

Navigation Constants MBN-D.A 170

Last Edited on 18-OCT-2019,12:13

Magnetic Declination	10.39	degrees	East

Magnetometer Parameters MBN-D.A 170

Date Of Last Magnetometer Calibration 14-OCT-2013,10:52

	X Magnetometer	Y Magnetometer	Z Magnetometer
Slope	-1.000000	1.003752	1.001092
Offset	0.001123	0.014769	-0.000632

Magnetometer Constants MBN-D.A 170

Magnetometer Calibrator Number 000

Accelerometer Parameters MBN-D.A 170

Date Of Last Accelerometer Calibration 10-OCT-2013,11:36

	X Accelerometer	Y Accelerometer	Z Accelerometer
Slope	-1.094953	-1.100967	-1.098665
Offset	0.003562	0.001559	0.005569

Accelerometer Constants MBN-D.A 170

Last Edited on 15-SEP-2019,17:36

Accelerometer Calibrator Number 000

Accelerometer Temperature Characterisation

X Accelerometer

Serial Number	1295			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.35051e-05	7.52948e-10	1.11593e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.64265e-04	3.57391e-07	7.78280e-10

Y Accelerometer

Serial Number	1239			
Calibration Date	11-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.67211e-05	2.44017e-08	7.98178e-11
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.80123e-04	3.19145e-07	8.20741e-10

Z Accelerometer

Serial Number	1310			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.13686e-05	-1.00367e-08	1.99621e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.55395e-04	3.67997e-07	9.83969e-10

Neutron Calibration MDN-B.A 270

Base Calibration on 14-OCT-2019 09:17
Field Check on 16-OCT-2019 09:38

Base Calibration

	Measured		Calibrated (cps)	
	Near	Far	Near	Far
	2976	88	3714	110
Ratio	33.984		33.764	

Field Calibrator at Base

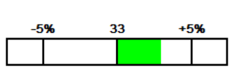
	Calibrated (cps)	
	2227	3352
Ratio	0.664	

Field Check

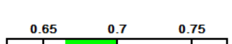
	Calibrated (cps)	
	2234	3344
Ratio	0.668	

Neutron Calibration Tolerances MDN-B.A 270

Ratio 33.984



Base Check 0.664



		Scale
Field Check	0.668	0.644 / 0.664 / 0.684

Neutron Constants MDN-B.A 270

Last Edited on 18-OCT-2019,12:14

Neutron Source Id	16150B	
Neutron Jig Number	NECD200	
Air Hole Processing	Legacy	
Caliper Source for Processing	Density Caliper	
Stand-off	0.00	mm
Mud Density	1.01	gm/cc
Limestone Sigma	7.10	cu
Sandstone Sigma	4.26	cu
Dolomite Sigma	4.70	cu
Formation Pressure Source	None	
Formation Pressure	N/A	kpsi
Temperature Source	MCG External Temperature	
Temperature	N/A	degrees C
Mud Salinity	0.00	kppm
Salinity Correction	Not Applied	
Formation Fluid Salinity Source	None	
Formation Fluid Salinity	N/A	kppm
Barite Mud Correction	Not Applied	

Laterolog Calibration MLE-C.A 109

Base Calibration on 11-OCT-2019 10:01
Field Check on 11-OCT-2019 10:06

Resistor 1 (ohm)	Resistor 2 (ohm)
0.0	1000.0

Base Calibration

	Measured		Calibrated (ohm-m)	
Channel	Resistor 1	Resistor 2	Resistor 1	Resistor 2
Shallow	0.0	999.4	0.0	1284.4
Deep	0.0	999.3	0.0	795.7
Groningen	0.0	999.1	0.0	808.4

Channel	Base Check (ohm-m)	Field Check (ohm-m)
Shallow	46.2	46.2
Deep	28.6	28.6
Groningen	232.8	232.8

Laterolog Calibration Tolerances MLE-C.A 109

	Value	Scale	Unit
Shallow Resistor 2	999.4	-3% / 985.0 / +3%	ohm
Deep Resistor 2	999.3	-3% / 985.0 / +3%	ohm
Groningen Resistor 2	999.1	-3% / 985.0 / +3%	ohm
Shallow Base Check	46.2	-2% / 46.23 / +2%	ohm-m
Deep Base Check	28.6	-2% / 28.64 / +2%	ohm-m
Groningen Base Check	232.8	-2% / 232.83 / +2%	ohm-m
Shallow Field Check	46.2	-2% / 46.2 / +2%	ohm-m
Deep Field Check	28.6	-2% / 28.6 / +2%	ohm-m
Groningen Field Check	232.8	-2% / 232.8 / +2%	ohm-m

Laterolog Constants MLE-C.A 109

Last Edited on 12-OCT-2019,13:26

Profiling Laterolog Type	Type 0	
Laterolog Output Filter	Logarithmic	
Profiling Limiter	On	
Median Filter	On	
Squasher Start	40000	ohm-m
Shallow Laterolog K Factor	1.2844	
Deep Laterolog K Factor	0.7957	
Groningen Laterolog K Factor	0.8084	
Interference Rejection	50 Hz	
SP Connection	SP Bridle Electrode (Lower)	
Groningen Connection	Groningen Electrode (Upper)	

Groningen Connection	Groningen Electrode (Upper)	
Profiling Deep Input	N/A	

Borehole Correction Constants		
Bridle Type	Standard	
Sonde Position	12.7	millimetres
Hole Size Source	Density Caliper	
Hole Size Constant Value	N/A	millimetres
Rm Source	Global Value: Temperature Corrected	
Temp. for Rm Corr.	MCG External Temperature	

Apparent Porosity and Water Saturation Constants		
Archie Constant (A)	1.00	
Cementation Exponent (M)	2.00	
Saturation Exponent (N)	2.00	
Saturation of Water for Apor	100.00	percent
Resistivity of Water for Apor and Sw	0.05	ohm-m
Resistivity of Mud Filtrate for Sw	0.00	ohm-m
Source for Rt	0.00	
Source for Rxo	0.00	

SP Calibration MLE-C.A 109

Field Calibration on 12-OCT-2019,13:26

	Measured	Calibrated (mV)
Reference 1	100.0	100.0
Reference 2	-100.0	-100.0

Micro Laterolog Calibration MMR-B.A 126

Base Calibration on 10-OCT-2019 16:30
Field Check on 10-OCT-2019 16:33

Resistor 1 (ohm)	Resistor 2 (ohm)
0.0	10000.0

Base Calibration

Measured		Calibrated (ohm-m)	
Ref 1	Ref 2	Ref 1	Ref 2
0.0	9867.3	0.0	128.0

Base Check (ohm-m)	Field Check (ohm-m)
5.2	5.2

Micro Laterolog Calibration Tolerances MMR-B.A 126

	Value	-Tol	Nominal	+Tol	Unit
Ref 2	9867.3	-3%	9900.00	+3%	ohm
Base Check	5.2	-2%	5.2	+2%	ohm-m
Field Check	5.2	-2%	5.2	+2%	ohm-m

Micro Laterolog Constants MMR-B.A 126

Last Edited on 13-OCT-2019,17:01

Pad Type	6 in Solid Nylon B23059	
Standoff Offset	12.7000	millimetres
Micro Laterolog K Factor	0.0128	
Micro Laterolog Rm K Factor	N/A	

Mudcake Thickness Correction Constants		
Mud Cake Source	Constant Value	
Mud Cake Thickness	10.1600	millimetres
Mud Cake Thickness Caliper	N/A	
Mud Cake Resistivity	0.1500	ohm-m
Mud Cake Resistivity Temp.	20.00	Degrees C
Mud Cake Resistivity Source	Temperature Corr	
Temp. for Rmc Corr.	MCG Internal Temperature	

Caliper Calibration MMR-B.A 126

Base Calibration on 10-OCT-2019 16:39
Field Calibration on 13-OCT-2019 16:50

Base Calibration

Reading No	Measured	Calibrator Size (mm)
1	11571	101.83
2	14686	151.74
3	17781	202.31
4	21116	250.47
5	25245	302.77

6	N/A	N/A

Field Calibration

Measured Caliper (mm)	Actual Caliper (mm)
209.37	213.89

Caliper Calibration Tolerances MMR-B.A 126

Short Arm Field Cal. 209.4 [208.8 | 213.9 | 219.0] mm

Caliper Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:43
Field Calibration on 16-OCT-2019 09:31

Base Calibration

Reading No	Measured	Calibrator Size (in)
1	16321	4.01
2	26493	5.97
3	36763	7.96
4	46427	9.86
5	57546	11.92
6	N/A	N/A

Field Calibration

Measured Caliper (in)	Actual Caliper (in)
7.95	7.96

Caliper Calibration Tolerances MPD-D.A 449

Short Arm Field Cal. 7.95 [7.76 | 7.96 | 8.16] in

Photo Density Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:30
Field Check on 16-OCT-2019 09:28

Density Calibration

Base Calibration	Measured Near	Measured Far	Calibrated (sdu) Near	Calibrated (sdu) Far
Background	1134	1304		
Reference 1	45330	22156	59605	30991
Reference 2	18639	2314	24756	2530

Field Check at Base

Near	Far
1134.5	1304.4

Field Check

Near	Far
1139.8	1311.8

PE Calibration

Base Calibration	Measured WS	Measured WH	Measured Ratio	Calibrated Ratio
Background	215	1016		
Reference 1	20159	45158	0.452	0.372
Reference 2	5757	18514	0.317	0.271

Field Check at Base

WS	WH
215.2	1016.3

Field Check

WS	WH
214.2	1018.6

Photo Density Calibration Tolerances MPD-D.A 449

Near Density Ratio 2.52 [-5% | 2.52 | +5%]

Far Density Ratio 20.65 [-5% | 21.00 | +5%]

PE Calibration 0.126 [0.089 | 0.110 | 0.131]

Near Den. Field Check 1139.8 [-3% | 1134.5 | +3%]

Far Den. Field Check 1311.8 [-3% | 1304.4 | +3%]

PE WS Field Check 214.2 [-6% | 215.2 | +6%]

PE WH Field Check 1018.6 [-6% | 1016.3 | +6%]

Density Constants MPD-D.A 449

Last Edited on 18-OCT-2019,12:13

Density Source Id	15756B
Nylon Calibrator Number	DCE-667
Aluminium Calibrator Number	DACD-674

Aluminium Calibrator Number	DACD-674	
Density Shoe Profile	8 inch	
Caliper Source for Processing	Density Caliper	
PE Correction to Density	Not Applied	
Mud Density	1.01	gm/cc
Mud Density Type		
Mud Filtrate Density	1.00	gm/cc
Dry Hole Mud Filtrate Density	1.00	gm/cc
DNCT	0.00	gm/cc
CRCT	0.00	gm/cc
Density Z/A Correction	Hybrid	
Precision Enhanced Density Processing	Not Applied	

Matrix density (gm/cc)	Depth (m)
2.71	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	

DOWNHOLE EQUIPMENT

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018a.dta

11C Tension Cablehead
MCC-A 1 LG: 0.73 m WT: 19.8 lb OD: 57.0 mm

11C-11B Compact Tool Adaptor
MTA-A 70 LG: 0.47 m WT: 13.2 lb OD: 57.0 mm

Compact Stiff Bridle Electrode Sub.
MBE-C.B 179 LG: 3.76 m WT: 77.2 lb OD: 58.0 mm

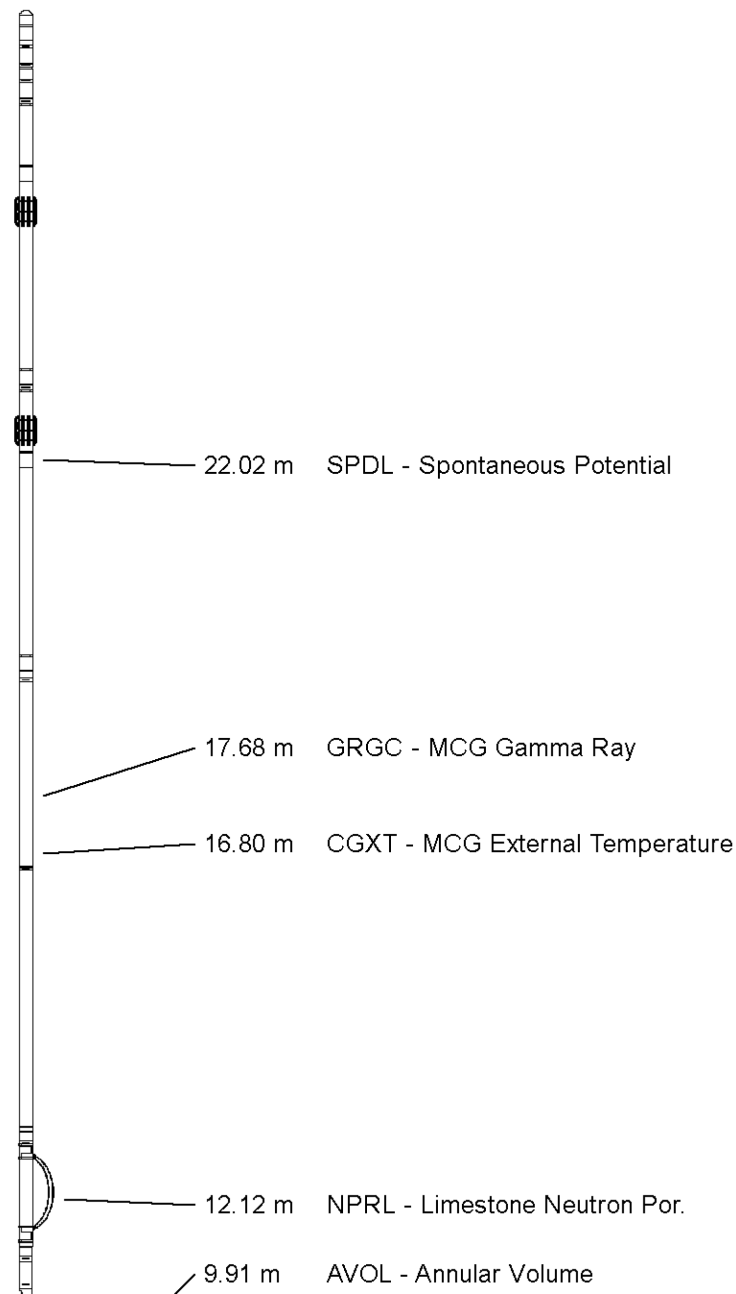
Compact Stiff Bridle Electrode Sub.
MBE-A 54 LG: 3.76 m WT: 77.2 lb OD: 57.0 mm

Compact Comms Gamma
MCG-E.A 580 LG: 2.65 m WT: 63.9 lb OD: 57.0 mm

Compact Navigation
MBN-D.A 170 LG: 3.60 m WT: 70.5 lb OD: 57.0 mm

Compact Neutron
MDN-B.A 270 LG: 1.53 m WT: 50.7 lb OD: 57.0 mm

Compact Density/Caliper
MPD-D.A 449 LG: 2.92 m WT: 90.4 lb OD: 62.2 mm



MPD-D.A 449   LG: 2.92 m   WT: 90.4 lb   OD: 62.2 mm

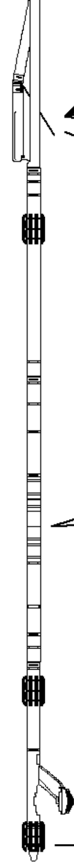
Compact Upper Guard Sub.
MUG-B.A 144   LG: 2.74 m   WT: 68.3 lb   OD: 57.0 mm

Compact Laterolog Electrode Sub.
MLE-C.A 109   LG: 3.76 m   WT: 92.6 lb   OD: 57.0 mm

Compact Micro-Resistivity
MMR-B.A 126   LG: 2.62 m   WT: 81.6 lb   OD: 97.0 mm

Pressure Bung + Hole Finder
HFS 3   LG: 0.28 m   WT: 6.6 lb   OD: 57.0 mm

Total   Length: 28.82 m   Weight: 712.1 lb



9.91 m	HVOL - Hole Volume
9.91 m	CLDC - Density Caliper
9.32 m	DEN - Compensated Density
9.32 m	DCOR - Density Correction
9.30 m	PDPE - PE
3.93 m	DSLL - Shallow Laterolog
3.93 m	DDLL - Deep Laterolog
0.30 m	MRRS - MicroRes Resistance (S)
0.30 m	MATC - MMR Caliper
Tool Zero	(0.85m from bottom)
-0.85 m	SMTU - DST Uphole Tension

All measurements relative to tool zero.

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA

Elevation Kelly Bushing	349.55	metres	First Reading		metres
Elevation Drill Floor	349.55	metres	Depth Driller	565.55	metres
Elevation Ground Level	345.1	metres	Depth Logger	560.75	metres

Weatherford®

MLL- SLL - DLL - SONIC
DENSITY NEUTRON
1:100

MLL - SLL - DLL
DENSITY - NEUTRON
1:200

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA
LOCATION	PL198

Latitude	27°23'18.4689" S	Other Services
Longitude	151°6'16.9779" E	BOREHOLE NAVIGATION
UTM Easting	312 589.65	
UTM Northing	6 969 111.4	

Permanent Datum M.S.L., Elevation 0.00 metres	Elevations:	metres
Log Measured From GL, 345.10 metres above Permanent Datum	KB	349.55
Drilling Measured From KB	DF	349.55
	GL	345.10

Date	18-OCT-2019		
Run Number	1		
Service Order	90518		
Depth Driller	565.55	metres	
Depth Logger	560.75	metres	
First Reading	560.23	metres	
Last Reading	0.00	metres	
Casing Driller	164.55	metres	
Casing Logger	164.40	metres	
Bit Size	215.900	inches	
Hole Fluid Type	KCL		
Density / Viscosity	1.01 g/c3	26.00	sec/qt
PH / Fluid Loss	8.50	0.00	ml/30Min
Sample Source	MUD TANK		
Rm @ Measured Temp	0.302 @ 25.0	ohm-m	
Rmf @ Measured Temp	0.275 @ 25.0	ohm-m	
Rmc @ Measured Temp	0.453 @ 25.0	ohm-m	
Source Rmf / Rmc	CAL	CAL	
Rm @ BHT	0.218 @ 43.0	ohm-m	
Time Since Circulation	8 HRS 37 MIN		
Max Recorded Temp	43.00	deg C	
Equipment / Base	13354	ROMA	
Recorded By	K. SHIBESHI		
Witnessed By	M. GALBRAITH		
Stop Circulation	06:00 / 18 OCT 2019		

BOREHOLE RECORD

Last Edited: 18-OCT-2019 11:45

Bit Size millimetres	Depth From metres	Depth To metres
311.150	0.00	164.95
215.900	164.95	565.55

CASING RECORD

Type	Size millimetres	Depth From metres	Shoe Depth metres	Weight pounds/ft
SURFACE	244.475	0.00	164.55	36.00

REMARKS

RUN NUMBER 1 IS THE PRIMARY DEPTH REFERENCE LOG. ALL OTHER RUNS ARE CORRELATED BACK TO THIS LOG.

SOFTWARE ISSUE: VERSION 19.01.8879 RELEASED DATE APRIL 1, 2019.

CUSTOMER SCALES AND INTERVALS LOGGED.

RUN 1: MMR, MLE,MUG, MPD, MDN, MBN, MCG, MBE, MBE MTA, CBH TOOLS RAN IN COMBINATION.

HARDWARE
RUN 1:
- MBE: 1 X 0.5" STANDOFF ON EACH.
- MUG: 1 X 0.5" STANDOFF.
- MMR: 2 X 0.5" STANDOFF.
- MDN: 1 X DUAL ECCENTRALIZER.

TIME ON BOTTOM: 14:35 / 18 OCT 2019.

ALL DEPTH REFERENCE TO GROUND LEVEL.RT TO GL IS 4.45 M.

LOGGER TOTAL DEPTH: 560.75 mGL.

MMR CALIPER CLOSED 5 METRES BELOW CASING SHOE TO AVOID DAMAGE PAD.

MPD AND MDN CORRECTED FOR DENSITY CALIPER AND MUD DENSITY.

TOTAL HOLE VOLUME (HVOL) FROM TD M TO 9.625" SURFACE CASING SHOE = 14.2 CUBIC METRES.

TOTAL ANNULAR VOLUME (AVOL) FROM TD M TO CASING SHOE WITH 7" CASING = 4.66 CUBIC METRES.

MAXIMUM TEMPERATURE RECORDED 1.59 DEG AT 167.1 METRES.

REPEAT SECTION RECORDED FROM 545 M TO 385 M.

CONVEYANCE TYPE: WIRELINE.

BOREHOLE STATUS: OPEN HOLE.

RIG: SLR 167.

SERVICE REPORT NUMBER: 90518.

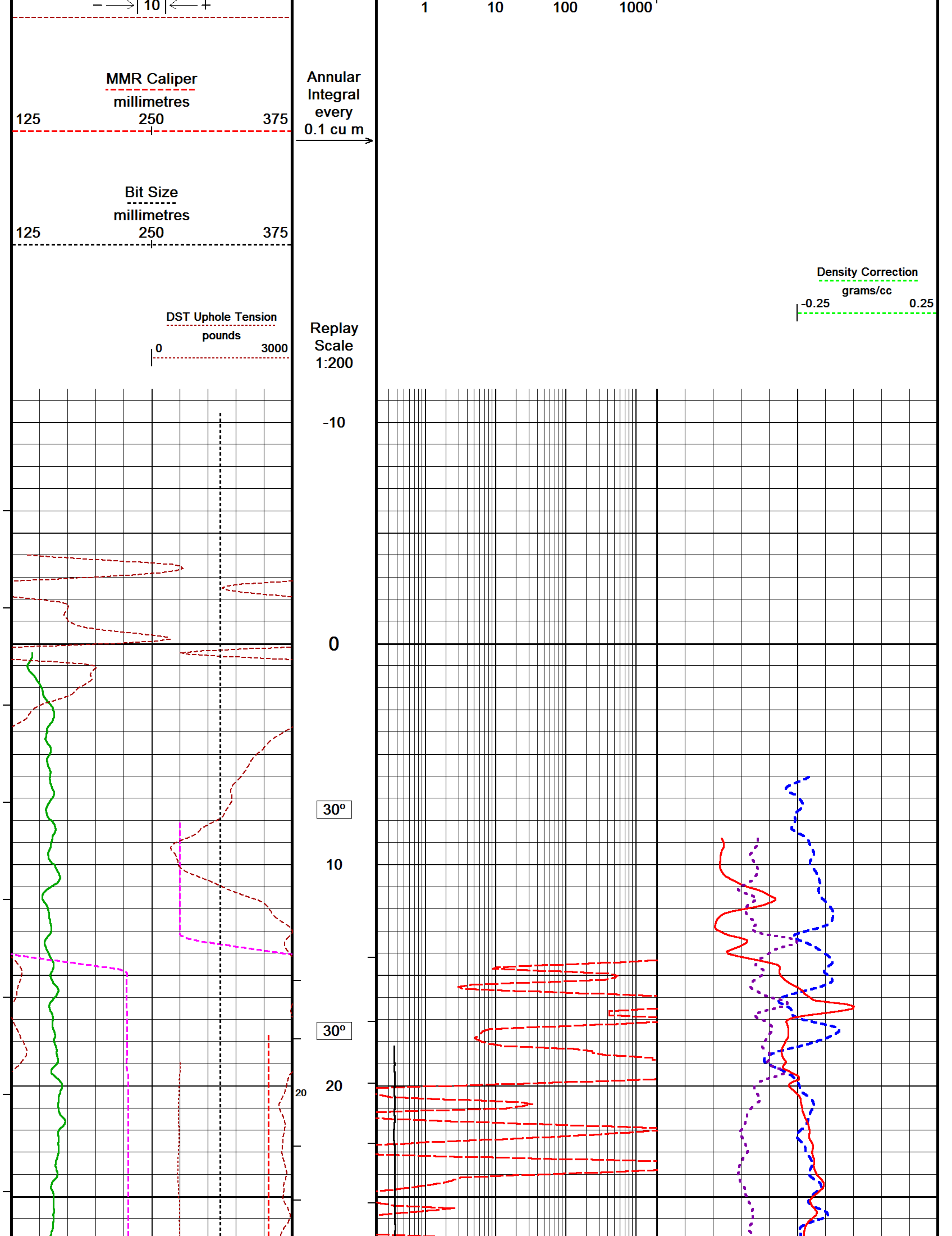
LOGGING CREW: ENGINEERS - K. SHIBESHI; OPERATOR: R. FRAMPTON, E. PAVIA

MAINLOG 1:200

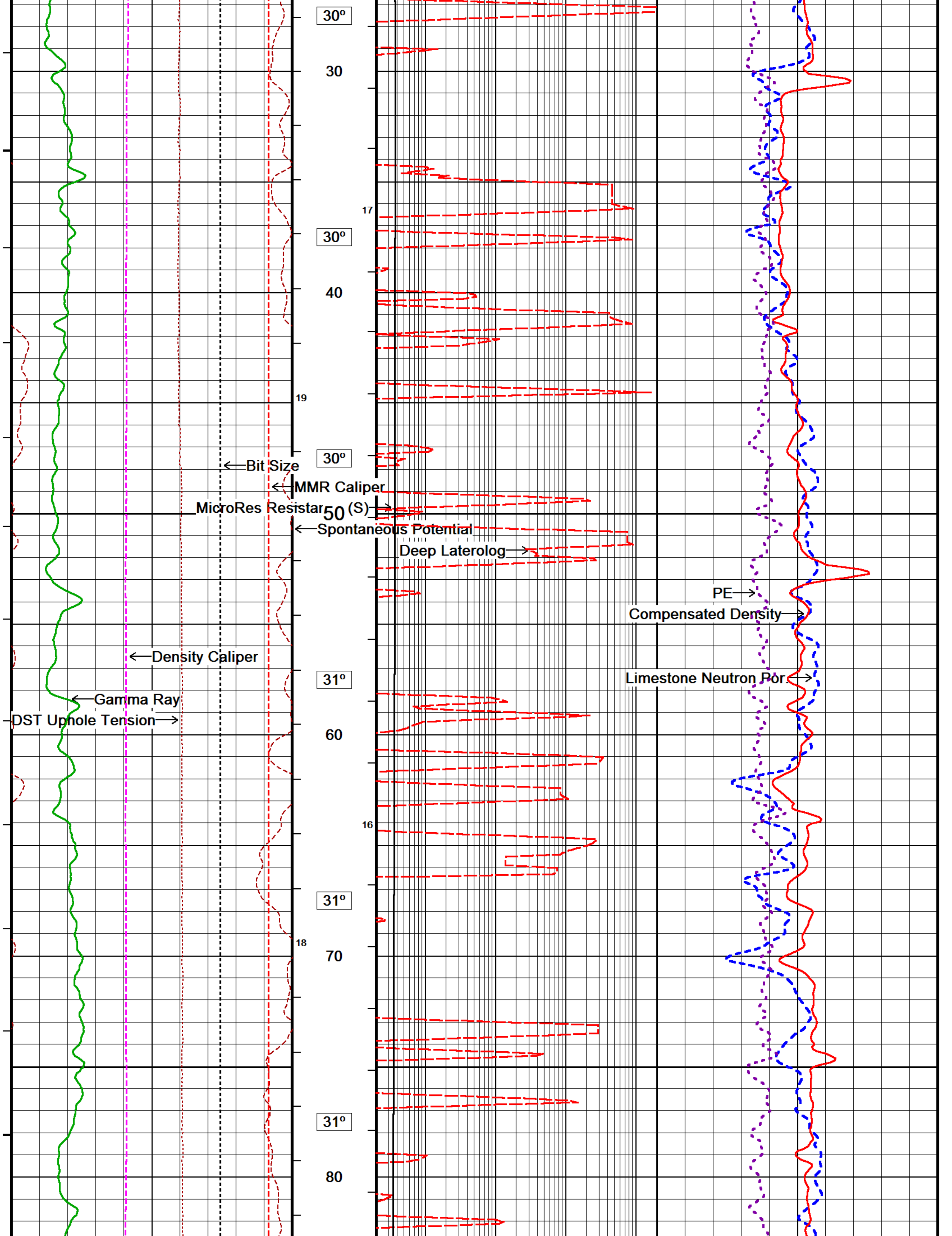
Depth Based Data - Maximum Sampling Increment 10.0cm
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\W...\TIPTON_268_10cm_CMP_OH_20191018.dta
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

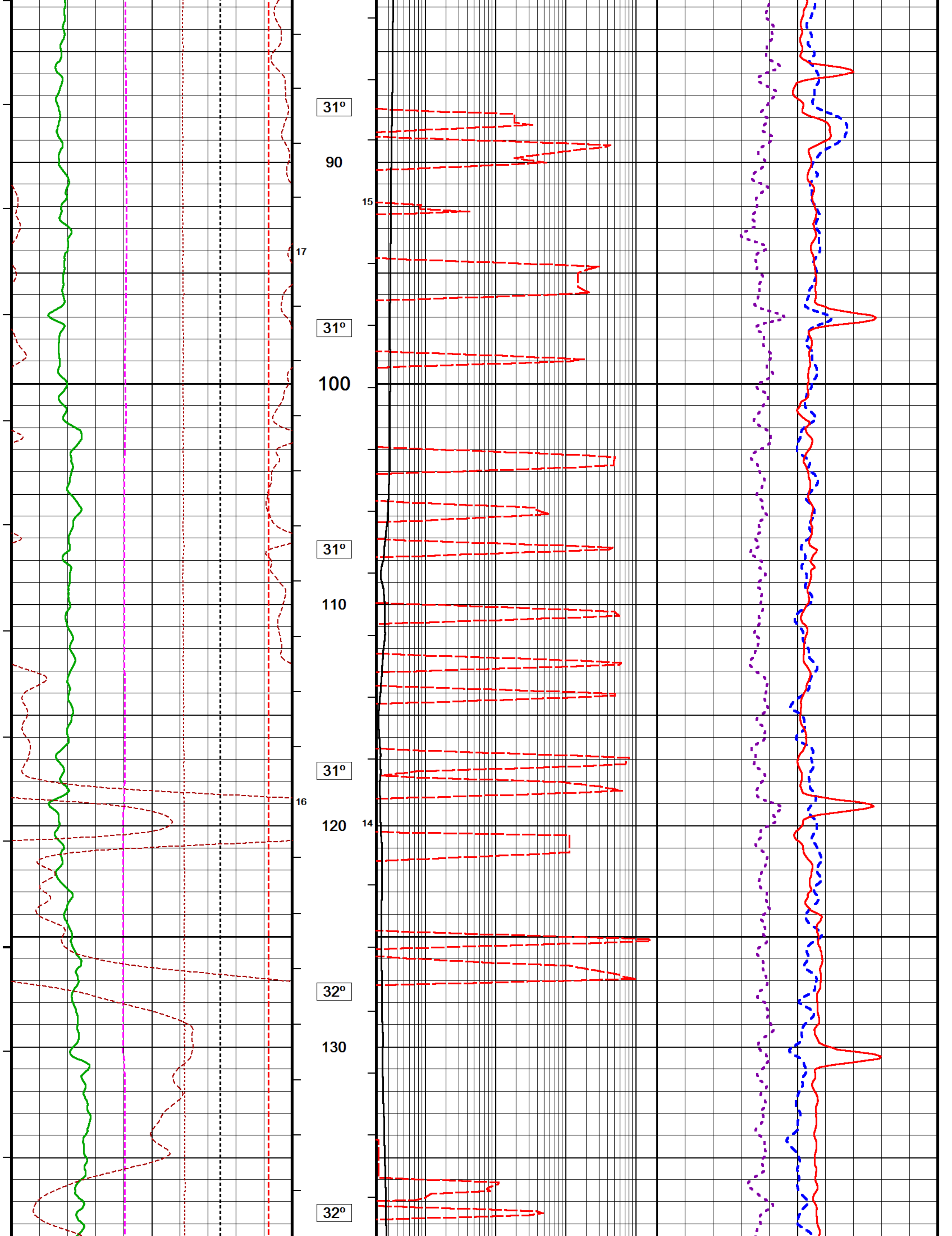
Plotted on 18-OCT-2019 17:45
Recorded on 18-OCT-2019 14:38

Track 1	Depth	Track 2
Timing Marks every 60.0 sec	Depth in Metres	Shallow Laterolog ohm metres 0.20 — 2000 (1, 10, 100, 1000) Limestone Neutron Por. v/v 1 — 0.40 — -0.20
Gamma Ray API 0 — 100 — 200 200 — 300 — 400	Borehole Temp in deg C	Deep Laterolog ohm metres 0.20 — 2000 (1, 10, 100, 1000) Compensated Density grams/cc 1 — 2 — 3
Density Caliper millimetres 125 — 250 — 375		
Spontaneous Potential millivolts	HVI every 0.1 cu m	MicroRes Resistance (S) ohm metres 0.20 — 2000 PE barns/electron 0 — 5 — 10

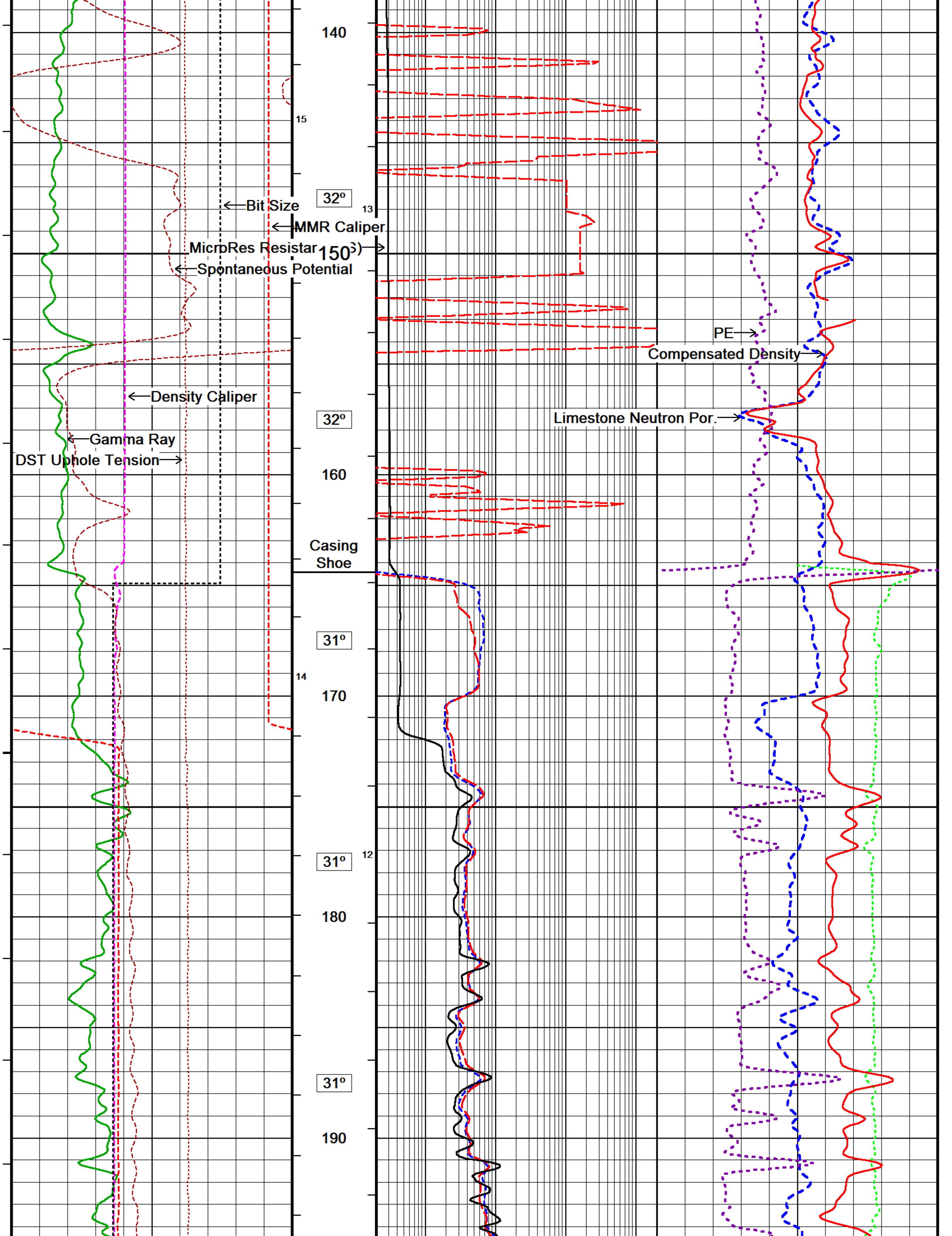


10
1
10
100
1000
MMR Caliper
millimetres
125
250
375
Annular
Integral
every
0.1 cu m
Bit Size
millimetres
125
250
375
Density Correction
grams/cc
-0.25
0.25
DST Uphole Tension
pounds
0
3000
Replay
Scale
1:200
-10
0
30º
10
30º
20
20

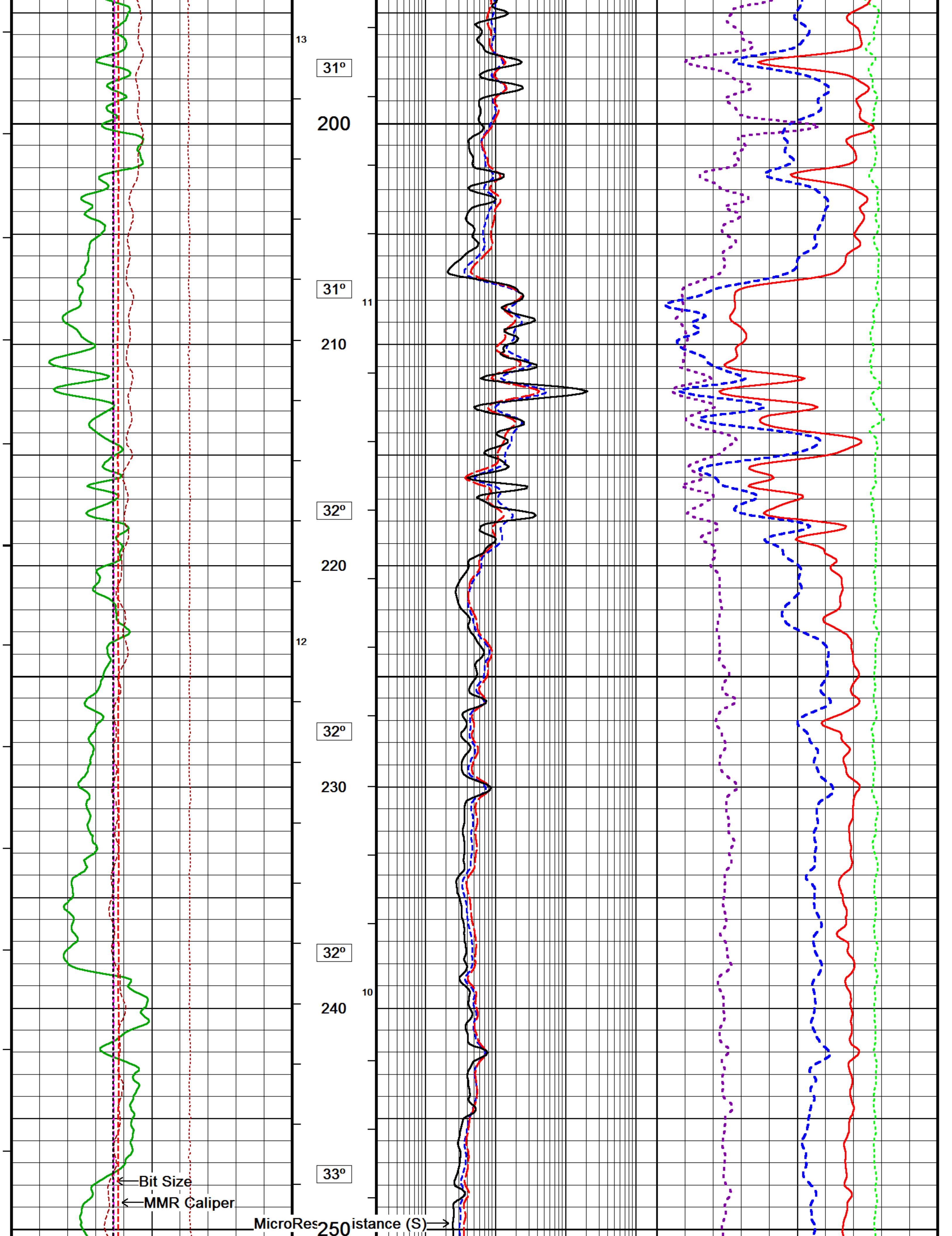

30º
30
17
30º
40
19
30º
Bit Size
MMR Caliper
MicroRes Resistar
50
(S)
Spontaneous Potential
Deep Laterolog
PE
Compensated Density
Density Caliper
31º
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
60
16
31º
18
70
31º
80



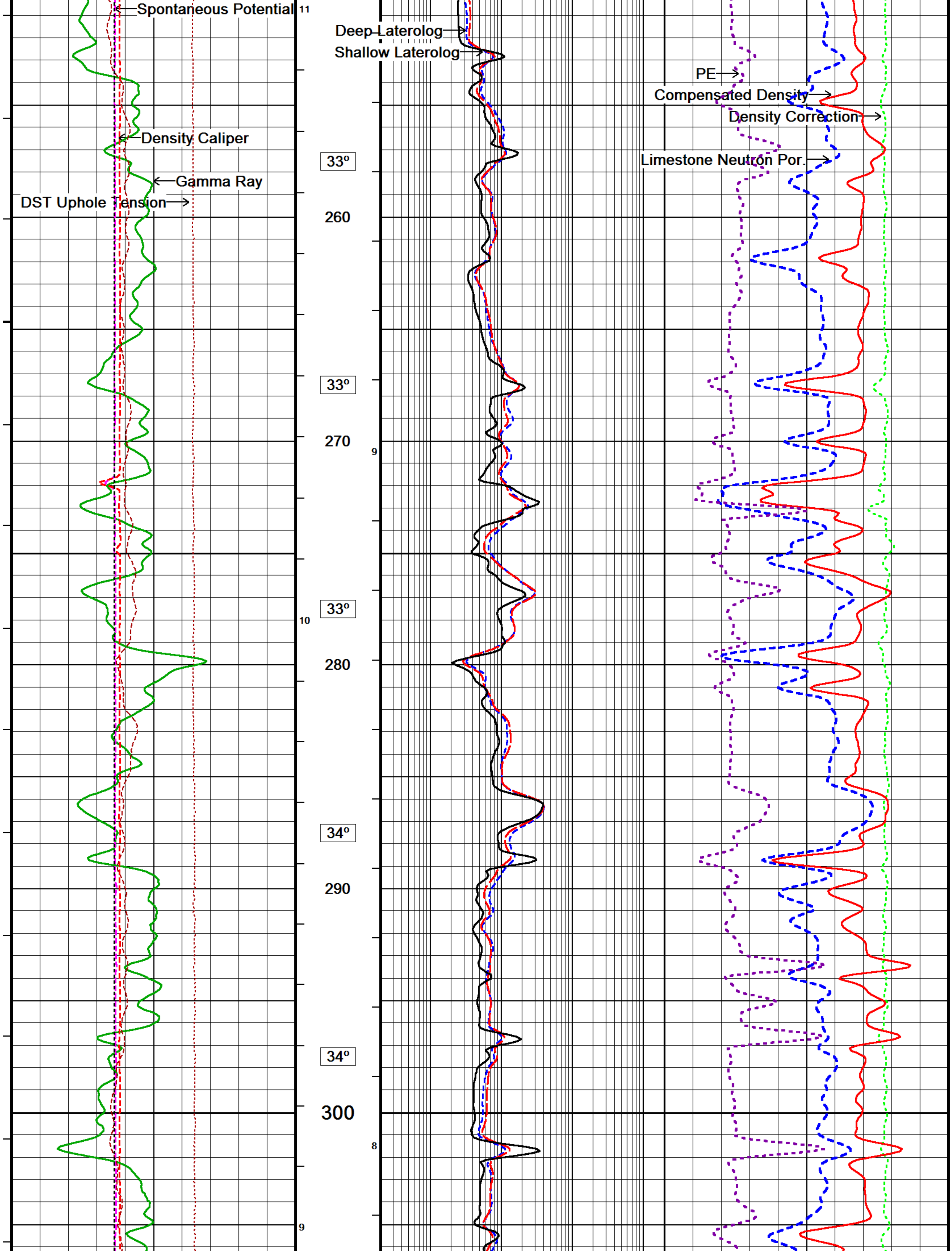
31°
90
15
17
31°
100
31°
110
31°
16
120
14
32°
130
32°



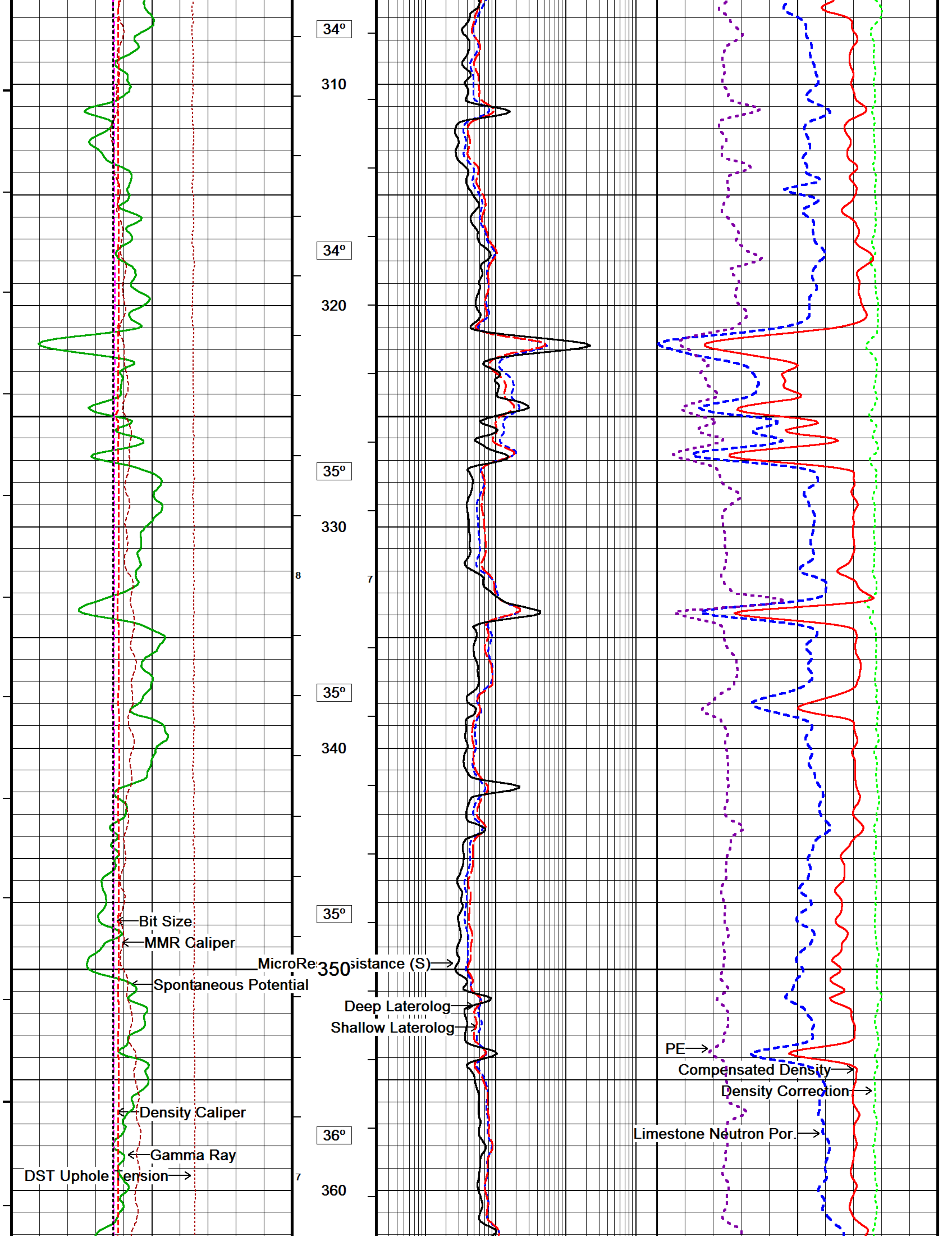
140
15
32°
13
←Bit Size
←MMR Caliper
MicroRes Resistar
150
←Spontaneous Potential
PE→
Compensated Density→
←Density Caliper
32°
Limestone Neutron Por.→
←Gamma Ray
DST Uphole Tension→
160
Casing
Shoe
31°
14
170
31°
12
180
31°
190

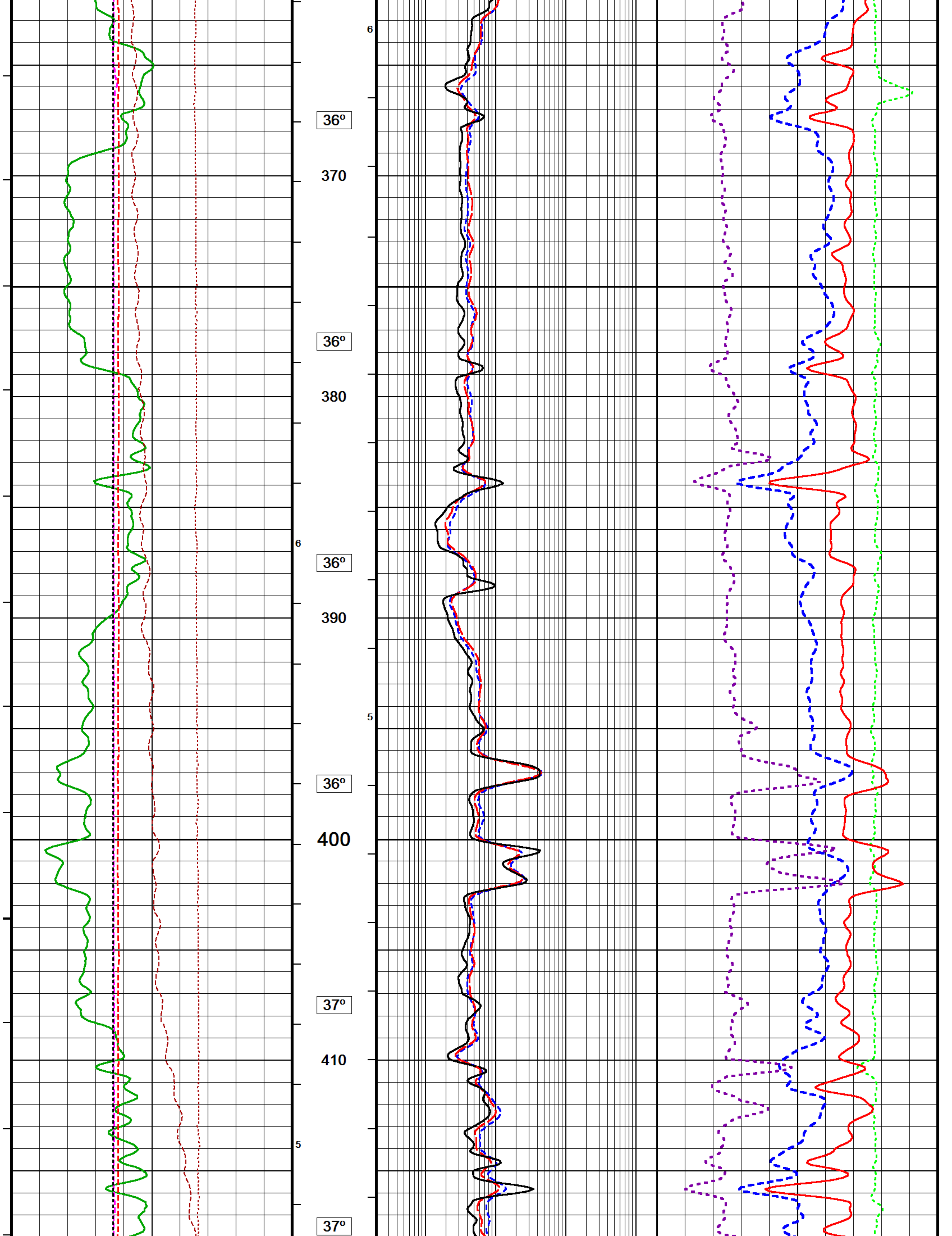


13
31°
200
31°
11
210
32°
220
12
32°
230
32°
10
240
33°
Bit Size
MMR Caliper
MicroResistance (S)
250

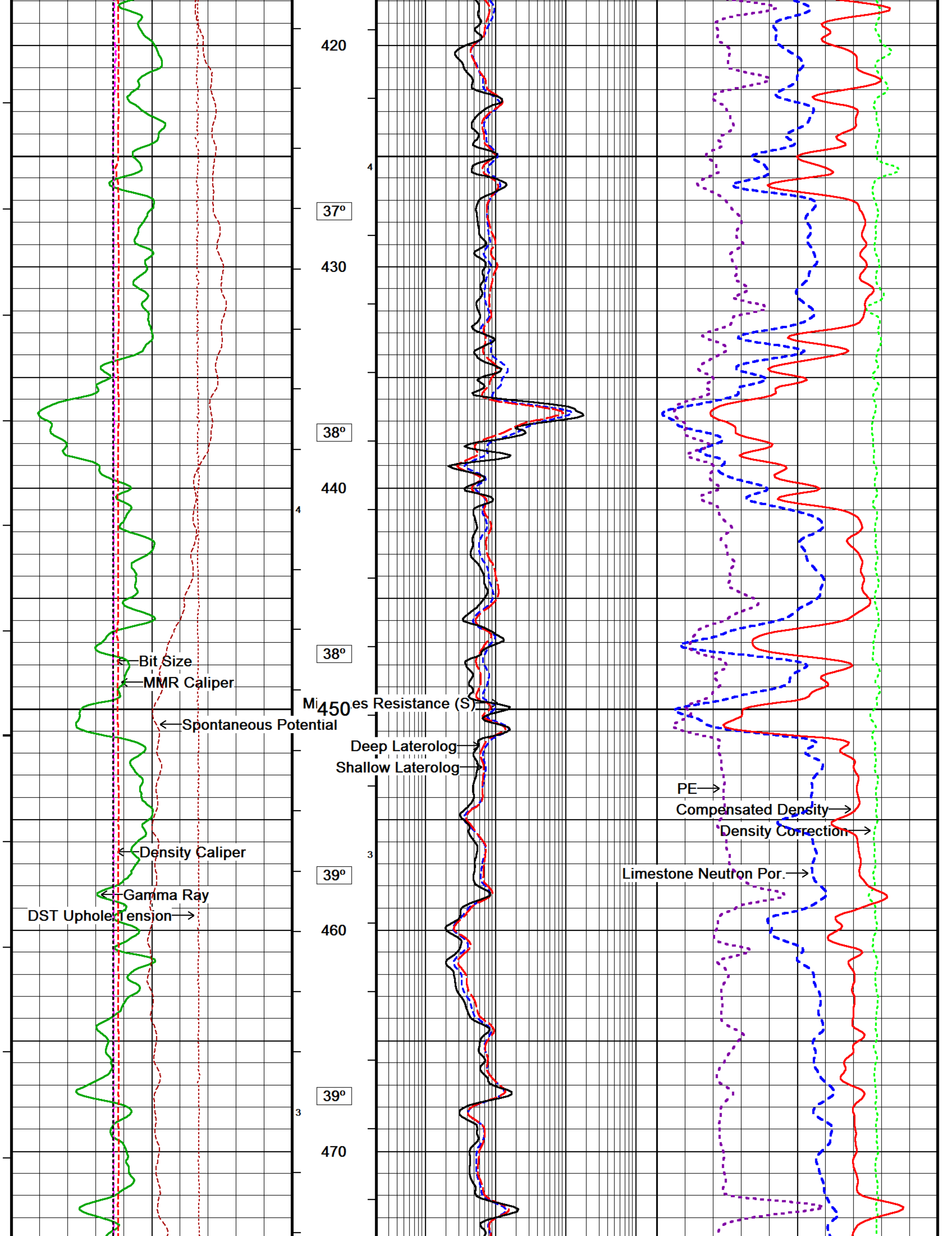


Spontaneous Potential
11
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
Density Caliper
33º
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
260
33º
270
9
33º
10
280
34º
290
34º
300
8
9

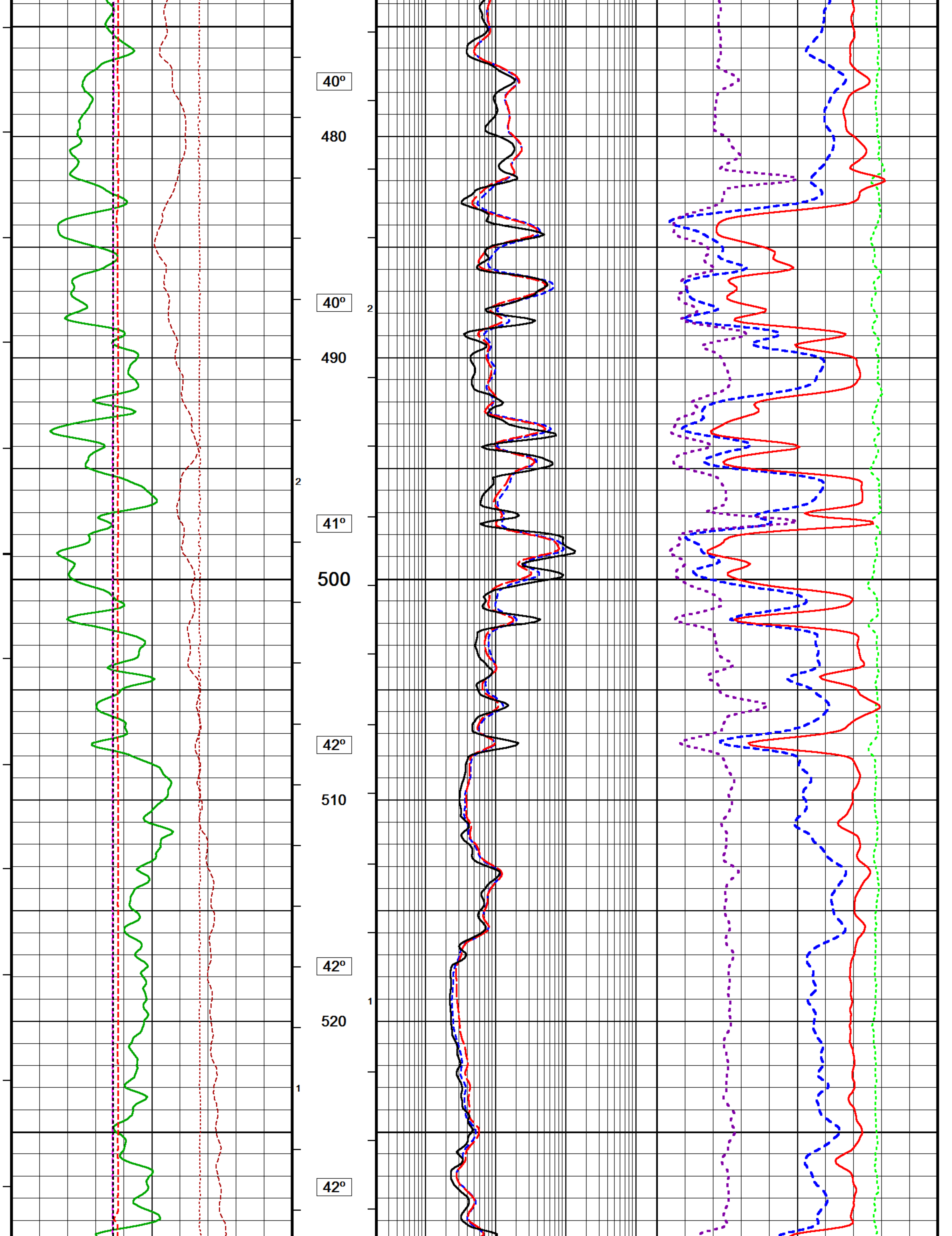

34º
310
34º
320
35º
330
8
7
35º
340
35º
Bit Size
MMR Caliper
MicroRe
350
sistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
Density Caliper
36º
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
7
360



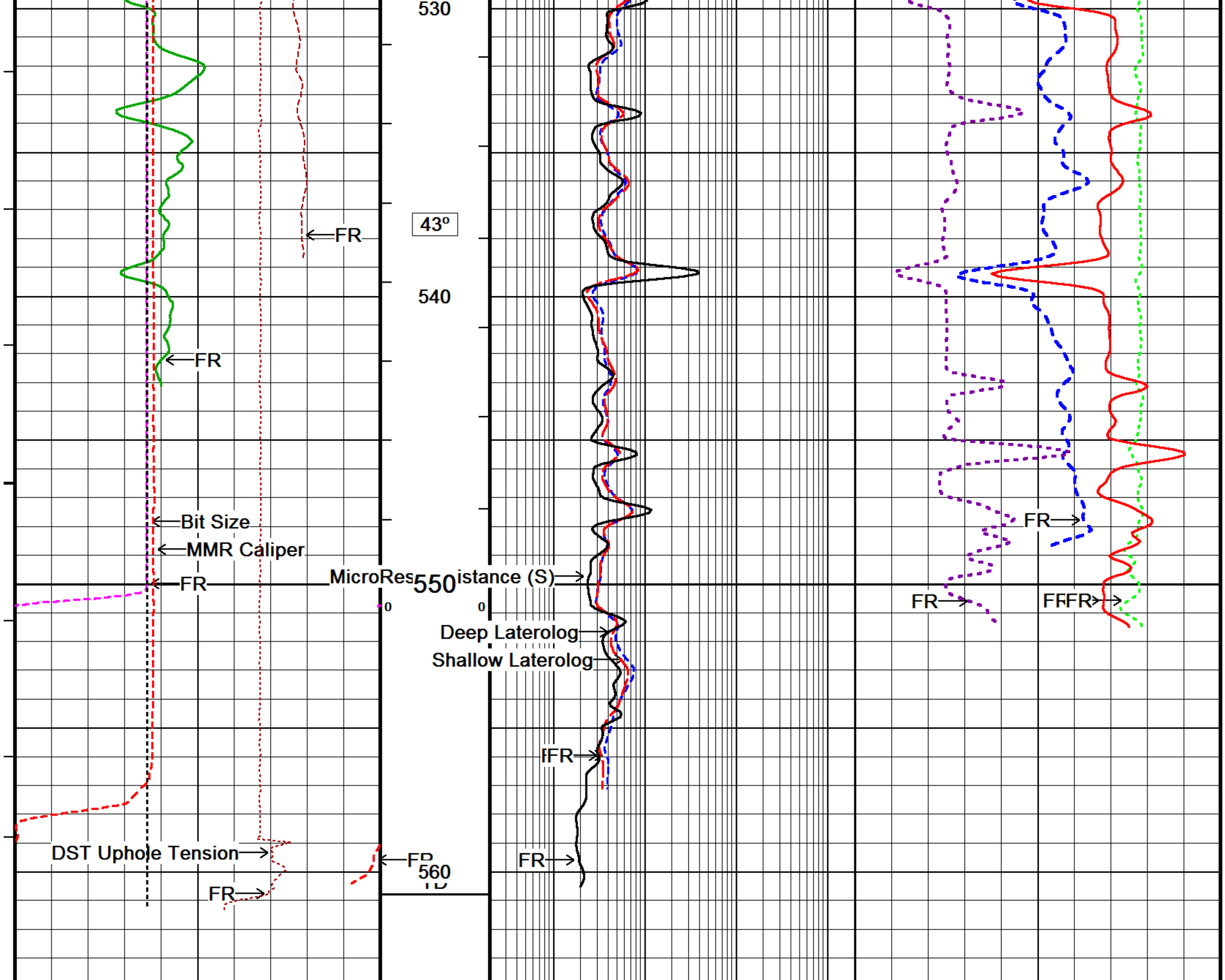
36º
370
36º
380
36º
390
36º
400
37º
410
37º
6
5
6
5



420
37º
430
38º
440
4
38º
Bit Size
MMR Caliper
450
es Resistance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Compensated Density
Density Correction
Density Caliper
39º
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
460
39º
3
470



40º
480
40º
2
490
2
41º
500
42º
510
42º
1
520
1
42º



Timing Marks
every 60.0 sec

Gamma Ray
API
0 100 200
200 300 400

Density Caliper
millimetres
125 250 375

Spontaneous Potential
millivolts
− ⟶| 10 |⟵ +

Depth in Metres

Borehole Temp in deg C

HVI every 0.1 cu m

Shallow Laterolog
ohm metres
0.20 2000
1 10 100 1000

Deep Laterolog
ohm metres
0.20 2000
1 10 100 1000

MicroRes Resistance (S)
ohm metres
0.20 2000
1 10 100 1000

Limestone Neutron Por.
v/v
1 0.40 -0.20

Compensated Density
grams/cc
1 2 3

PE
barns/electron
0 5 10

MMR Caliper
millimetres
125 250 375

Annular Integral every 0.1 cu m

Bit Size
millimetres
125 250 375

Density Correction
grams/cc
-0.25 0.25

DST Uphole Tension
pounds
0 3000

Replay Scale 1:200

Depth Based Data - Maximum Sampling Increment 10.0cm
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\W...\TIPTON_268_10cm_CMP_OH_20191018.dta
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

Plotted on 18-OCT-2019 17:45
Recorded on 18-OCT-2019 14:38

MAINLOG 1:200

HIGH RESOLUTION 1:200

Depth Based Data - Maximum Sampling Increment 2.5cm
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_2.5cm_CMP_OH_20191018.dta
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

Plotted on 18-OCT-2019 17:45
Recorded on 18-OCT-2019 14:38

Timing Marks
every 60.0 sec

Depth in Metres

Shallow Laterolog
ohm metres
0.20 2000
1 10 100 1000

Limestone Neutron Por.
v/v
1 0.40 -0.20

Gamma Ray
API
0 100 200
200 300 400

Borehole Temp in deg C

Deep Laterolog
ohm metres
0.20 2000
1 10 100 1000

Vectar Processed Density
grams/cc
1 2 3

Density Caliper
millimetres
125 250 375

HVI every 0.1 cu m

Spontaneous Potential
millivolts
– → | 10 | ← +

MicroRes Resistance (S)
ohm metres
0.20 2000
1 10 100 1000

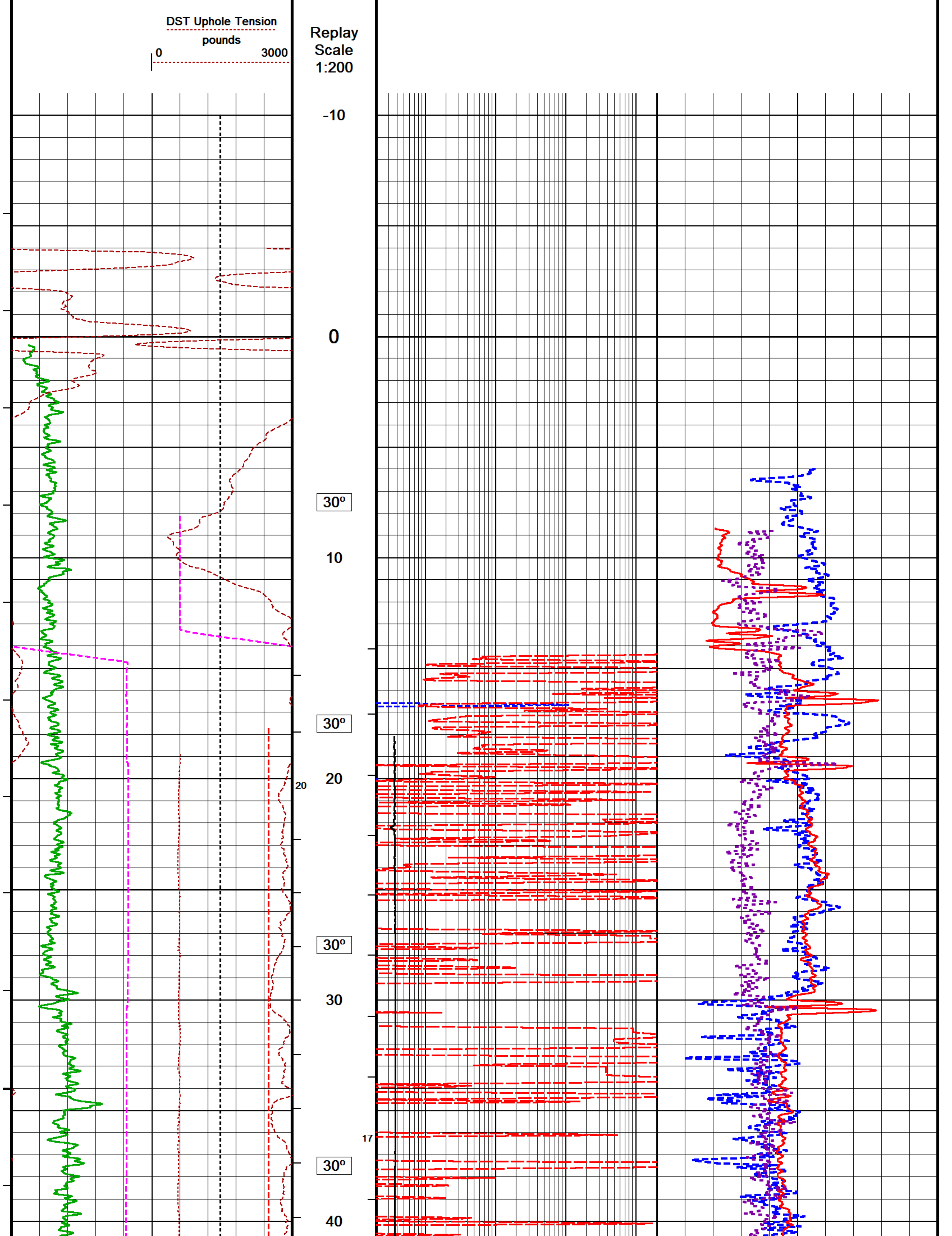
PE
barns/electron
0 5 10

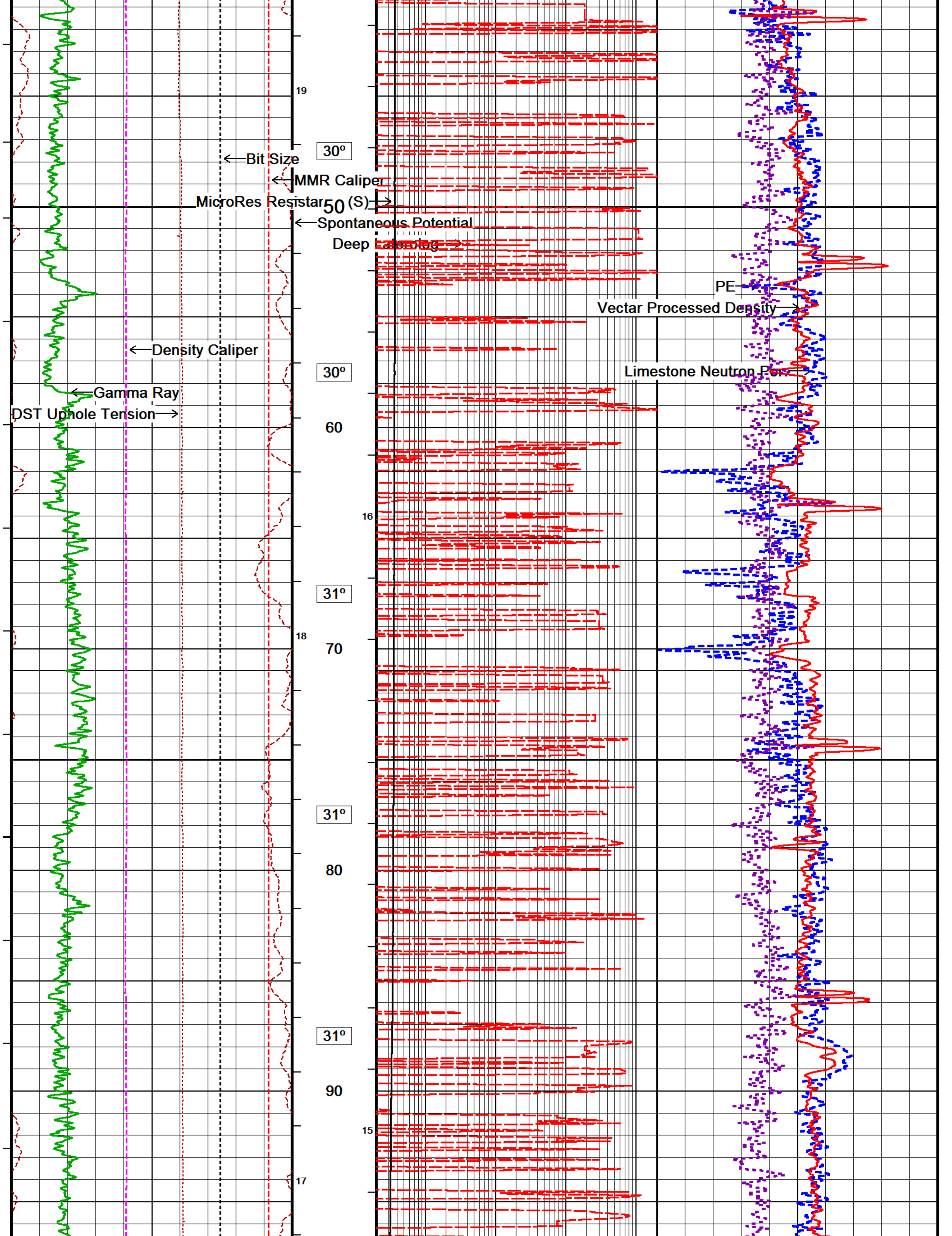
MMR Caliper
millimetres
125 250 375

Annular Integral every 0.1 cu m

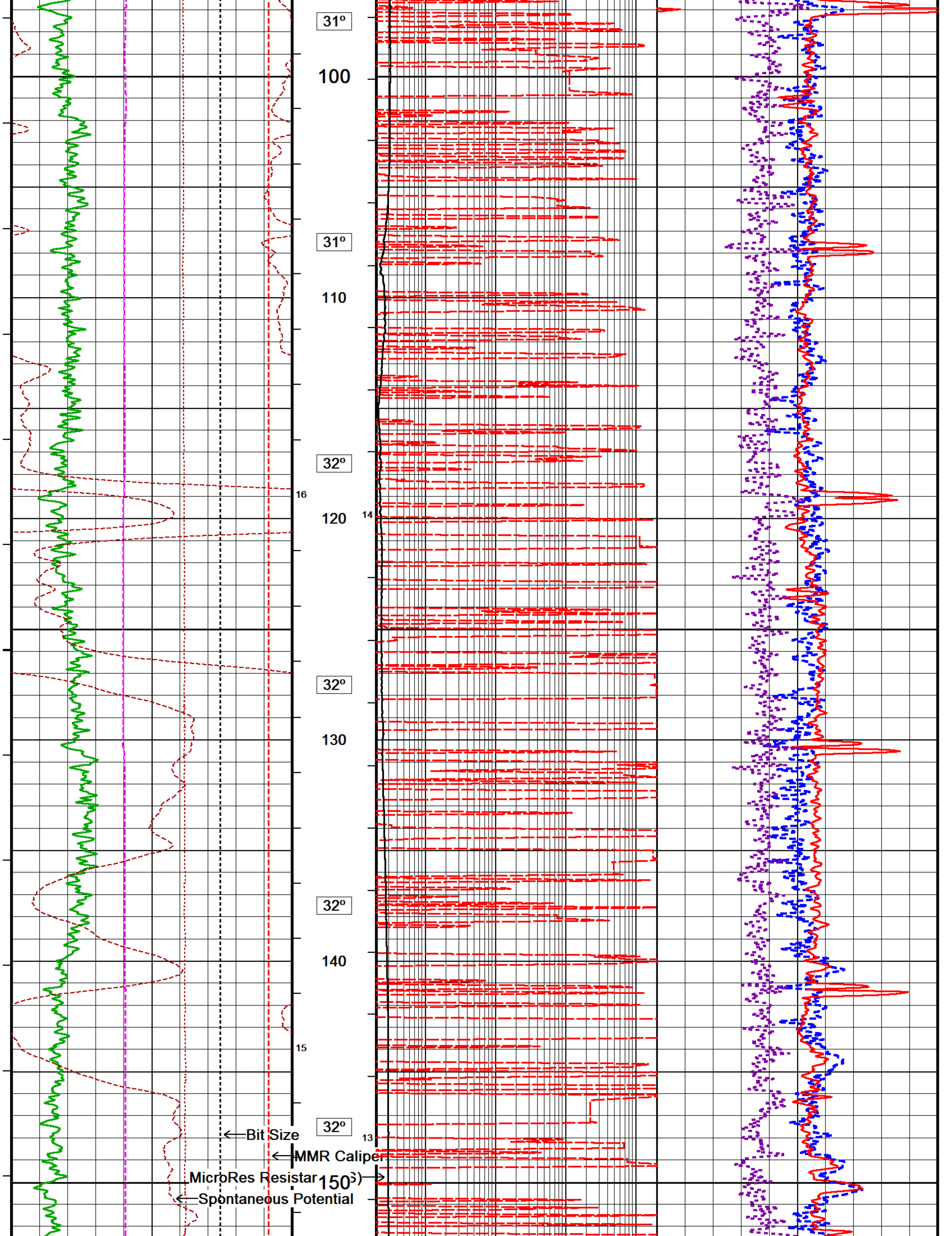
Bit Size
millimetres
125 250 375

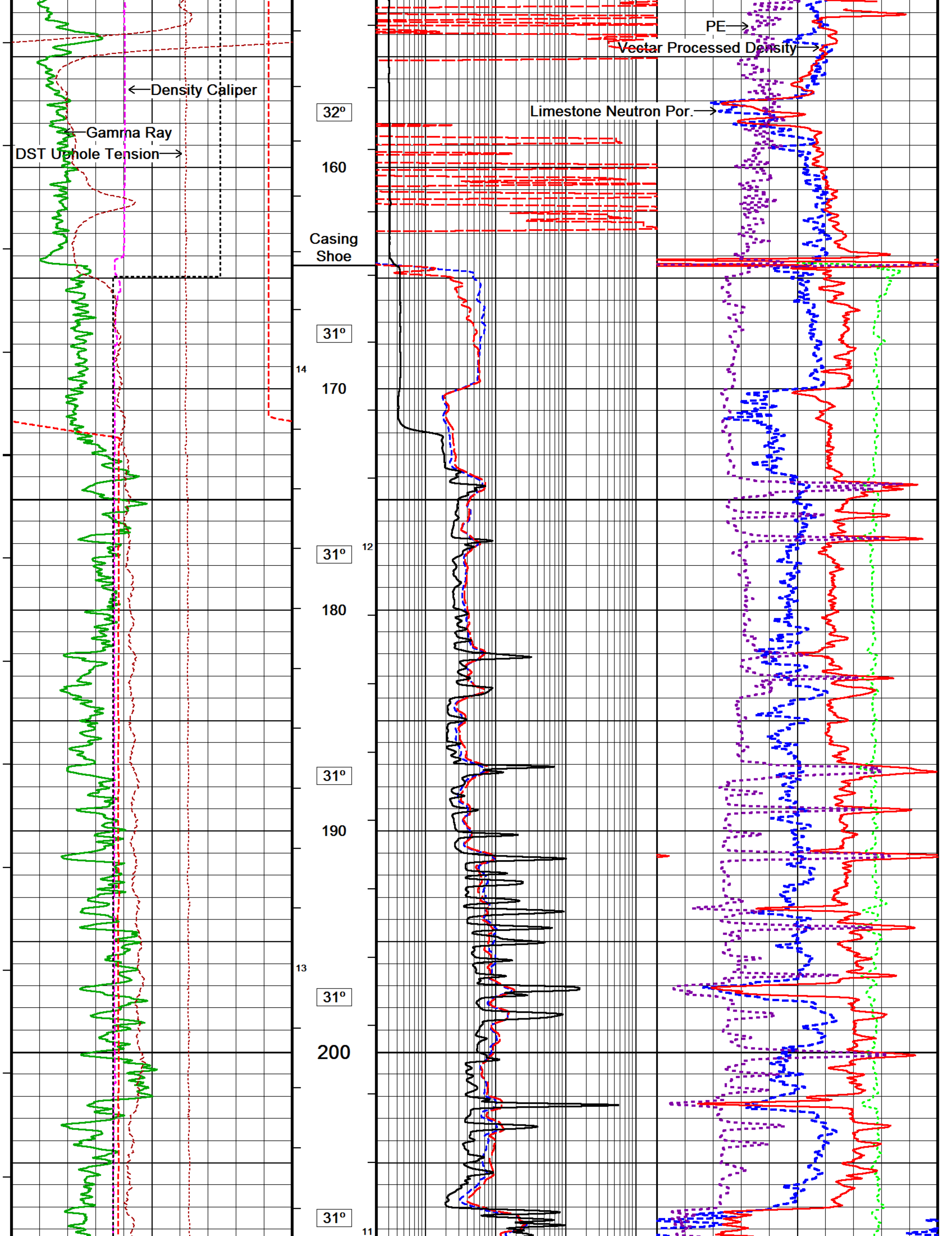
Density Correction
grams/cc
-0.25 0.25


DST Uphole Tension
pounds
0
3000
Replay
Scale
1:200
-10
0
30º
10
30º
20
20
30º
30
17
30º
40

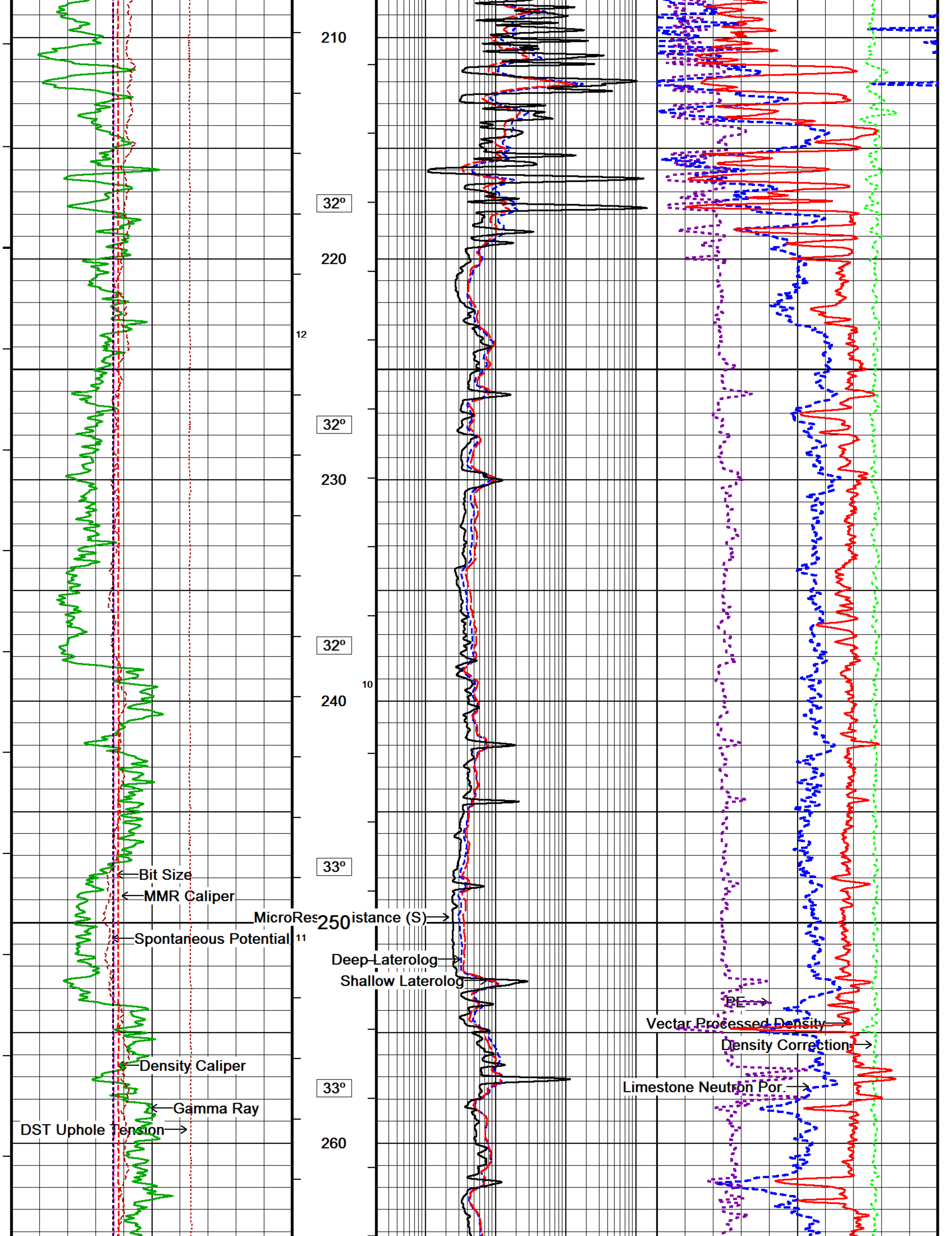


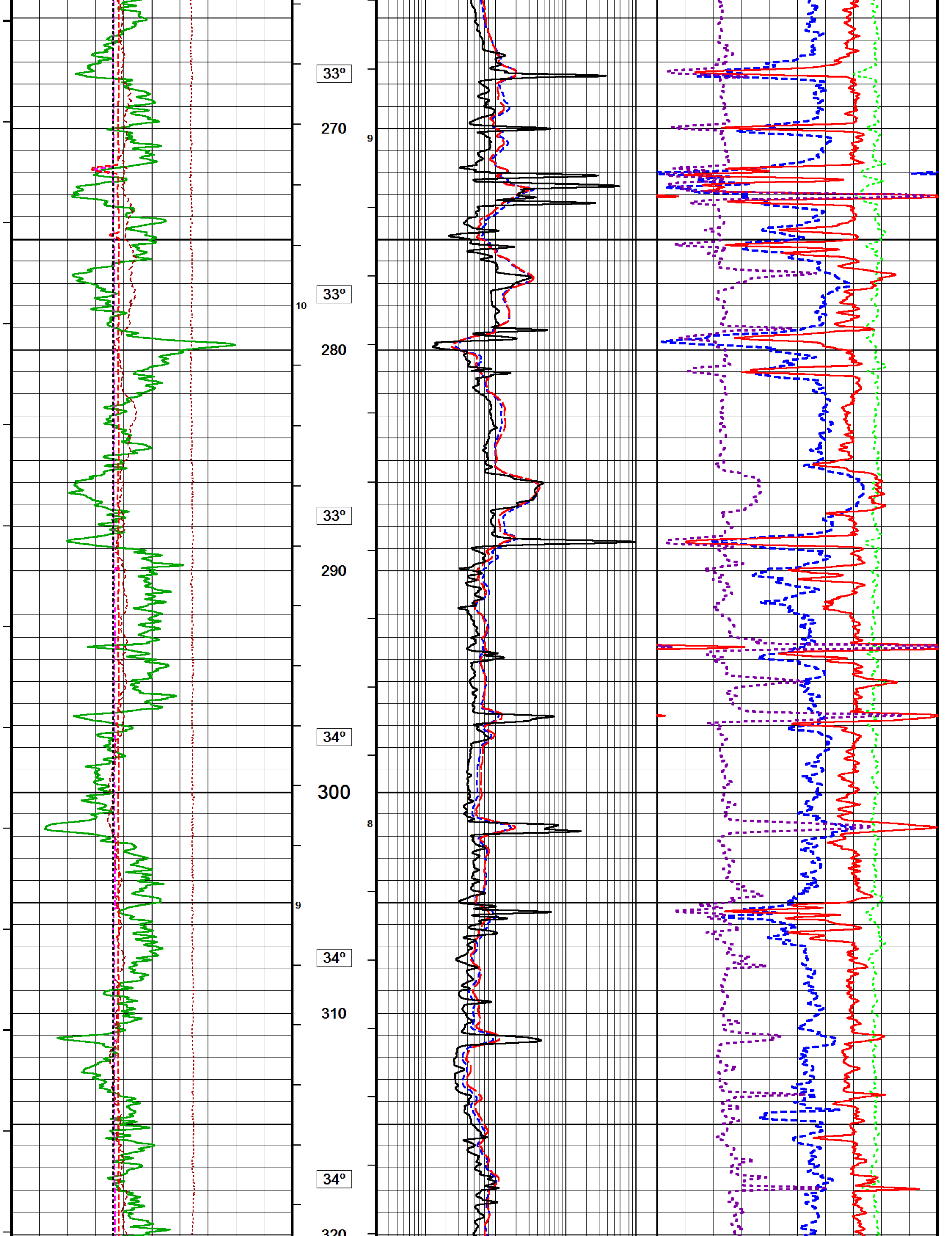
19
30°
Bit Size
MMR Caliper
MicroRes Resistar
(S)
50
Spontaneous Potential
Deep Laterolog
PE
Vectar Processed Density
Density Caliper
30°
Limestone Neutron Por
Gamma Ray
DST Uphole Tension
60
16
31°
18
70
31°
80
31°
90
15
17

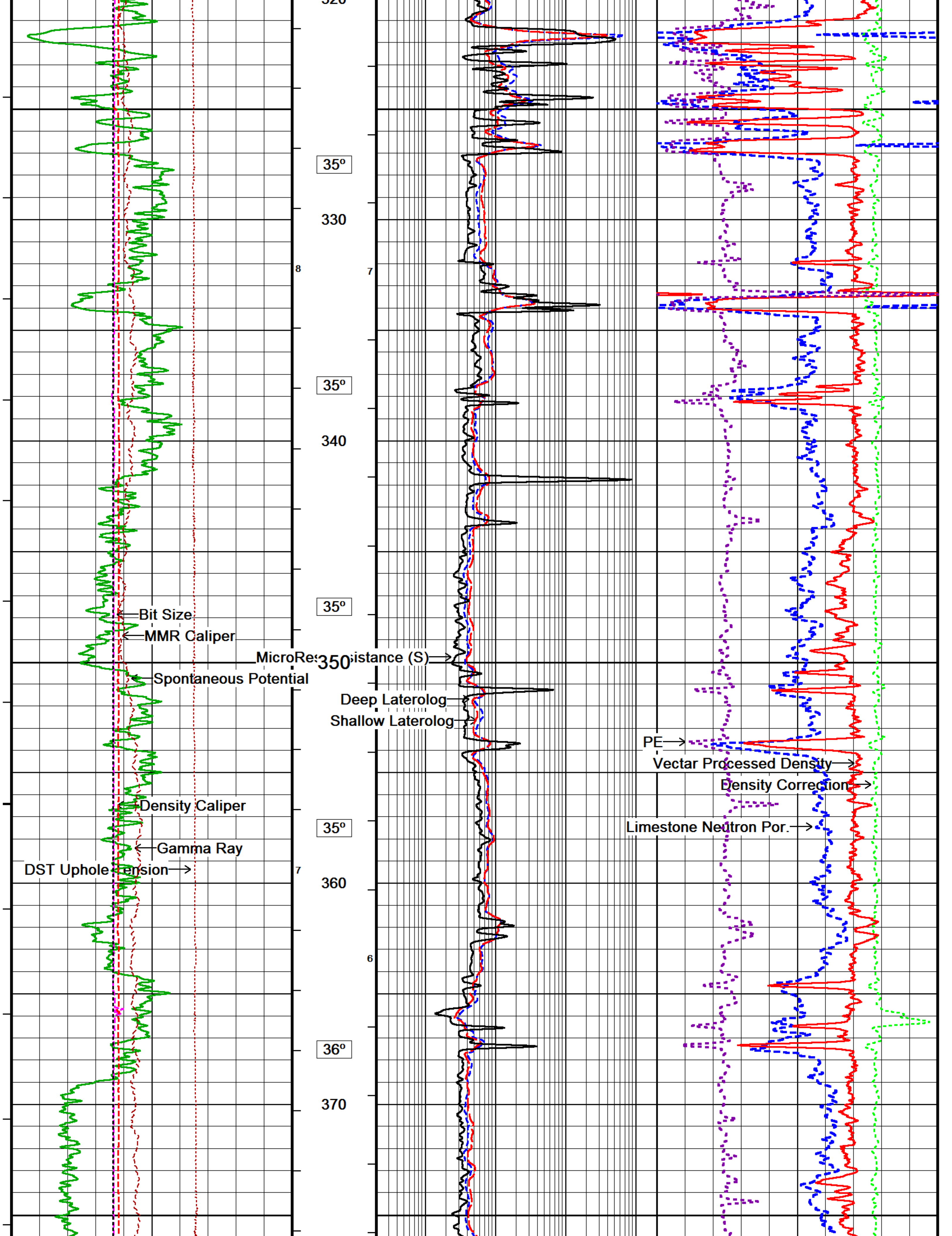

31°
100
31°
110
32°
16
120
14
32°
130
32°
140
15
32°
13
Bit Size
MMR Caliper
MicroRes Resistar
150
Spontaneous Potential

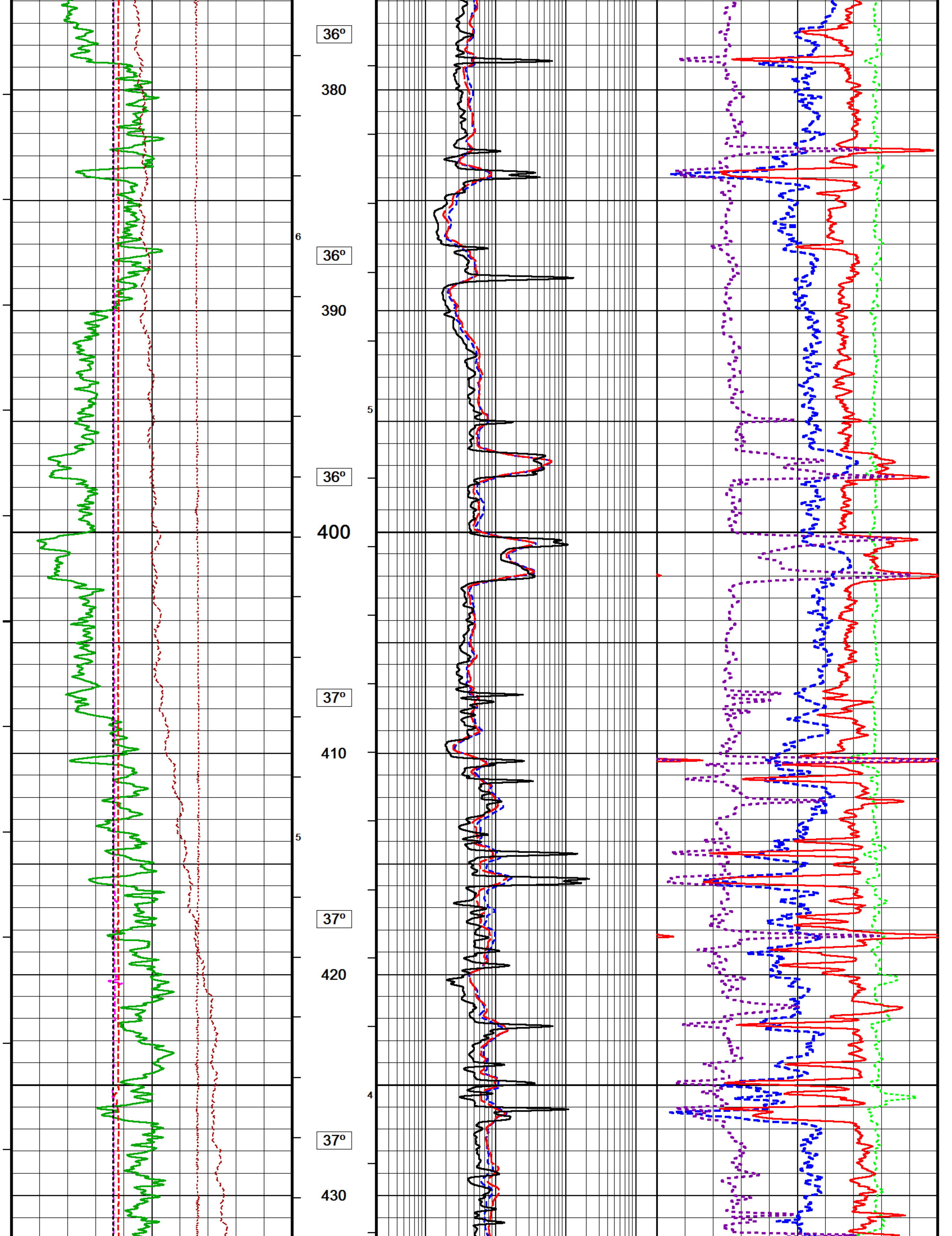


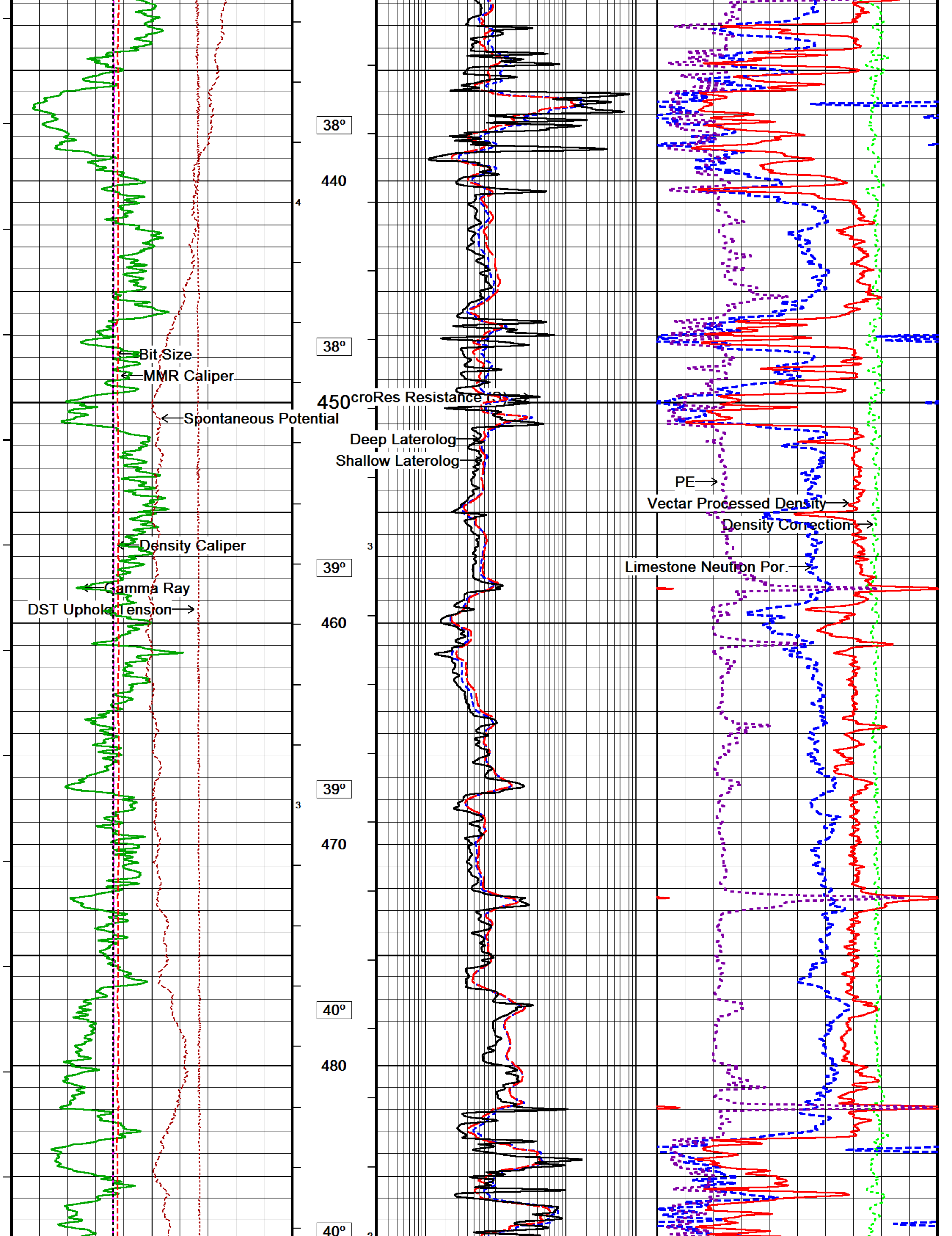
PE
Vectar Processed Density
Density Caliper
32º
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
160
Casing
Shoe
31º
14
170
31º
12
180
31º
190
13
31º
200
31º
11


210
220
230
240
250
260
32º
32º
32º
33º
33º
12
10
11
Bit Size
MMR Caliper
MicroRes
istance (S)
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Vectar Processed Density
Density Correction
Limestone Neutron Por.
Density Caliper
Gamma Ray
DST Uphole Tension

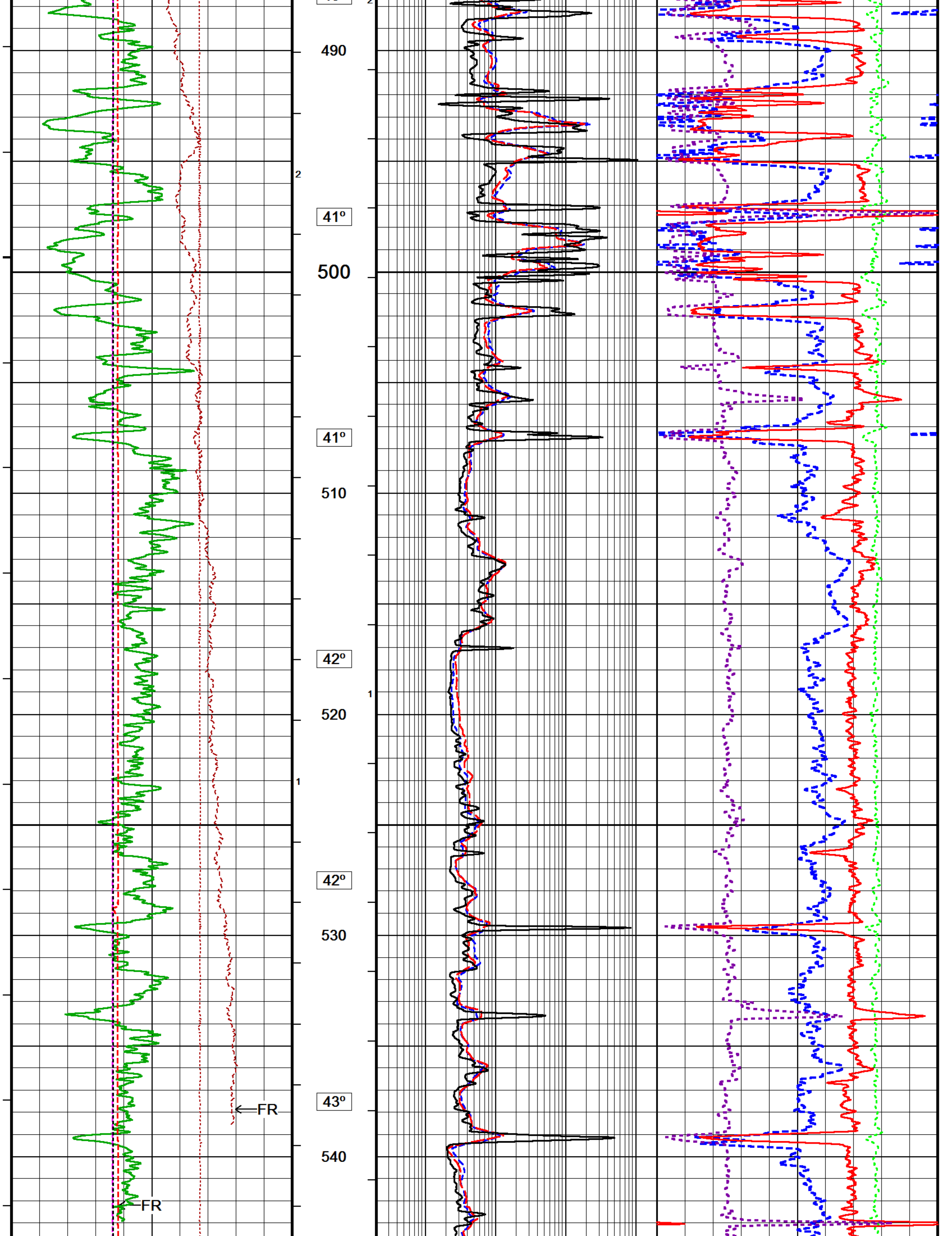

33º
270
9
10
33º
280
33º
290
34º
300
8
9
34º
310
34º
320


35º
330
8
7
35º
340
35º
Bit Size
MMR Caliper
MicroResistance (S)
350
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Vectar Processed Density
Density Correction
Density Caliper
35º
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
7
360
6
36º
370

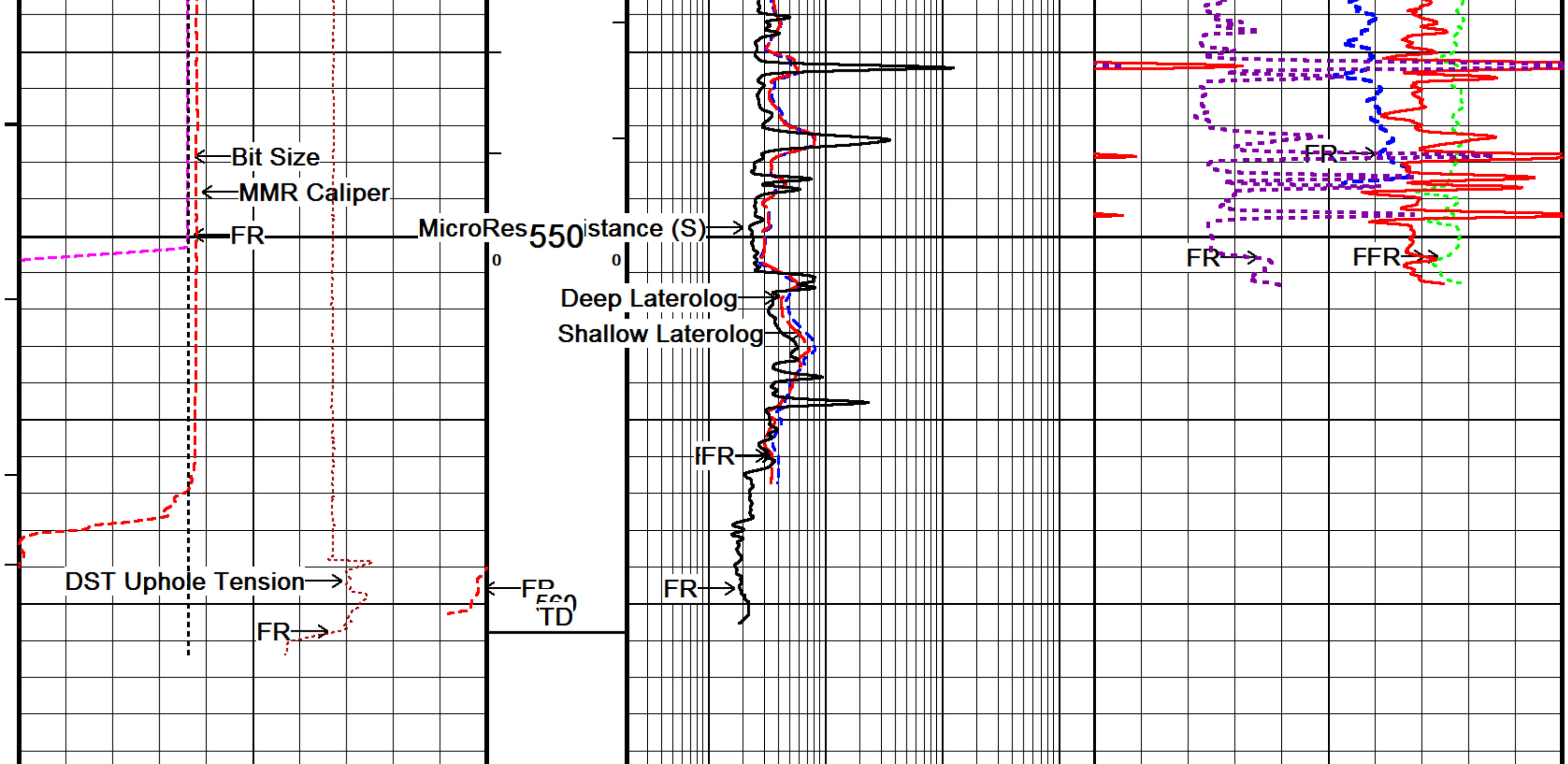

36º
380
6
36º
390
5
36º
400
37º
410
5
37º
420
4
37º
430



38°
440
4
38°
Bit Size
MMR Caliper
450
croRes Resistance
Spontaneous Potential
Deep Laterolog
Shallow Laterolog
PE
Vectar Processed Density
Density Correction
Density Caliper
3
39°
Limestone Neutron Por.
Gamma Ray
DST Uphole Tension
460
39°
3
470
40°
480
40°



490
41º
500
41º
510
42º
520
42º
530
43º
540
FR
FR
2
1



Timing Marks
every 60.0 sec

Gamma Ray
API
0 100 200
200 300 400

Density Caliper
millimetres
125 250 375

Spontaneous Potential
millivolts
– ⟶| 10 |⟵ +

MMR Caliper
millimetres
125 250 375

Bit Size
millimetres
125 250 375

DST Uphole Tension
pounds
0 3000

Depth
in
Metres

Borehole
Temp in
deg C

HVI
every
0.1 cu m

Annular
Integral
every
0.1 cu m

Replay
Scale
1:200

Shallow Laterolog
ohm metres
0.20 2000
1 10 100 1000

Deep Laterolog
ohm metres
0.20 2000
1 10 100 1000

MicroRes Resistance (S)
ohm metres
0.20 2000
1 10 100 1000

Limestone Neutron Por.
v/v
1 0.40 -0.20

Vectar Processed Density
grams/cc
1 2 3

PE
barns/electron
0 5 10

Density Correction
grams/cc
-0.25 0.25

Depth Based Data - Maximum Sampling Increment 2.5cm
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_2.5cm_CMP_OH_20191018.dta
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

Plotted on 18-OCT-2019 17:45
Recorded on 18-OCT-2019 14:38

HIGH RESOLUTION 1:200

REPEAT SECTION

Depth Based Data - Maximum Sampling Increment 10.0cm
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_10cm_CMP_OH_20191018a.dta
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

Plotted on 18-OCT-2019 17:45
Recorded on 18-OCT-2019 14:13

Timing Marks
every 60.0 sec

Gamma Ray
API
0 100 200
200 300 400

Density Caliper
millimetres
125 250 375

Spontaneous Potential
millivolts
– ⟶| 10 |⟵ +

MMR Caliper
millimetres
125 250 375

Bit Size
millimetres
125 250 375

DST Uphole Tension
pounds
0 3000

Depth
in
Metres

Borehole
Temp in
deg C

HVI
every
0.1 cu m

Annular
Integral
every
0.1 cu m

Replay
Scale
1:200

Shallow Laterolog
ohm metres
0.20 2000
1 10 100 1000

Limestone Neutron Por.
v/v
1 0.40 -0.20

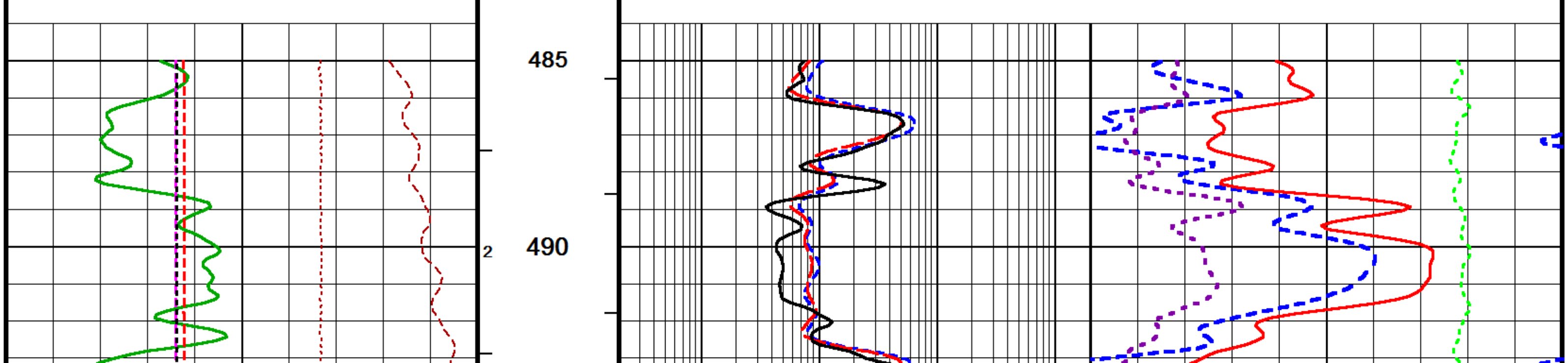
Deep Laterolog
ohm metres
0.20 2000
1 10 100 1000

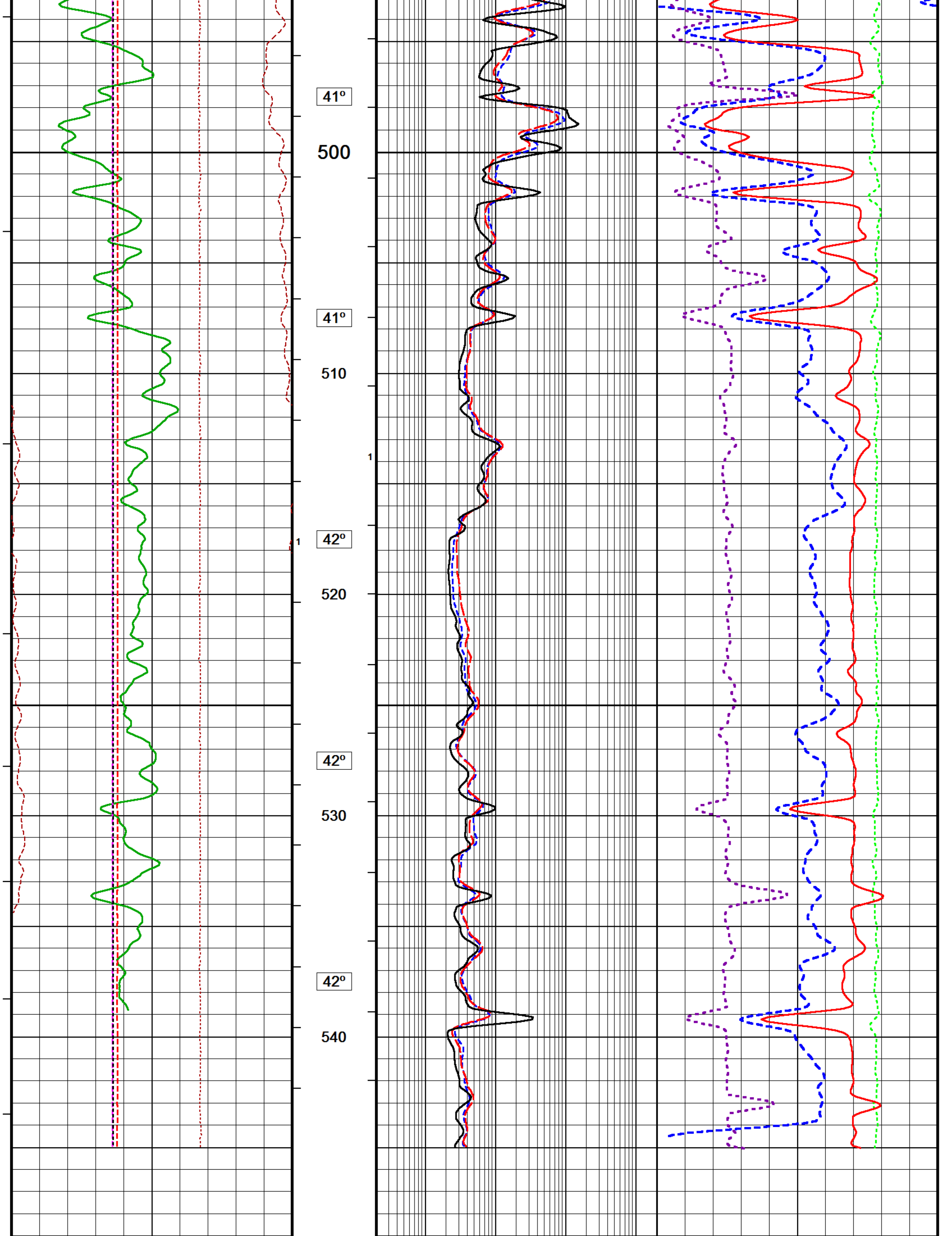
Compensated Density
grams/cc
1 2 3

MicroRes Resistance (S)
ohm metres
0.20 2000
1 10 100 1000

PE
barns/electron
0 5 10

Density Correction
grams/cc
-0.25 0.25




41º
500
41º
510
1
1
42º
520
42º
530
42º
540

Timing Marks every 60.0 sec	Depth in Metres	Shallow Laterolog ohm metres 0.20 – 2000 (1, 10, 100, 1000); Limestone Neutron Por. v/v 1 – 0.40 – -0.20
Gamma Ray API 0 – 100 – 200; 200 – 300 – 400	Borehole Temp in deg C	Deep Laterolog ohm metres 0.20 – 2000 (1, 10, 100, 1000); Compensated Density grams/cc 1 – 2 – 3
Density Caliper millimetres 125 – 250 – 375		
Spontaneous Potential millivolts – → | 10 | ← +	HVI every 0.1 cu m	MicroRes Resistance (S) ohm metres 0.20 – 2000 (1, 10, 100, 1000); PE barns/electron 0 – 5 – 10
MMR Caliper millimetres 125 – 250 – 375	Annular Integral every 0.1 cu m	
Bit Size millimetres 125 – 250 – 375		
DST Uphole Tension pounds 0 – 3000	Replay Scale 1:200	Density Correction grams/cc -0.25 – 0.25

Depth Based Data - Maximum Sampling Increment 10.0cm — Plotted on 18-OCT-2019 17:45

Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_10cm_CMP_OH_20191018a.dta — Recorded on 18-OCT-2019 14:13

System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

REPEAT SECTION

BEFORE SURVEY CALIBRATION

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018a.dta

General Constants All 000 — Last Edited on 18-OCT-2019,12:12

General Parameters		
Mud Resistivity	0.302	ohm-metres
Mud Resistivity Temperature	25.000	degrees C
Water Level	0.000	metres
Borehole Fluid Processing	Water Level Switch	
Hole/Annular Volume and Differential Caliper Parameters		
HVOL Method	Single Caliper	
HVOL Caliper 1	Density Caliper	
HVOL Caliper 2	N/A	
Annular Volume Diameter	76.200	mm
Caliper for Differential Caliper	Density Caliper	

Caliper for Differential Caliper	Density Caliper

Rwa Parameters	
Porosity used	Base Density Porosity
Resistivity used	Deep Laterolog
RWA Constant A	0.620
RWA Constant M	2.150
SW/APOR Tool Source	0.000

Gamma Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:35

	Measured	Calibrated (API)
Background	65	43
Calibrator (Gross)	849	565
Calibrator (Net)	784	522

Gamma Calibration Tolerances MCG-E.A 580

1.40 1.475 1.55

Ratio 1.502 Counts/API

Gamma Constants MCG-E.A 580

Last Edited on 18-OCT-2019,12:13

Gamma Calibrator Number	120	
GRC-M Calibrator Jig in Use?	NO	
Inactive Background Jig in Use?	NO	
Mud Density	1.01	gm/cc
Caliper Source for Processing	Density Caliper	
Tool Position	Eccentred	
Potassium Equivalence	Chloride	
K Mud Concentration	0.00	%

High Resolution Temperature Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:36

	Measured	Calibrated(Deg C)
Lower	0.00	0.00
Upper	100.00	100.00

High Resolution Temperature Constants MCG-E.A 580

Last Edited on 12-OCT-2019,13:36

Pre-filter Length	11

Navigation Constants MBN-D.A 170

Last Edited on 18-OCT-2019,12:13

Magnetic Declination	10.39	degrees	East

Magnetometer Parameters MBN-D.A 170

Date Of Last Magnetometer Calibration 14-OCT-2013,10:52

	X Magnetometer	Y Magnetometer	Z Magnetometer
Slope	-1.000000	1.003752	1.001092
Offset	0.001123	0.014769	-0.000632

Magnetometer Constants MBN-D.A 170

Magnetometer Calibrator Number	000

Accelerometer Parameters MBN-D.A 170

Date Of Last Accelerometer Calibration 10-OCT-2013,11:36

	X Accelerometer	Y Accelerometer	Z Accelerometer
Slope	-1.094953	-1.100967	-1.098665
Offset	0.003562	0.001559	0.005569

Accelerometer Constants MBN-D.A 170

Last Edited on 15-SEP-2019,17:36

Accelerometer Calibrator Number	000

Accelerometer Temperature Characterisation

X Accelerometer

Serial Number	1295			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.35051e-05	7.52948e-10	1.11593e-10
	SF0	SF1	SF2	SF3

Scale Factor(mA/g)	3.00000e+00	2.64265e-04	3.57391e-07	7.78280e-10

Y Accelerometer

Serial Number	1239			
Calibration Date	11-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.67211e-05	2.44017e-08	7.98178e-11
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.80123e-04	3.19145e-07	8.20741e-10

Z Accelerometer

Serial Number	1310			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.13686e-05	-1.00367e-08	1.99621e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.55395e-04	3.67997e-07	9.83969e-10

Neutron Calibration MDN-B.A 270

Base Calibration on 14-OCT-2019 09:17
Field Check on 16-OCT-2019 09:38

Base Calibration

	Measured		Calibrated (cps)	
	Near	Far	Near	Far
	2976	88	3714	110
Ratio	33.984		33.764	

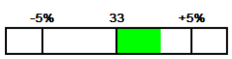
Field Calibrator at Base

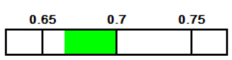
	Calibrated (cps)	
	2227	3352
Ratio	0.664	

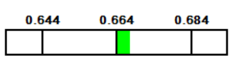
Field Check

	Calibrated (cps)	
	2234	3344
Ratio	0.668	

Neutron Calibration Tolerances MDN-B.A 270

Ratio 33.984 

Base Check 0.664 

Field Check 0.668 

Neutron Constants MDN-B.A 270

Last Edited on 18-OCT-2019,12:14

Neutron Source Id	16150B	
Neutron Jig Number	NECD200	
Air Hole Processing	Legacy	
Caliper Source for Processing	Density Caliper	
Stand-off	0.00	mm
Mud Density	1.01	gm/cc
Limestone Sigma	7.10	cu
Sandstone Sigma	4.26	cu
Dolomite Sigma	4.70	cu
Formation Pressure Source	None	
Formation Pressure	N/A	kpsi
Temperature Source	MCG External Temperature	
Temperature	N/A	degrees C
Mud Salinity	0.00	kppm
Salinity Correction	Not Applied	
Formation Fluid Salinity Source	None	
Formation Fluid Salinity	N/A	kppm
Barite Mud Correction	Not Applied	

Laterolog Calibration MLE-C.A 109

Base Calibration on 11-OCT-2019 10:01
Field Check on 11-OCT-2019 10:06

Resistor 1 (ohm)	Resistor 2 (ohm)
0.0	1000.0

Base Calibration

	Measured		Calibrated (ohm-m)	
Channel	Resistor 1	Resistor 2	Resistor 1	Resistor 2
Shallow	0.0	999.4	0.0	1284.4
Deep	0.0	[illegible]	0.0	795.7

Deep	0.0	999.3	0.0	795.7
Groningen	0.0	999.1	0.0	808.4

Channel	Base Check (ohm-m)	Field Check (ohm-m)
Shallow	46.2	46.2
Deep	28.6	28.6
Groningen	232.8	232.8

Laterolog Calibration Tolerances MLE-C.A 109

	Value	-3%	Nominal	+3%	Unit
Shallow Resistor 2	999.4	-3%	985.0	+3%	ohm
Deep Resistor 2	999.3	-3%	985.0	+3%	ohm
Groningen Resistor 2	999.1	-3%	985.0	+3%	ohm

	Value	-2%	Nominal	+2%	Unit
Shallow Base Check	46.2	-2%	46.23	+2%	ohm-m
Deep Base Check	28.6	-2%	28.64	+2%	ohm-m
Groningen Base Check	232.8	-2%	232.83	+2%	ohm-m
Shallow Field Check	46.2	-2%	46.2	+2%	ohm-m
Deep Field Check	28.6	-2%	28.6	+2%	ohm-m
Groningen Field Check	232.8	-2%	232.8	+2%	ohm-m

Laterolog Constants MLE-C.A 109

Last Edited on 12-OCT-2019,13:26

Profiling Laterolog Type	Type 0	
Laterolog Output Filter	Logarithmic	
Profiling Limiter	On	
Median Filter	On	
Squasher Start	40000	ohm-m
Shallow Laterolog K Factor	1.2844	
Deep Laterolog K Factor	0.7957	
Groningen Laterolog K Factor	0.8084	
Interference Rejection	50 Hz	
SP Connection	SP Bridle Electrode (Lower)	
Groningen Connection	Groningen Electrode (Upper)	
Profiling Deep Input	N/A	

Borehole Correction Constants

Bridle Type	Standard	
Sonde Position	12.7	millimetres
Hole Size Source	Density Caliper	
Hole Size Constant Value	N/A	millimetres
Rm Source	Global Value: Temperature Corrected	
Temp. for Rm Corr.	MCG External Temperature	

Apparent Porosity and Water Saturation Constants

Archie Constant (A)	1.00	
Cementation Exponent (M)	2.00	
Saturation Exponent (N)	2.00	
Saturation of Water for Apor	100.00	percent
Resistivity of Water for Apor and Sw	0.05	ohm-m
Resistivity of Mud Filtrate for Sw	0.00	ohm-m
Source for Rt	0.00	
Source for Rxo	0.00	

SP Calibration MLE-C.A 109

Field Calibration on 12-OCT-2019,13:26

	Measured	Calibrated (mV)
Reference 1	100.0	100.0
Reference 2	-100.0	-100.0

Micro Laterolog Calibration MMR-B.A 126

Base Calibration on 10-OCT-2019 16:30
Field Check on 10-OCT-2019 16:33

Resistor 1 (ohm)	Resistor 2 (ohm)
0.0	10000.0

Base Calibration

Measured		Calibrated (ohm-m)	
Ref 1	Ref 2	Ref 1	Ref 2

0.0	9867.3	0.0	128.0

Base Check (ohm-m)	Field Check (ohm-m)
5.2	5.2

Micro Laterolog Calibration Tolerances MMR-B.A 126

	Value	-Tol	Nominal	+Tol	Unit
Ref 2	9867.3	-3%	9900.00	+3%	ohm
Base Check	5.2	-2%	5.2	+2%	ohm-m
Field Check	5.2	-2%	5.2	+2%	ohm-m

Micro Laterolog Constants MMR-B.A 126

Last Edited on 13-OCT-2019,17:01

Pad Type	6 in Solid Nylon B23059	
Standoff Offset	12.7000	millimetres
Micro Laterolog K Factor	0.0128	
Micro Laterolog Rm K Factor	N/A	

Mudcake Thickness Correction Constants		
Mud Cake Source	Constant Value	
Mud Cake Thickness	10.1600	millimetres
Mud Cake Thickness Caliper	N/A	
Mud Cake Resistivity	0.1500	ohm-m
Mud Cake Resistivity Temp.	20.00	Degrees C
Mud Cake Resistivity Source	Temperature Corr	
Temp. for Rmc Corr.	MCG Internal Temperature	

Caliper Calibration MMR-B.A 126

Base Calibration on 10-OCT-2019 16:39
Field Calibration on 13-OCT-2019 16:50

Base Calibration

Reading No	Measured	Calibrator Size (mm)
1	11571	101.83
2	14686	151.74
3	17781	202.31
4	21116	250.47
5	25245	302.77
6	N/A	N/A

Field Calibration

Measured Caliper (mm)	Actual Caliper (mm)
209.37	213.89

Caliper Calibration Tolerances MMR-B.A 126

	Value	Min	Nominal	Max	Unit
Short Arm Field Cal.	209.4	208.8	213.9	219.0	mm

Caliper Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:43
Field Calibration on 16-OCT-2019 09:31

Base Calibration

Reading No	Measured	Calibrator Size (in)
1	16321	4.01
2	26493	5.97
3	36763	7.96
4	46427	9.86
5	57546	11.92
6	N/A	N/A

Field Calibration

Measured Caliper (in)	Actual Caliper (in)
7.95	7.96

Caliper Calibration Tolerances MPD-D.A 449

	Value	Min	Nominal	Max	Unit
Short Arm Field Cal.	7.95	7.76	7.96	8.16	in

Photo Density Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:30
Field Check on 16-OCT-2019 09:28

Density Calibration

Base Calibration	Measured	Calibrated (sdu)

Base Calibration	Measured Near	Measured Far	Calibrated (cdu) Near	Calibrated (cdu) Far
Background	1134	1304		
Reference 1	45330	22156	59605	30991
Reference 2	18639	2314	24756	2530

Field Check at Base	Near	Far
	1134.5	1304.4

Field Check	Near	Far
	1139.8	1311.8

PE Calibration

Base Calibration	WS	Measured WH	Ratio	Calibrated Ratio
Background	215	1016		
Reference 1	20159	45158	0.452	0.372
Reference 2	5757	18514	0.317	0.271

Field Check at Base	WS	WH
	215.2	1016.3

Field Check	WS	WH
	214.2	1018.6

Photo Density Calibration Tolerances MPD-D.A 449

	Value	Tolerance		Value	Tolerance
Near Density Ratio	2.52	-5% 2.52 +5%	Far Density Ratio	20.65	-5% 21.00 +5%
PE Calibration	0.126	0.089 0.110 0.131			
Near Den. Field Check	1139.8	-3% 1134.5 +3%	Far Den. Field Check	1311.8	-3% 1304.4 +3%
PE WS Field Check	214.2	-6% 215.2 +6%	PE WH Field Check	1018.6	-6% 1016.3 +6%

Density Constants MPD-D.A 449

Last Edited on 18-OCT-2019,12:13

Density Source Id	15756B	
Nylon Calibrator Number	DCE-667	
Aluminium Calibrator Number	DACD-674	
Density Shoe Profile	8 inch	
Caliper Source for Processing	Density Caliper	
PE Correction to Density	Not Applied	
Mud Density	1.01	gm/cc
Mud Density Type		
Mud Filtrate Density	1.00	gm/cc
Dry Hole Mud Filtrate Density	1.00	gm/cc
DNCT	0.00	gm/cc
CRCT	0.00	gm/cc
Density Z/A Correction	Hybrid	
Precision Enhanced Density Processing	Not Applied	

Matrix density (gm/cc)	Depth (m)
2.71	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	

DOWNHOLE EQUIPMENT

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018a.dta

11C Tension Cablehead
MCC-A 1 LG: 0.73 m WT: 19.8 lb OD: 57.0 mm

11C-11B Compact Tool Adaptor
MTA-A 70 LG: 0.47 m WT: 13.2 lb OD: 57.0 mm

MTA-A 70 LG: 0.47 m WT: 13.2 lb OD: 57.0 mm

Compact Stiff Bridle Electrode Sub.
MBE-C.B 179 LG: 3.76 m WT: 77.2 lb OD: 58.0 mm

Compact Stiff Bridle Electrode Sub.
MBE-A 54 LG: 3.76 m WT: 77.2 lb OD: 57.0 mm

Compact Comms Gamma
MCG-E.A 580 LG: 2.65 m WT: 63.9 lb OD: 57.0 mm

Compact Navigation
MBN-D.A 170 LG: 3.60 m WT: 70.5 lb OD: 57.0 mm

Compact Neutron
MDN-B.A 270 LG: 1.53 m WT: 50.7 lb OD: 57.0 mm

Compact Density/Caliper
MPD-D.A 449 LG: 2.92 m WT: 90.4 lb OD: 62.2 mm

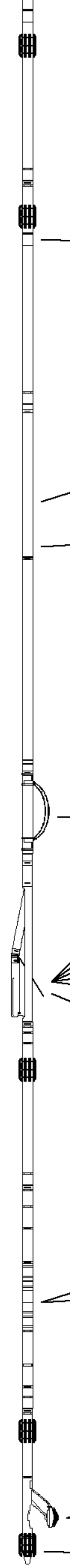
Compact Upper Guard Sub.
MUG-B.A 144 LG: 2.74 m WT: 68.3 lb OD: 57.0 mm

Compact Laterolog Electrode Sub.
MLE-C.A 109 LG: 3.76 m WT: 92.6 lb OD: 57.0 mm

Compact Micro-Resistivity
MMR-B.A 126 LG: 2.62 m WT: 81.6 lb OD: 97.0 mm

Pressure Bung + Hole Finder
HFS 3 LG: 0.28 m WT: 6.6 lb OD: 57.0 mm

Total Length: 28.82 m Weight: 712.1 lb



22.02 m	SPDL - Spontaneous Potential
17.68 m	GRGC - MCG Gamma Ray
16.80 m	CGXT - MCG External Temperature
12.12 m	NPRL - Limestone Neutron Por.
9.91 m	AVOL - Annular Volume
9.91 m	HVOL - Hole Volume
9.91 m	CLDC - Density Caliper
9.32 m	DEN - Compensated Density
9.32 m	DCOR - Density Correction
9.30 m	PDPE - PE
3.93 m	DSLL - Shallow Laterolog
3.93 m	DDLL - Deep Laterolog
0.30 m	MRRS - MicroRes Resistance (S)
0.30 m	MATC - MMR Caliper
Tool Zero	(0.85m from bottom)
-0.85 m	SMTU - DST Uphole Tension

All measurements relative to tool zero.

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA

Elevation Kelly Bushing	349.55	metres	First Reading	560.23	metres
Elevation Drill Floor	349.55	metres	Depth Driller	565.55	metres
Elevation Ground Level	345.1	metres	Depth Logger	560.75	metres

Weatherford®

MLL - SLL - DLL

DENSITY - NEUTRON

1:200

MLL- SLL - DLL - SONIC
DENSITY NEUTRON
1:500

Weatherford®

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA
LOCATION	PL198

FIELD PRINT

Latitude	27°23'18.4689" S	Other Services
Longitude	151°6'16.9779" E	BOREHOLE NAVIGATION
UTM Easting	312 589.65	
UTM Northing	6 969 111.4	

Permanent Datum M.S.L., Elevation 0.00 metres	Elevations:	metres
Log Measured From GL, 345.10 metres above Permanent Datum	KB	349.55
Drilling Measured From KB	DF	349.55
	GL	345.10

Date	18-OCT-2019		
Run Number	1		
Service Order	90518		
Depth Driller	565.55	metres	
Depth Logger	560.75	metres	
First Reading	560.23	metres	
Last Reading	0.00	metres	
Casing Driller	164.55	metres	
Casing Logger	164.40	metres	
Bit Size	215.900	inches	
Hole Fluid Type	KCL		
Density / Viscosity	1.01 g/c3	26.00 sec/qt	
PH / Fluid Loss	8.50	0.00 ml/30Min	
Sample Source	MUD TANK		
Rm @ Measured Temp	0.302 @ 25.0	ohm-m	
Rmf @ Measured Temp	0.275 @ 25.0	ohm-m	
Rmc @ Measured Temp	0.453 @ 25.0	ohm-m	
Source Rmf / Rmc	CAL	CAL	
Rm @ BHT	0.218 @ 43.0	ohm-m	
Time Since Circulation	8 HRS 37 MIN		
Max Recorded Temp	43.00	deg C	
Equipment / Base	13354	ROMA	
Recorded By	K. SHIBESHI		
Witnessed By	M. GALBRAITH		
Stop Circulation	06:00 / 18 OCT 2019		

BOREHOLE RECORD

Last Edited: 18-OCT-2019 11:45

Bit Size millimetres	Depth From metres	Depth To metres
311.150	0.00	164.95
215.900	164.95	565.55

CASING RECORD

Type	Size millimetres	Depth From metres	Shoe Depth metres	Weight pounds/ft
SURFACE	244.475	0.00	164.55	36.00

REMARKS

RUN NUMBER 1 IS THE PRIMARY DEPTH REFERENCE LOG. ALL OTHER RUNS ARE CORRELATED BACK TO THIS LOG.

SOFTWARE ISSUE: VERSION 19.01.8879 RELEASED DATE APRIL 1, 2019.

CUSTOMER SCALES AND INTERVALS LOGGED.

RUN 1: MMR, MLE,MUG, MPD, MDN, MBN, MCG, MBE, MBE MTA, CBH TOOLS RAN IN COMBINATION.

HARDWARE
RUN 1:
- MBE: 1 X 0.5" STANDOFF ON EACH.
- MUG: 1 X 0.5" STANDOFF.
- MMR: 2 X 0.5" STANDOFF.
- MDN: 1 X DUAL ECCENTRALIZER.

TIME ON BOTTOM: 14:35 / 18 OCT 2019.

ALL DEPTH REFERENCE TO GROUND LEVEL.RT TO GL IS 4.45 M.

LOGGER TOTAL DEPTH: 560.75 mGL.

MMR CALIPER CLOSED 5 METRES BELOW CASING SHOE TO AVOID DAMAGE PAD.

MPD AND MDN CORRECTED FOR DENSITY CALIPER AND MUD DENSITY.

TOTAL HOLE VOLUME (HVOL) FROM TD M TO 9.625" SURFACE CASING SHOE = 14.2 CUBIC METRES.

TOTAL ANNULAR VOLUME (AVOL) FROM TD M TO CASING SHOE WITH 7" CASING = 4.66 CUBIC METRES.

MAXIMUM TEMPERATURE RECORDED 1.59 DEG AT 167.1 METRES.

REPEAT SECTION RECORDED FROM 545 M TO 385 M.

CONVEYANCE TYPE: WIRELINE.

BOREHOLE STATUS: OPEN HOLE.

RIG: SLR 167.

SERVICE REPORT NUMBER: 90518.

LOGGING CREW: ENGINEERS - K. SHIBESHI; OPERATOR: R. FRAMPTON, E. PAVIA

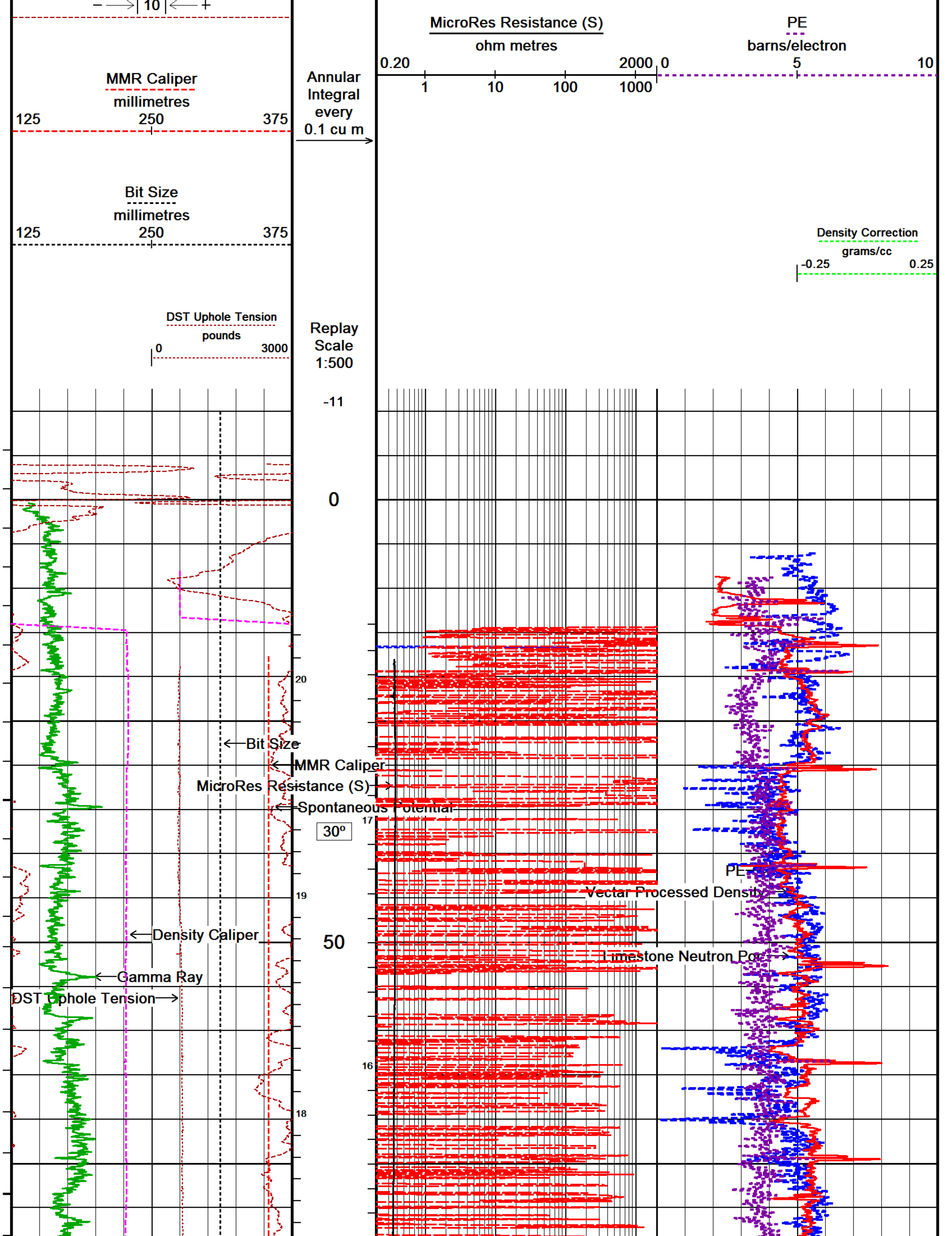
MAINLOG 1:500

Depth Based Data - Maximum Sampling Increment 2.5cm — Plotted on 18-OCT-2019 17:45

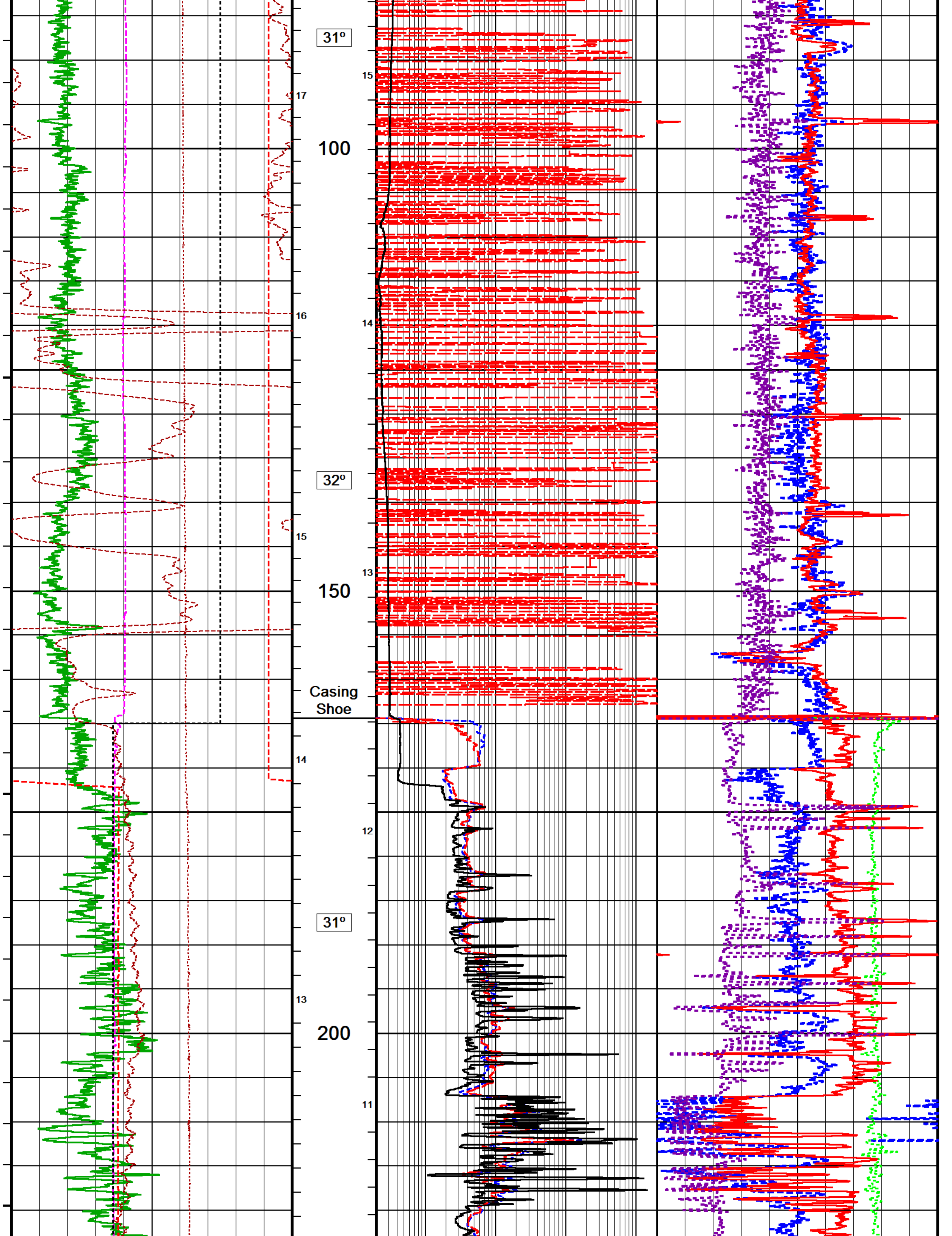
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_2.5cm_CMP_OH_20191018.dta — Recorded on 18-OCT-2019 14:38

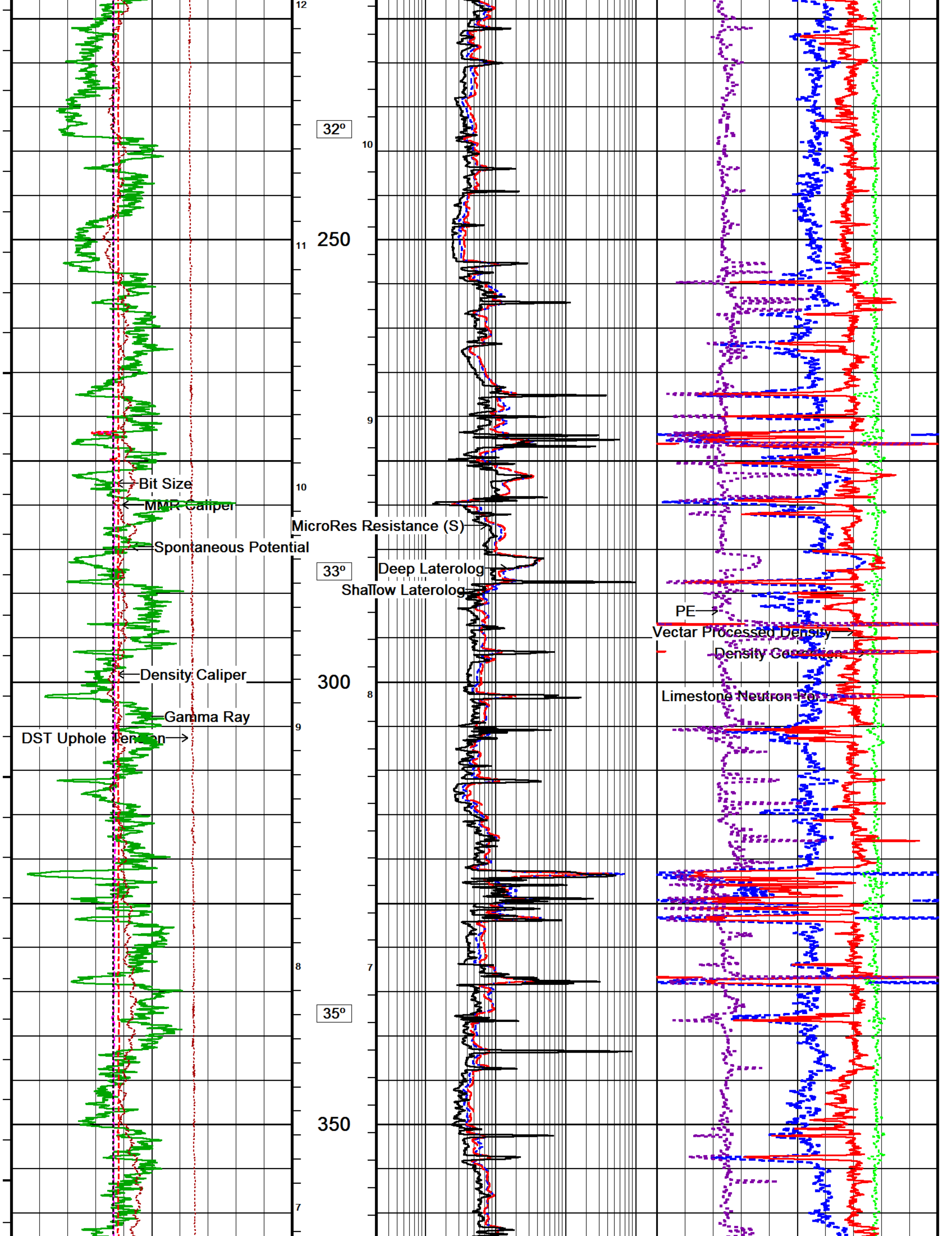
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

Track 1	Depth track	Track 2
Timing Marks, every 60.0 sec	Depth in Metres	Shallow Laterolog, ohm metres: 0.20 – 2000 (1, 10, 100, 1000)
		Limestone Neutron Por., v/v: 1 – 0.40 – -0.20
Gamma Ray, API: 0 – 100 – 200; 200 – 300 – 400	Borehole Temp in deg C	Deep Laterolog, ohm metres: 0.20 – 2000 (1, 10, 100, 1000)
Density Caliper, millimetres: 125 – 250 – 375		Vectar Processed Density, grams/cc: 1 – 2 – 3
Spontaneous Potential, millivolts	HVI every 0.1 cu m	

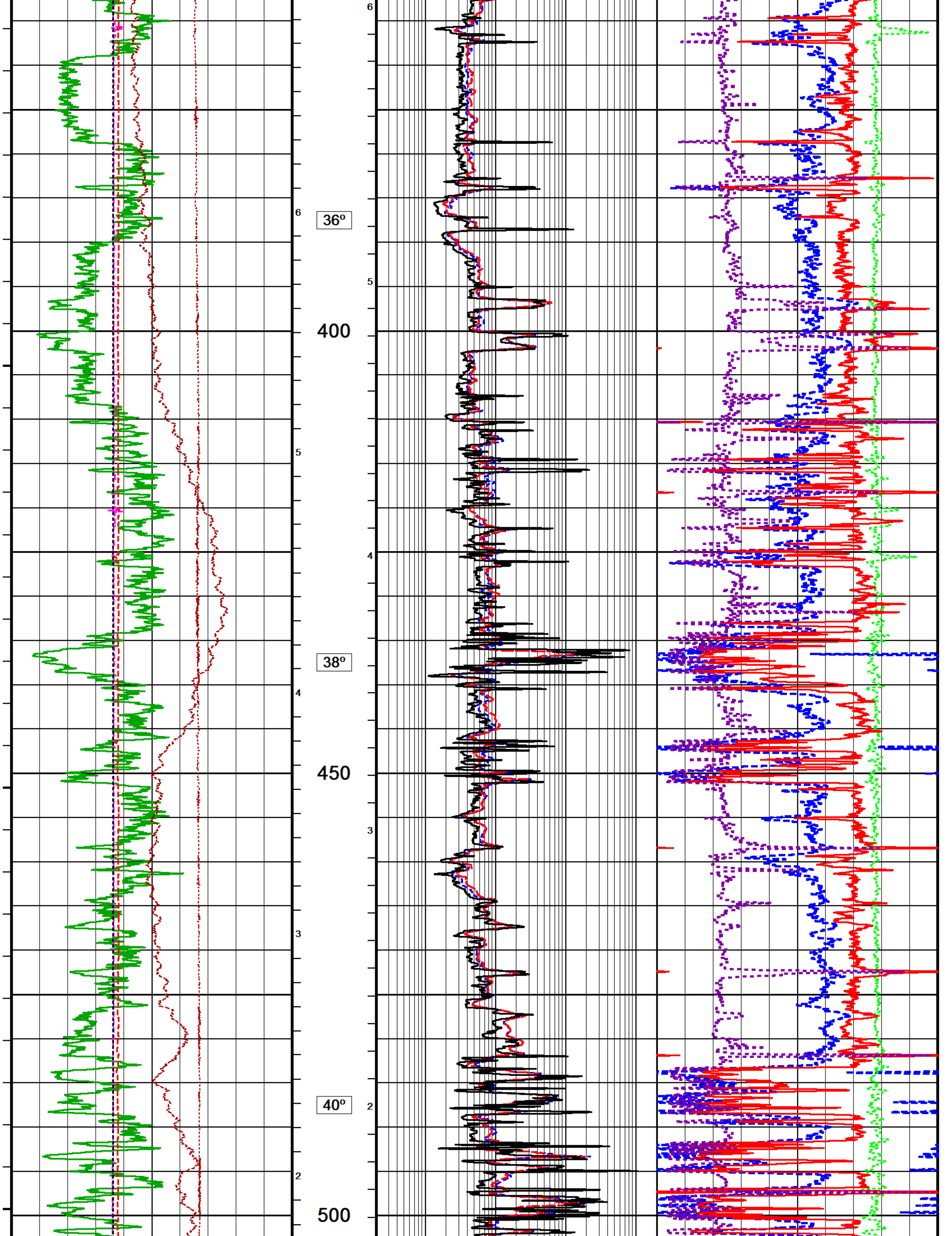


10
MMR Caliper
millimetres
125
250
375
Bit Size
millimetres
125
250
375
DST Uphole Tension
pounds
0
3000
Annular
Integral
every
0.1 cu m
Replay
Scale
1:500
-11
0
30º
50
MicroRes Resistance (S)
ohm metres
0.20
2000
1
10
100
1000
PE
barns/electron
0
5
10
Density Correction
grams/cc
-0.25
0.25
Bit Size
MMR Caliper
MicroRes Resistance (S)
Spontaneous Potential
Density Caliper
Gamma Ray
DST Uphole Tension
PE
Vectar Processed Density
Limestone Neutron Po
20
19
18
17
16

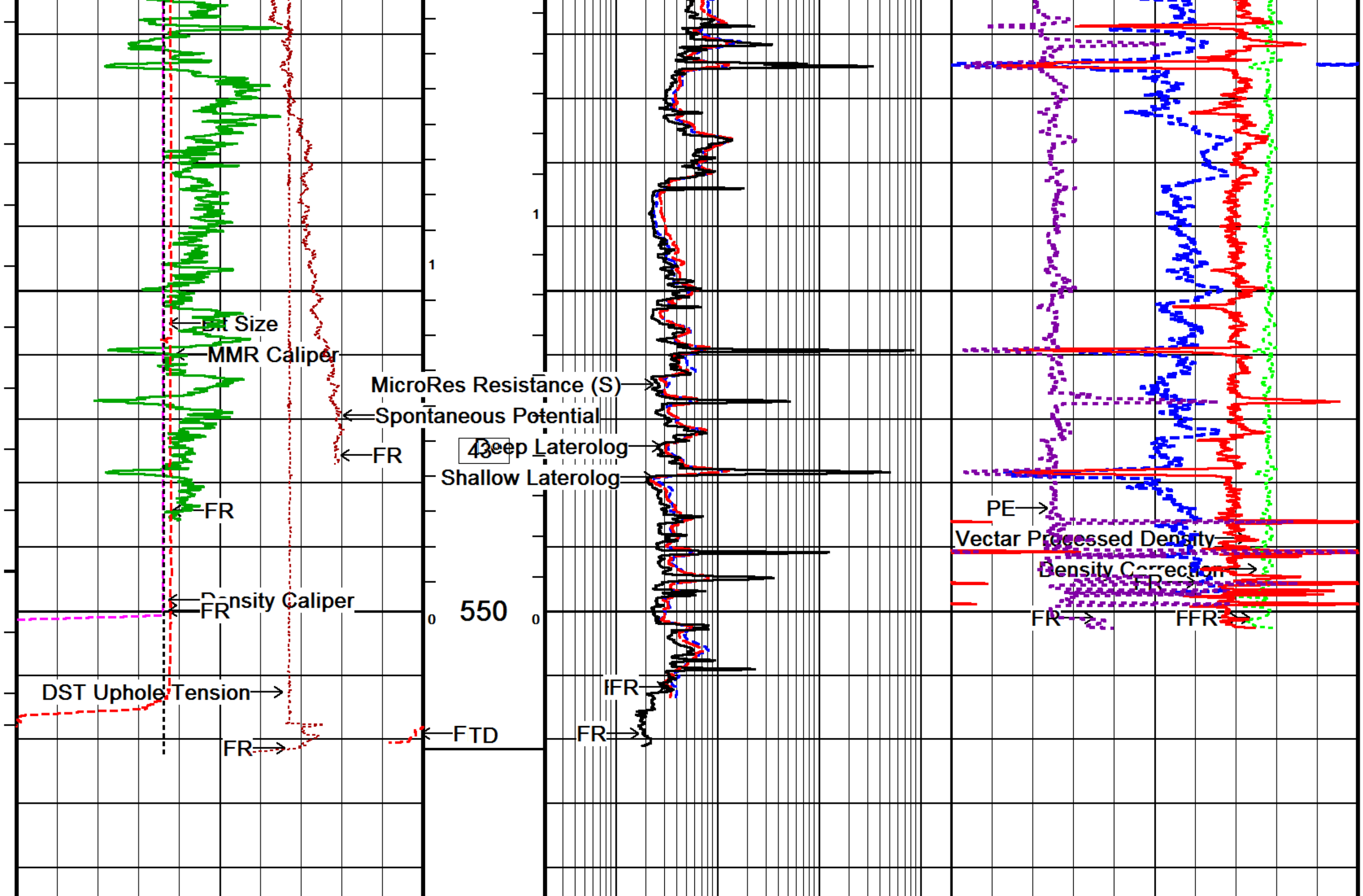

31º
100
32º
150
Casing Shoe
31º
200
17
16
15
14
13
15
14
13
12
11



32º
250
Bit Size
MMR Caliper
Spontaneous Potential
MicroRes Resistance (S)
33º
Deep Laterolog
Shallow Laterolog
PE
Vectar Processed Density
Density Correction
Limestone Neutron Por
Density Caliper
Gamma Ray
DST Uphole Tension
300
35º
350



6
36º
5
400
4
38º
450
3
40º
2
500



Timing Marks
every 60.0 sec

Gamma Ray
API
0 100 200
200 300 400

Density Caliper
millimetres
125 250 375

Spontaneous Potential
millivolts
− | 10 | +

MMR Caliper
millimetres
125 250 375

Bit Size

Depth
in
Metres

Borehole
Temp in
deg C

HVI
every
0.1 cu m

Annular
Integral
every
0.1 cu m

Shallow Laterolog
ohm metres
0.20 2000
1 10 100 1000

Deep Laterolog
ohm metres
0.20 2000
1 10 100 1000

MicroRes Resistance (S)
ohm metres
0.20 2000
1 10 100 1000

Limestone Neutron Por.
v/v
1 0.40 -0.20

Vectar Processed Density
grams/cc
1 2 3

PE
barns/electron
0 5 10

millimetres				Density Correction
125 250 375		Replay Scale 1:500		grams/cc -0.25 0.25
DST Uphole Tension pounds 0 3000				

Depth Based Data - Maximum Sampling Increment 2.5cm
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\...\TIPTON_268_2.5cm_CMP_OH_20191018.dta
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

Plotted on 18-OCT-2019 17:45
Recorded on 18-OCT-2019 14:38

MAINLOG 1:500

BEFORE SURVEY CALIBRATION

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018a.dta

General Constants All 000

Last Edited on 18-OCT-2019,12:12

General Parameters		
Mud Resistivity	0.302	ohm-metres
Mud Resistivity Temperature	25.000	degrees C
Water Level	0.000	metres
Borehole Fluid Processing	Water Level Switch	

Hole/Annular Volume and Differential Caliper Parameters		
HVOL Method	Single Caliper	
HVOL Caliper 1	Density Caliper	
HVOL Caliper 2	N/A	
Annular Volume Diameter	76.200	mm
Caliper for Differential Caliper	Density Caliper	

Rwa Parameters	
Porosity used	Base Density Porosity
Resistivity used	Deep Laterolog
RWA Constant A	0.620
RWA Constant M	2.150
SW/APOR Tool Source	0.000

Gamma Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:35

	Measured	Calibrated (API)
Background	65	43
Calibrator (Gross)	849	565
Calibrator (Net)	784	522

Gamma Calibration Tolerances MCG-E.A 580

		Tolerance (1.40 / 1.475 / 1.55)	
Ratio	1.502		Counts/API

Gamma Constants MCG-E.A 580

Last Edited on 18-OCT-2019,12:13

Gamma Calibrator Number	120	
GRC-M Calibrator Jig in Use?	NO	
Inactive Background Jig in Use?	NO	
Mud Density	1.01	gm/cc
Caliper Source for Processing	Density Caliper	
Tool Position	Eccentred	
Potassium Equivalence	Chloride	
K Mud Concentration	0.00	%

High Resolution Temperature Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:36

	Measured	Calibrated(Deg C)
Lower	0.00	0.00
Upper	100.00	100.00

High Resolution Temperature Constants MCG-E.A 580

Last Edited on 12-OCT-2019,13:36

Pre-filter Length	11

Navigation Constants MBN-D.A 170

Last Edited on 18-OCT-2019,12:13

Magnetic Declination	10.39	degrees	East

Magnetometer Parameters MBN-D.A 170

Date Of Last Magnetometer Calibration 14-OCT-2013,10:52

	X Magnetometer	Y Magnetometer	Z Magnetometer
Slope	-1.000000	1.003752	1.001092
Offset	0.001123	0.014769	-0.000632

Magnetometer Constants MBN-D.A 170

Magnetometer Calibrator Number 000

Accelerometer Parameters MBN-D.A 170

Date Of Last Accelerometer Calibration 10-OCT-2013,11:36

	X Accelerometer	Y Accelerometer	Z Accelerometer
Slope	-1.094953	-1.100967	-1.098665
Offset	0.003562	0.001559	0.005569

Accelerometer Constants MBN-D.A 170

Last Edited on 15-SEP-2019,17:36

Accelerometer Calibrator Number 000

Accelerometer Temperature Characterisation

X Accelerometer

Serial Number	1295			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.35051e-05	7.52948e-10	1.11593e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.64265e-04	3.57391e-07	7.78280e-10

Y Accelerometer

Serial Number	1239			
Calibration Date	11-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.67211e-05	2.44017e-08	7.98178e-11
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.80123e-04	3.19145e-07	8.20741e-10

Z Accelerometer

Serial Number	1310			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.13686e-05	-1.00367e-08	1.99621e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.55395e-04	3.67997e-07	9.83969e-10

Neutron Calibration MDN-B.A 270

Base Calibration on 14-OCT-2019 09:17
Field Check on 16-OCT-2019 09:38

Base Calibration

	Measured		Calibrated (cps)	
	Near	Far	Near	Far
	2976	88	3714	110
Ratio	33.984		33.764	

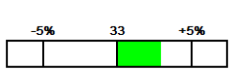
Field Calibrator at Base

	Calibrated (cps)	
	2227	3352
Ratio	0.664	

Field Check

	Calibrated (cps)	
	2234	3344
Ratio	0.668	

Neutron Calibration Tolerances MDN-B.A 270

Ratio 33.984 

Base Check	0.664	0.65 / 0.7 / 0.75	
Field Check	0.668	0.644 / 0.664 / 0.684	

Neutron Constants MDN-B.A 270

Last Edited on 18-OCT-2019,12:14

Neutron Source Id	16150B	
Neutron Jig Number	NECD200	
Air Hole Processing	Legacy	
Caliper Source for Processing	Density Caliper	
Stand-off	0.00	mm
Mud Density	1.01	gm/cc
Limestone Sigma	7.10	cu
Sandstone Sigma	4.26	cu
Dolomite Sigma	4.70	cu
Formation Pressure Source	None	
Formation Pressure	N/A	kpsi
Temperature Source	MCG External Temperature	
Temperature	N/A	degrees C
Mud Salinity	0.00	kppm
Salinity Correction	Not Applied	
Formation Fluid Salinity Source	None	
Formation Fluid Salinity	N/A	kppm
Barite Mud Correction	Not Applied	

Laterolog Calibration MLE-C.A 109

Base Calibration on 11-OCT-2019 10:01
Field Check on 11-OCT-2019 10:06

Resistor 1 (ohm)	Resistor 2 (ohm)
0.0	1000.0

Base Calibration

	Measured		Calibrated (ohm-m)	
Channel	Resistor 1	Resistor 2	Resistor 1	Resistor 2
Shallow	0.0	999.4	0.0	1284.4
Deep	0.0	999.3	0.0	795.7
Groningen	0.0	999.1	0.0	808.4

Channel	Base Check (ohm-m)	Field Check (ohm-m)
Shallow	46.2	46.2
Deep	28.6	28.6
Groningen	232.8	232.8

Laterolog Calibration Tolerances MLE-C.A 109

Shallow Resistor 2	999.4	-3% / 985.0 / +3%	ohm
Deep Resistor 2	999.3	-3% / 985.0 / +3%	ohm
Groningen Resistor 2	999.1	-3% / 985.0 / +3%	ohm
Shallow Base Check	46.2	-2% / 46.23 / +2%	ohm-m
Deep Base Check	28.6	-2% / 28.64 / +2%	ohm-m
Groningen Base Check	232.8	-2% / 232.83 / +2%	ohm-m
Shallow Field Check	46.2	-2% / 46.2 / +2%	ohm-m
Deep Field Check	28.6	-2% / 28.6 / +2%	ohm-m
Groningen Field Check	232.8	-2% / 232.8 / +2%	ohm-m

Laterolog Constants MLE-C.A 109

Last Edited on 12-OCT-2019,13:26

Profiling Laterolog Type	Type 0	
Laterolog Output Filter	Logarithmic	
Profiling Limiter	On	
Median Filter	On	
Squasher Start	40000	ohm-m
Shallow Laterolog K Factor	1.2844	
Deep Laterolog K Factor	0.7957	
Groningen Laterolog K Factor	0.8084	
Interference Rejection	50 Hz	

Interference Rejection	50 Hz	
SP Connection	SP Bridle Electrode (Lower)	
Groningen Connection	Groningen Electrode (Upper)	
Profiling Deep Input	N/A	
Borehole Correction Constants		
Bridle Type	Standard	
Sonde Position	12.7	millimetres
Hole Size Source	Density Caliper	
Hole Size Constant Value	N/A	millimetres
Rm Source	Global Value: Temperature Corrected	
Temp. for Rm Corr.	MCG External Temperature	
Apparent Porosity and Water Saturation Constants		
Archie Constant (A)	1.00	
Cementation Exponent (M)	2.00	
Saturation Exponent (N)	2.00	
Saturation of Water for Apor	100.00	percent
Resistivity of Water for Apor and Sw	0.05	ohm-m
Resistivity of Mud Filtrate for Sw	0.00	ohm-m
Source for Rt	0.00	
Source for Rxo	0.00	

SP Calibration MLE-C.A 109

Field Calibration on 12-OCT-2019,13:26

	Measured	Calibrated (mV)
Reference 1	100.0	100.0
Reference 2	-100.0	-100.0

Micro Laterolog Calibration MMR-B.A 126

Base Calibration on 10-OCT-2019 16:30
Field Check on 10-OCT-2019 16:33

Resistor 1 (ohm)	Resistor 2 (ohm)
0.0	10000.0

Base Calibration

Measured		Calibrated (ohm-m)	
Ref 1	Ref 2	Ref 1	Ref 2
0.0	9867.3	0.0	128.0

Base Check (ohm-m)	Field Check (ohm-m)
5.2	5.2

Micro Laterolog Calibration Tolerances MMR-B.A 126

	Value	-Tol	Nominal	+Tol	Unit
Ref 2	9867.3	-3%	9900.00	+3%	ohm
Base Check	5.2	-2%	5.2	+2%	ohm-m
Field Check	5.2	-2%	5.2	+2%	ohm-m

Micro Laterolog Constants MMR-B.A 126

Last Edited on 13-OCT-2019,17:01

Pad Type	6 in Solid Nylon B23059	
Standoff Offset	12.7000	millimetres
Micro Laterolog K Factor	0.0128	
Micro Laterolog Rm K Factor	N/A	
Mudcake Thickness Correction Constants		
Mud Cake Source	Constant Value	
Mud Cake Thickness	10.1600	millimetres
Mud Cake Thickness Caliper	N/A	
Mud Cake Resistivity	0.1500	ohm-m
Mud Cake Resistivity Temp.	20.00	Degrees C
Mud Cake Resistivity Source	Temperature Corr	
Temp. for Rmc Corr.	MCG Internal Temperature	

Caliper Calibration MMR-B.A 126

Base Calibration on 10-OCT-2019 16:39
Field Calibration on 13-OCT-2019 16:50

Base Calibration

Reading No	Measured	Calibrator Size (mm)
1	11571	101.83
2	14686	151.74
3	17781	202.31

3	17781	202.31
4	21116	250.47
5	25245	302.77
6	N/A	N/A

Field Calibration

Measured Caliper (mm)	Actual Caliper (mm)
209.37	213.89

Caliper Calibration Tolerances MMR-B.A 126

	Value	208.8	213.9	219.0	
Short Arm Field Cal.	209.4				mm

Caliper Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:43
Field Calibration on 16-OCT-2019 09:31

Base Calibration

Reading No	Measured	Calibrator Size (in)
1	16321	4.01
2	26493	5.97
3	36763	7.96
4	46427	9.86
5	57546	11.92
6	N/A	N/A

Field Calibration

Measured Caliper (in)	Actual Caliper (in)
7.95	7.96

Caliper Calibration Tolerances MPD-D.A 449

	Value	7.76	7.96	8.16	
Short Arm Field Cal.	7.95				in

Photo Density Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:30
Field Check on 16-OCT-2019 09:28

Density Calibration

Base Calibration

	Measured Near	Measured Far	Calibrated (sdu) Near	Calibrated (sdu) Far
Background	1134	1304		
Reference 1	45330	22156	59605	30991
Reference 2	18639	2314	24756	2530

Field Check at Base

Near	Far
1134.5	1304.4

Field Check

Near	Far
1139.8	1311.8

PE Calibration

Base Calibration

	Measured WS	Measured WH	Measured Ratio	Calibrated Ratio
Background	215	1016		
Reference 1	20159	45158	0.452	0.372
Reference 2	5757	18514	0.317	0.271

Field Check at Base

WS	WH
215.2	1016.3

Field Check

WS	WH
214.2	1018.6

Photo Density Calibration Tolerances MPD-D.A 449

	Value	Low	Nominal	High
Near Density Ratio	2.52	-5%	2.52	+5%
Far Density Ratio	20.65	-5%	21.00	+5%
PE Calibration	0.126	0.089	0.110	0.131
Near Den. Field Check	1139.8	-3%	1134.5	+3%
Far Den. Field Check	1311.8	-3%	1304.4	+3%
PE WS Field Check	214.2	-6%	215.2	+6%
PE WH Field Check	1018.6	-6%	1016.3	+6%

Density Constants MPD-D.A 449

Last Edited on 18-OCT-2019,12:13

Density Source Id 15756B

Density Source Id	15756B	
Nylon Calibrator Number	DCE-667	
Aluminium Calibrator Number	DACD-674	
Density Shoe Profile	8 inch	
Caliper Source for Processing	Density Caliper	
PE Correction to Density	Not Applied	
Mud Density	1.01	gm/cc
Mud Density Type		
Mud Filtrate Density	1.00	gm/cc
Dry Hole Mud Filtrate Density	1.00	gm/cc
DNCT	0.00	gm/cc
CRCT	0.00	gm/cc
Density Z/A Correction	Hybrid	
Precision Enhanced Density Processing	Not Applied	

Matrix density (gm/cc)	Depth (m)
2.71	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	

DOWNHOLE EQUIPMENT

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018a.dta

11C Tension Cablehead
MCC-A 1 LG: 0.73 m WT: 19.8 lb OD: 57.0 mm

11C-11B Compact Tool Adaptor
MTA-A 70 LG: 0.47 m WT: 13.2 lb OD: 57.0 mm

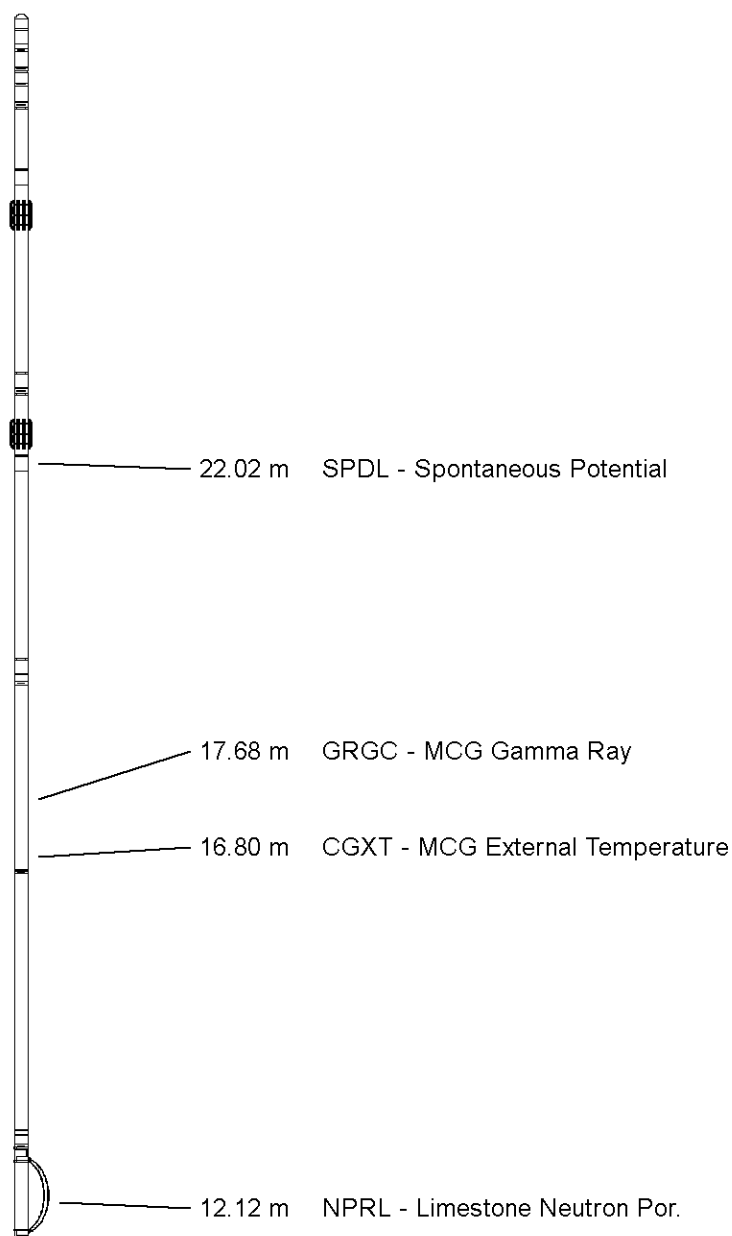
Compact Stiff Bridle Electrode Sub.
MBE-C.B 179 LG: 3.76 m WT: 77.2 lb OD: 58.0 mm

Compact Stiff Bridle Electrode Sub.
MBE-A 54 LG: 3.76 m WT: 77.2 lb OD: 57.0 mm

Compact Comms Gamma
MCG-E.A 580 LG: 2.65 m WT: 63.9 lb OD: 57.0 mm

Compact Navigation
MBN-D.A 170 LG: 3.60 m WT: 70.5 lb OD: 57.0 mm

Compact Neutron
MDN-B.A 270 LG: 1.53 m WT: 50.7 lb OD: 57.0 mm



Compact Density/Caliper
MPD-D.A 449 LG: 2.92 m WT: 90.4 lb OD: 62.2 mm

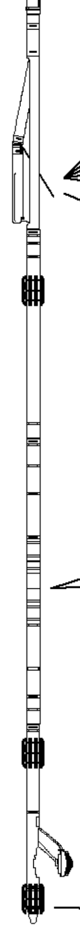
Compact Upper Guard Sub.
MUG-B.A 144 LG: 2.74 m WT: 68.3 lb OD: 57.0 mm

Compact Laterolog Electrode Sub.
MLE-C.A 109 LG: 3.76 m WT: 92.6 lb OD: 57.0 mm

Compact Micro-Resistivity
MMR-B.A 126 LG: 2.62 m WT: 81.6 lb OD: 97.0 mm

Pressure Bung + Hole Finder
HFS 3 LG: 0.28 m WT: 6.6 lb OD: 57.0 mm

Total Length: 28.82 m Weight: 712.1 lb



9.91 m	AVOL - Annular Volume
9.91 m	HVOL - Hole Volume
9.91 m	CLDC - Density Caliper
9.32 m	DEN - Compensated Density
9.32 m	DCOR - Density Correction
9.30 m	PDPE - PE
3.93 m	DSLL - Shallow Laterolog
3.93 m	DDLL - Deep Laterolog
0.30 m	MRRS - MicroRes Resistance (S)
0.30 m	MATC - MMR Caliper
Tool Zero	(0.85m from bottom)
-0.85 m	SMTU - DST Uphole Tension

All measurements relative to tool zero.

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA

Elevation Kelly Bushing	349.55	metres	First Reading		metres
Elevation Drill Floor	349.55	metres	Depth Driller	565.55	metres
Elevation Ground Level	345.1	metres	Depth Logger	560.75	metres

Weatherford®

MLL- SLL - DLL - SONIC
DENSITY NEUTRON
1:500

Weatherford

HOLE VOLUME LOG
1:500

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA
LOCATION	PL198

FIELD PRINT

Latitude	27°23'18.4689" S
Longitude	151°6'16.9779" E
UTM Easting	312 589.65
UTM Northing	6 969 111.4

Other Services
PHOTO DENSITY
DUAL LATEROLOG
MICRO LATEROLOG
COMPENSATED NEUTRON
COMPENSATED SONIC
BOREHOLE NAVIGATION

Permanent Datum M.S.L., Elevation 0.00 metres
Log Measured From GL, 345.10 metres above Permanent Datum
Drilling Measured From KB

Elevations:	metres
KB	349.55
DF	349.55
GL	345.10

Date	18-OCT-2019			
Run Number	1			
Service Order	90518			
Depth Driller	565.55	metres		
Depth Logger	560.75	metres		
First Reading	560.23	metres		
Last Reading	0.00	metres		
Casing Driller	164.55	metres		
Casing Logger	164.40	metres		
Bit Size	215.900	inches		
Hole Fluid Type	KCL			
Density / Viscosity	1.01 g/c3	26.00 sec/qt		
PH / Fluid Loss	8.50	0.00 ml/30Min		
Sample Source	MUD TANK			
Rm @ Measured Temp	0.302 @ 25.0	ohm-m		
Rmf @ Measured Temp	0.275 @ 25.0	ohm-m		
Rmc @ Measured Temp	0.453 @ 25.0	ohm-m		
Source Rmf / Rmc	CAL	CAL		
Rm @ BHT	0.218 @ 43.0	ohm-m		
Time Since Circulation	8 HRS 37 MIN			
Max Recorded Temp	43.00	deg C		
Equipment / Base	13354	ROMA		
Recorded By	K. SHIBESHI			
Witnessed By	M. GALBRAITH			
Stop Circulation	06:00 / 18 OCT 2019		.	

BOREHOLE RECORD

Last Edited: 18-OCT-2019 11:45

Bit Size millimetres	Depth From metres	Depth To metres
311.150	0.00	164.95
215.900	164.95	565.55

CASING RECORD

Type	Size millimetres	Depth From metres	Shoe Depth metres	Weight pounds/ft
SURFACE	244.475	0.00	164.55	36.00

REMARKS

RUN NUMBER 1 IS THE PRIMARY DEPTH REFERENCE LOG. ALL OTHER RUNS ARE CORRELATED BACK TO THIS LOG.

SOFTWARE ISSUE: VERSION 19.01.8879 RELEASED DATE APRIL 1, 2019.

CUSTOMER SCALES AND INTERVALS LOGGED.

RUN 1: MMR, MLE,MUG, MPD, MDN, MBN, MCG, MBE, MBE MTA, CBH TOOLS RAN IN COMBINATION.

HARDWARE
RUN 1:
- MBE: 1 X 0.5" STANDOFF ON EACH.
- MUG: 1 X 0.5" STANDOFF.
- MMR: 2 X 0.5" STANDOFF.
- MDN: 1 X DUAL ECCENTRALIZER.

TIME ON BOTTOM: 14:35 / 18 OCT 2019.

ALL DEPTH REFERENCE TO GROUND LEVEL.RT TO GL IS 4.45 M.

LOGGER TOTAL DEPTH: 560.75 mGL.

MMR CALIPER CLOSED 5 METRES BELOW CASING SHOE TO AVOID DAMAGE PAD.

MPD AND MDN CORRECTED FOR DENSITY CALIPER AND MUD DENSITY.

TOTAL HOLE VOLUME (HVOL) FROM TD M TO 9.625" SURFACE CASING SHOE = 14.2 CUBIC METRES.

TOTAL ANNULAR VOLUME (AVOL) FROM TD M TO CASING SHOE WITH 7" CASING = 4.66 CUBIC METRES.

MAXIMUM TEMPERATURE RECORDED 1.59 DEG AT 167.1 METRES.

REPEAT SECTION RECORDED FROM 545 M TO 385 M.

CONVEYANCE TYPE: WIRELINE.

BOREHOLE STATUS: OPEN HOLE.

RIG: SLR 167.

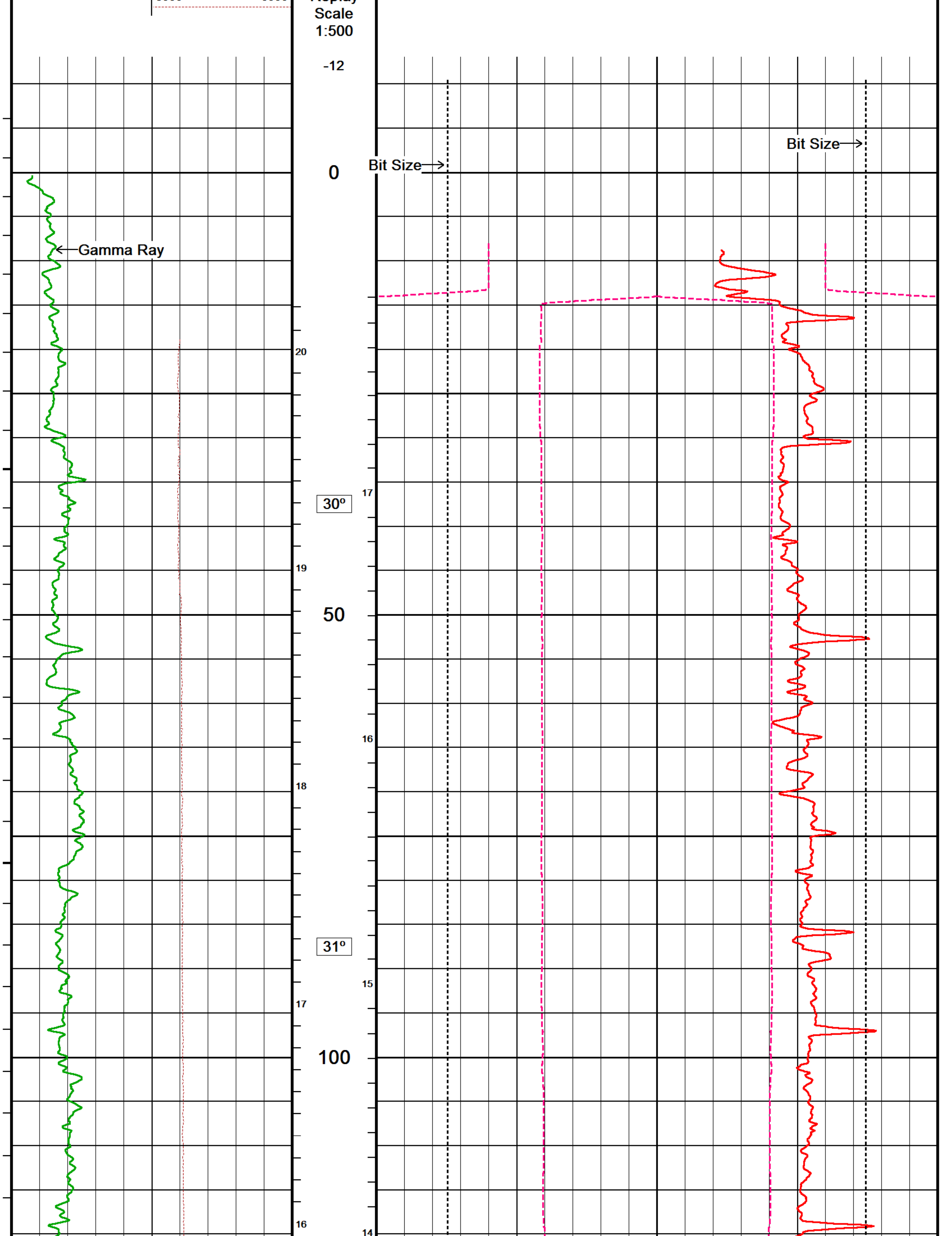
SERVICE REPORT NUMBER: 90518.

LOGGING CREW: ENGINEERS - K. SHIBESHI; OPERATOR: R. FRAMPTON, E. PAVIA

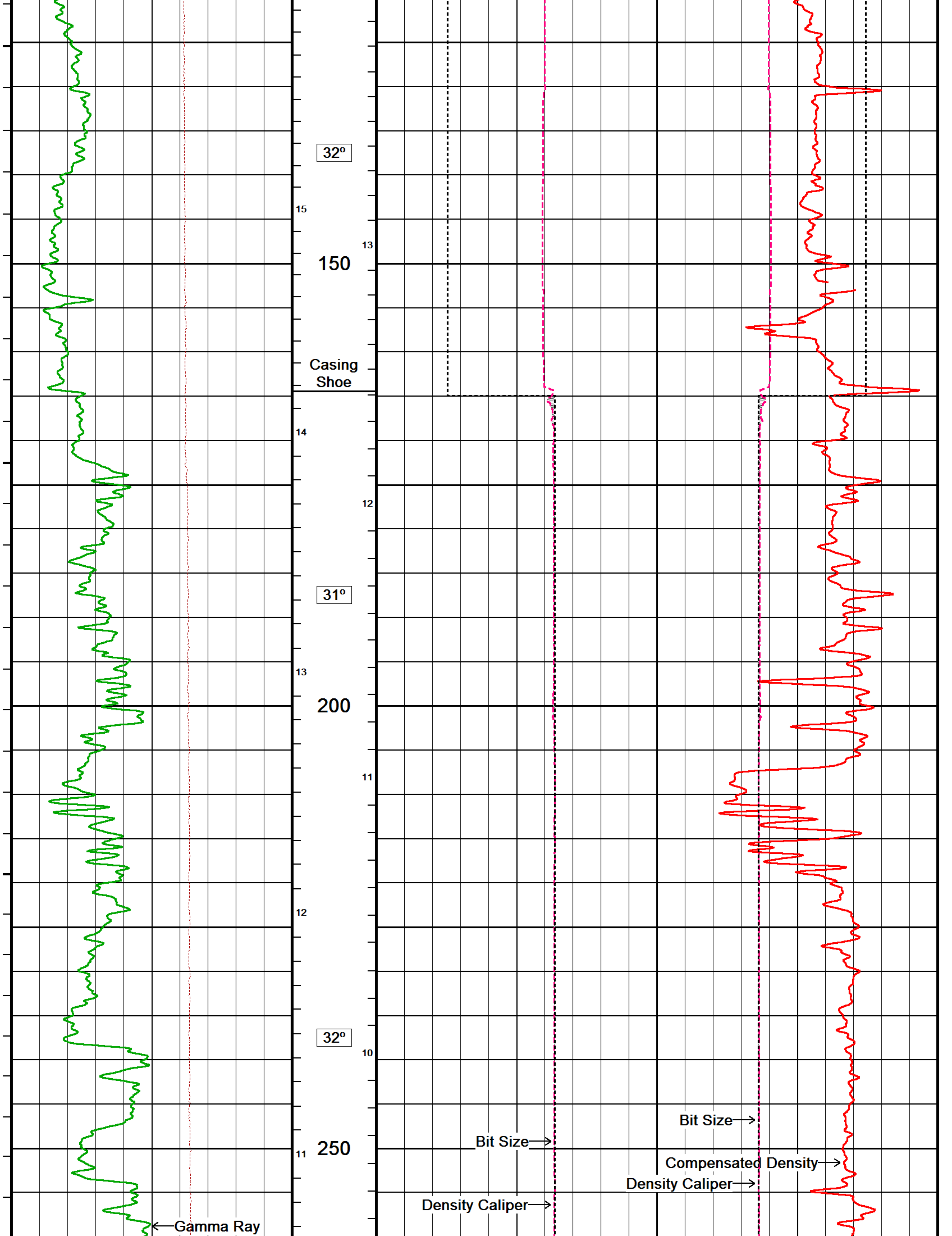
MAINLOG 1:500

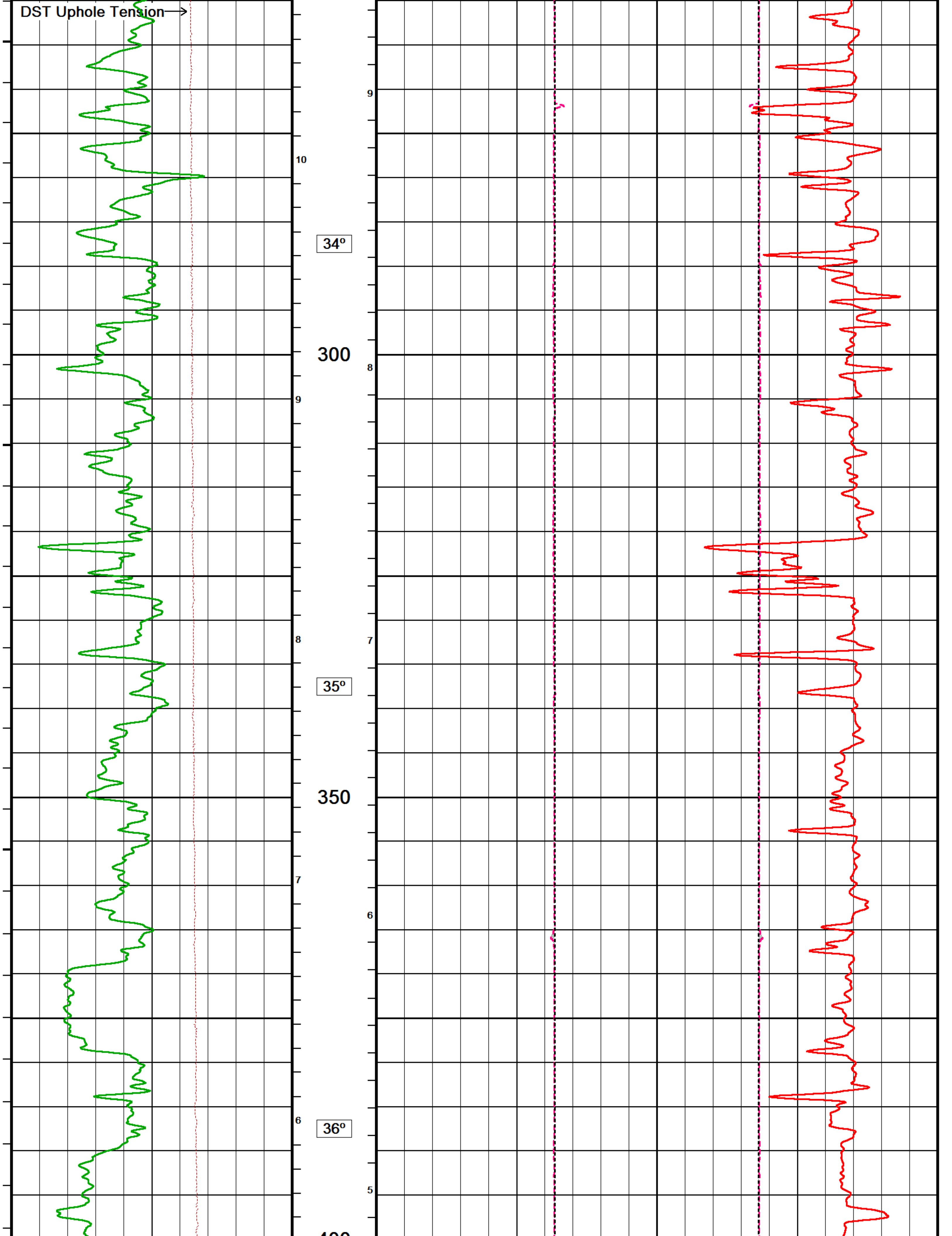
Depth Based Data - Maximum Sampling Increment 10.0cm
Plotted on 18-OCT-2019 17:47
Filename: C:\Users\ADMINI~2\AppData\Local\Temp\W...\TIPTON_268_10cm_CMP_OH_20191018.dta
Recorded on 18-OCT-2019 14:38
System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

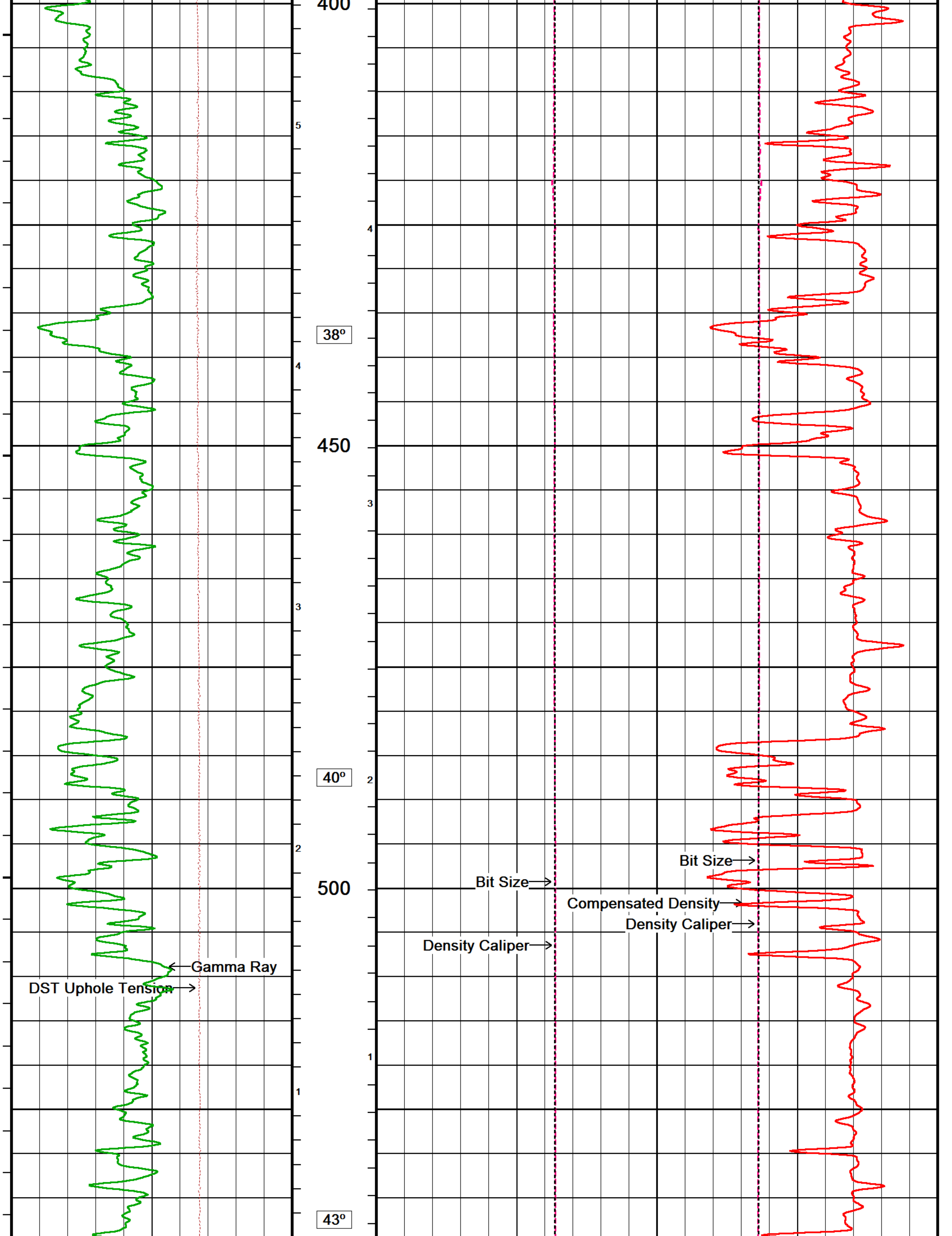
Track 1	Depth	Track 2	Track 3
Timing Marks every 60.0 sec	Depth in Metres	Density Caliper millimetres 375 – 250 – 125	Density Caliper millimetres 125 – 250 – 375
Gamma Ray API 0 – 100 – 200	HVI every 0.1 cu m	Bit Size millimetres 375 – 250 – 125	Compensated Density grams/cc 1 – 2 – 3
DST Uphole Tension pounds 0 – 3000; 3000 – 6000	Annular Integral every 0.1 cu m		Bit Size millimetres 125 – 250 – 375
	Borehole Temp in deg C		
	Replay		



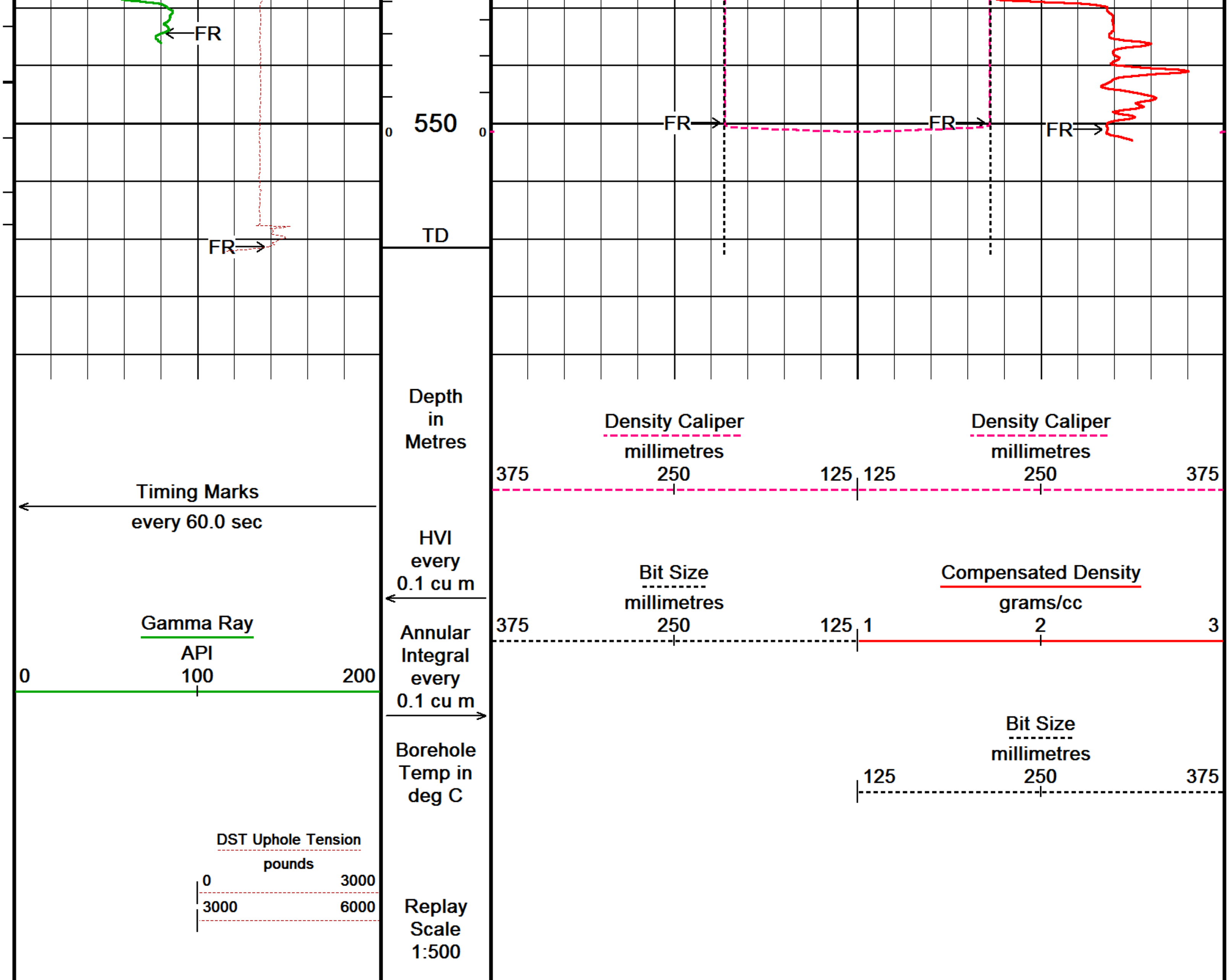
Scale
1:500
-12
0
Bit Size
Bit Size
Gamma Ray
20
17
30º
19
50
16
18
31º
15
17
100
16
14


32º
15
13
150
Casing
Shoe
14
12
31º
13
200
11
12
32º
10
11
250
Bit Size
Bit Size
Compensated Density
Density Caliper
Density Caliper
Gamma Ray


DST Uphole Tension
10
34º
300
9
8
35º
350
7
6
36º
9
8
7
6
5



400
450
500
38º
40º
43º
5
4
3
2
1
Gamma Ray
DST Uphole Tension
Bit Size
Bit Size
Compensated Density
Density Caliper
Density Caliper



Depth Based Data - Maximum Sampling Increment 10.0cm | Plotted on 18-OCT-2019 17:47

Filename: C:\Users\ADMINI~2\AppData\Local\Temp\W...\TIPTON_268_10cm_CMP_OH_20191018.dta | Recorded on 18-OCT-2019 14:38

System Versions: Logged with 19.01.8879 Processed with 19.01.8879 Plotted with 18.05.4651

MAINLOG 1:500

BEFORE SURVEY CALIBRATION

C:\Users\ADMINI~2\AppData\Local\Temp\Weatherford PreView2a\0\TIPTON_268_10cm_CMP_OH_20191018.dta

General Constants All 000 — Last Edited on 18-OCT-2019,12:12

General Parameters		
Mud Resistivity	0.302	ohm-metres
Mud Resistivity Temperature	25.000	degrees C
Water Level	0.000	metres
Borehole Fluid Processing	Water Level Switch	
Hole/Annular Volume and Differential Caliper Parameters		
HVOL Method	Single Caliper	
HVOL Caliper 1	Density Caliper	
HVOL Caliper 2	N/A	
Annular Volume Diameter	76.200	mm
Caliper for Differential Caliper	Density Caliper	
Rwa Parameters		
Porosity used	Base Density Porosity	

Resistivity used	Deep Laterolog
RWA Constant A	0.620
RWA Constant M	2.150
SW/APOR Tool Source	0.000

Gamma Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:35

	Measured	Calibrated (API)
Background	65	43
Calibrator (Gross)	849	565
Calibrator (Net)	784	522

Gamma Calibration Tolerances MCG-E.A 580

1.40 1.475 1.55

Ratio 1.502 Counts/API

Gamma Constants MCG-E.A 580

Last Edited on 18-OCT-2019,12:13

Gamma Calibrator Number	120	
GRC-M Calibrator Jig in Use?	NO	
Inactive Background Jig in Use?	NO	
Mud Density	1.01	gm/cc
Caliper Source for Processing	Density Caliper	
Tool Position	Eccentred	
Potassium Equivalence	Chloride	
K Mud Concentration	0.00	%

High Resolution Temperature Calibration MCG-E.A 580

Field Calibration on 12-OCT-2019,13:36

	Measured	Calibrated(Deg C)
Lower	0.00	0.00
Upper	100.00	100.00

High Resolution Temperature Constants MCG-E.A 580

Last Edited on 12-OCT-2019,13:36

Pre-filter Length	11

Navigation Constants MBN-D.A 170

Last Edited on 18-OCT-2019,12:13

Magnetic Declination	10.39	degrees	East

Magnetometer Parameters MBN-D.A 170

Date Of Last Magnetometer Calibration 14-OCT-2013,10:52

	X Magnetometer	Y Magnetometer	Z Magnetometer
Slope	-1.000000	1.003752	1.001092
Offset	0.001123	0.014769	-0.000632

Magnetometer Constants MBN-D.A 170

Magnetometer Calibrator Number	000

Accelerometer Parameters MBN-D.A 170

Date Of Last Accelerometer Calibration 10-OCT-2013,11:36

	X Accelerometer	Y Accelerometer	Z Accelerometer
Slope	-1.094953	-1.100967	-1.098665
Offset	0.003562	0.001559	0.005569

Accelerometer Constants MBN-D.A 170

Last Edited on 15-SEP-2019,17:36

Accelerometer Calibrator Number	000

Accelerometer Temperature Characterisation

X Accelerometer

Serial Number	1295			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.35051e-05	7.52948e-10	1.11593e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.64265e-04	3.57391e-07	7.78280e-10

Y Accelerometer

Serial Number	1239

Calibration Date	11-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.67211e-05	2.44017e-08	7.98178e-11
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.80123e-04	3.19145e-07	8.20741e-10

Z Accelerometer

Serial Number	1310			
Calibration Date	31-Jan-2013			
	B0	B1	B2	B3
Bias(g)	0.00000e+00	1.13686e-05	-1.00367e-08	1.99621e-10
	SF0	SF1	SF2	SF3
Scale Factor(mA/g)	3.00000e+00	2.55395e-04	3.67997e-07	9.83969e-10

Caliper Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:43
Field Calibration on 16-OCT-2019 09:31

Base Calibration

Reading No	Measured	Calibrator Size (in)
1	16321	4.01
2	26493	5.97
3	36763	7.96
4	46427	9.86
5	57546	11.92
6	N/A	N/A

Field Calibration

Measured Caliper (in)	Actual Caliper (in)
7.95	7.96

Caliper Calibration Tolerances MPD-D.A 449

	Value	Min	Nominal	Max	Unit
Short Arm Field Cal.	7.95	7.76	7.96	8.16	in

Photo Density Calibration MPD-D.A 449

Base Calibration on 14-OCT-2019 10:30
Field Check on 16-OCT-2019 09:28

Density Calibration

Base Calibration

	Measured Near	Measured Far	Calibrated (sdu) Near	Calibrated (sdu) Far
Background	1134	1304		
Reference 1	45330	22156	59605	30991
Reference 2	18639	2314	24756	2530

Field Check at Base

Near	Far
1134.5	1304.4

Field Check

Near	Far
1139.8	1311.8

PE Calibration

Base Calibration

	WS	Measured WH	Ratio	Calibrated Ratio
Background	215	1016		
Reference 1	20159	45158	0.452	0.372
Reference 2	5757	18514	0.317	0.271

Field Check at Base

WS	WH
215.2	1016.3

Field Check

WS	WH
214.2	1018.6

Photo Density Calibration Tolerances MPD-D.A 449

	Value	Min	Nominal	Max
Near Density Ratio	2.52	-5%	2.52	+5%
Far Density Ratio	20.65	-5%	21.00	+5%
PE Calibration	0.126	0.089	0.110	0.131
Near Den. Field Check	1139.8	-3%	1134.5	+3%
Far Den. Field Check	1311.8	-3%	1304.4	+3%
PE WS Field Check	214.2	-6%	215.2	+6%
PE WH Field Check	1018.6	-6%	1016.3	+6%

Density Constants MPD-D.A 449

Last Edited on 18-OCT-2019,12:13

Density Source Id	15756B	
Nylon Calibrator Number	DCE-667	
Aluminium Calibrator Number	DACD-674	
Density Shoe Profile	8 inch	
Caliper Source for Processing	Density Caliper	
PE Correction to Density	Not Applied	
Mud Density	1.01	gm/cc
Mud Density Type		
Mud Filtrate Density	1.00	gm/cc
Dry Hole Mud Filtrate Density	1.00	gm/cc
DNCT	0.00	gm/cc
CRCT	0.00	gm/cc
Density Z/A Correction	Hybrid	
Precision Enhanced Density Processing	Not Applied	

Matrix density (gm/cc)	Depth (m)
2.71	
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	

DOWNHOLE EQUIPMENT

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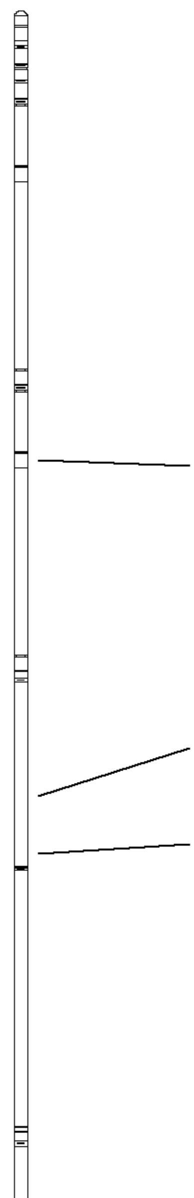

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MBE-C.B 179 LG: 3.76 m WT: 77.2 lb OD: 58.0 mm

Compact Stiff Bridle Electrode Sub.
MBE-A 54 LG: 3.76 m WT: 77.2 lb OD: 57.0 mm

22.02 m SPDL - Spontaneous Potential

Compact Comms Gamma
MCG-E.A 580 LG: 2.65 m WT: 63.9 lb OD: 57.0 mm

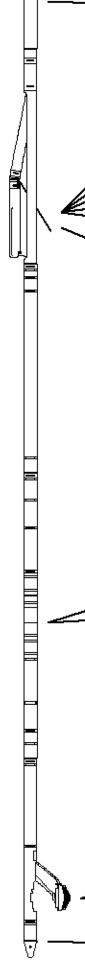
17.68 m GRGC - MCG Gamma Ray

16.80 m CGXT - MCG External Temperature

Compact Navigation
MBN-D.A 170 LG: 3.60 m WT: 70.5 lb OD: 57.0 mm

Compact Neutron
MDN-B.A 270 LG: 1.53 m WT: 50.7 lb OD: 57.0 mm

	12.12 m	NPRL - Limestone Neutron Por.
Compact Density/Caliper MPD-D.A 449 LG: 2.92 m WT: 90.4 lb OD: 62.2 mm	9.91 m	AVOL - Annular Volume
	9.91 m	HVOL - Hole Volume
	9.91 m	CLDC - Density Caliper
	9.32 m	DEN - Compensated Density
	9.32 m	DCOR - Density Correction
	9.30 m	PDPE - PE
Compact Upper Guard Sub. MUG-B.A 144 LG: 2.74 m WT: 68.3 lb OD: 57.0 mm		
Compact Laterolog Electrode Sub. MLE-C.A 109 LG: 3.76 m WT: 92.6 lb OD: 57.0 mm		
	3.93 m	DSLL - Shallow Laterolog
	3.93 m	DDLL - Deep Laterolog
Compact Micro-Resistivity MMR-B.A 126 LG: 2.62 m WT: 81.6 lb OD: 97.0 mm		
Pressure Bung + Hole Finder HFS 3 LG: 0.28 m WT: 6.6 lb OD: 57.0 mm	0.30 m	MRRS - MicroRes Resistance (S)
	0.30 m	MATC - MMR Caliper
Total Length: 28.82 m Weight: 712.1 lb	Tool Zero	(0.85m from bottom)
	-0.85 m	SMTU - DST Uphole Tension



All measurements relative to tool zero.

COMPANY	ARROW ENERGY
WELL	TIPTON 268
FIELD	TIPTON
PROVINCE/COUNTY	QUEENSLAND
COUNTRY/STATE	AUSTRALIA

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Weatherford®

HOLE VOLUME LOG

1:500